

Bio 340 Evolutionary Inference

Numbered Transcript

BIO 340 • Dr. Bret Weinstein

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Lecture 1: Curiosity in Evolution

At a glance — Weinstein opens on evolutionary inference and the role of curiosity in studying evolution.

- 1.1 Welcome. We're going to be talking today about Evolutionary Inference. We would like to make the world a simpler place for you to interpret. When you walk out into nature, it can feel incredibly complex, and the way that biology tends to be taught makes it seem like it's impossible to understand.
- 1.2 So our point here to you in these next few days is to give you some tools, an evolutionary toolkit, with which you can interpret what you're seeing in the world, when you go out and see organisms, including humans and their traits and what they're doing, and allow you to say, "I think I have some insight into what is going on there." So we're going to give three very short stories at the top of the hour here, and then I'm going to hand it over to Bret.
- 1.3 In 1994, when we had just started graduate school, we found ourselves in Costa Rica for the summer with a few other graduate students and Professor John Vandermeer, who is an ecologist with specialties in the tropics and agriculture and math. And he walked us into the lowland tropical rainforest of Costa Rica, colloquially known as "the jungle". It was dark, as it always is, even at midday in the jungle, because the trees cast so much shade. It was loud, as it is in the jungle, from the frogs and the insects and perhaps the monkeys overhead. And it was muddy, as it always is—we were in our jungle boots.
- 1.4 And he walked in front of us, and spread his arms wide, and said, "Look at all the questions!" And I looked around and all I saw was green. I couldn't see any questions at all. But over that field season, and the next, and the next, the questions started to reveal themselves. And that is what we hope to be able to help you do. On that same field trip, Brett was working on tent bats.
- 1.5 Well, I was hoping to work on tent bats. I had seen a very impressive lecture on these bats that modify understory leaves into these structures in which they roost. And I was excited. I had seen hundreds of pictures of these things and I was ready to go into the rainforest and find them and start learning about these animals. And I must have spent a week and a half looking for these things, and I got nothing. I just couldn't find them. I knew they were there, but they didn't show. And I was exasperated. And I went to John, our professor, and I said, "John, I'm not finding them. I know they're here, but I don't know what I'm doing wrong." And he said, "OK, stop looking for tents. What I want you to do is go take a hike, and I want you to look under every leaf big enough to have bats under it."
- 1.6 And I don't remember if I said to him or I just thought, "John, that is the dumbest idea I have ever heard. I've got to be better than just some automatic search of every leaf."

But anyway, I walked off down the trail, dejected, and I wasn't even out of sight of him at the point I looked under a leaf, and lo and behold, there were my animals. And there was a tent. And I learned what a tent looked like—not in a picture, but when you can move around it, when it's a three-dimensional structure. And as I continued to walk down the trail, I realized actually, they were everywhere. I just hadn't known how to see them.

- 1.7** So, third story to start us off here, this is a Moken village in the Andaman Sea in the Indian Ocean. The Moken people, who are known to outsiders as the sea gypsies of the Andaman Islands, live at the epicenter of the 2004 Boxing Day tsunami. This is the day after Christmas. An earthquake followed by a tsunami ravaged Indonesia and Thailand, the entire Indian Ocean, Sri Lanka, and Bangladesh, all the way to the east coast of Africa. And the Moken people live at the very center of it. And yet they did not die. Many people died. Hundreds of thousands of people died. They did not die, and they did not die for two broad reasons.
- 1.8** One, they retained a cultural memory of what to do when the sea recedes, and two, they know how to look. So 60 Minutes did a piece on the Moken people after the Boxing Day tsunami of 2004, and they interviewed one man who had been at sea. And so the people who were on land didn't die, and the people who were at sea didn't die, if you were Moken. And they interviewed one of the men who had been at sea, who had been right next to another boat of fisherpeople who were not Moken, who did die. The interviewer asked him, "What happened? Why didn't they die?"
- 1.9** And what he said was, "They were collecting squid. They were not looking at anything. They saw nothing. They looked at nothing. They don't know how to look." So what we want to do for you in this course is not teach you how to look. We can't do that. But we can show you how to look and show you what it's like to look and help you look through our eyes so that you can then go out into the world and look through your eyes and interpret what you're seeing.
- 1.10** So what we're going to be doing today is Bret's going to give you a whirlwind tour through microevolution, and then I'm going to give you a whirlwind tour through macroevolution. This should be fun. It should be revelatory. It shouldn't feel like medicine, although some of the macroevolution stuff might. But it will then provide the tools to do what we're going to do for the rest of the course, which is play what we're calling evolutionary jeopardy. Everything that you see, every organism that you see, every trait that you see, is an evolutionary answer to some ecological question. So start asking the why questions whenever you see organisms, and you will learn to think evolutionarily. I hand it off to you.
- 1.11** All right. Man, are we all lucky. Just think about this. Not only do you live on the only planet in the known universe that sustains life, but you are here as a member of the only species who can understand what that even means. And not only that, but you're here after we have begun to explain deeply what it all is about. That is an amazing

opportunity. You live on a planet full of miracles, and I should say they're not really miracles. They're miracles in every way but one. The one is that they're comprehensible. We know something about the process that produces them.

- 1.12** But if you can learn to see them as near miracles, if you can learn to understand in what way they function and why they behave in the ways that they do, then the world is your oyster. Everywhere you go, there are creatures doing fascinating things, creatures built in fascinating ways. And it is simply a joy to look at them and to understand something about what they must be up to. So that's the good news. That good news, that good luck, comes with one more piece of good luck, and then a bitter pill.
- 1.13** Second piece of good luck is that biology, unlike chemistry and physics and the math that underlies them, is intuitive. Creatures are absolutely obligated to make sense. The ones that don't make sense, they don't stick around. We don't see them. All the creatures that exist make nothing but sense all day long. So the fact that creatures make sense and make sense—yes, the universe, the physical universe, has to work, but it doesn't have to make sense in the same way, right? A quasar doesn't succeed or fail. A galaxy isn't trying to be a spiral; a star isn't trying to shine. But creatures are trying to do something. And you, even just sitting here, are trying to do something. You're trying to understand, right? So you know all about striving because you're doing it all the time. And that means that when you look at other things that are striving, you have some basis to understand them.
- 1.14** The bitter pill that goes with all that good luck about biology being comprehensible and you living at a marvelous time on a planet full of miracles—the bitter pill is this. All of those miracles, every single creature that has ever lived, and presumably every creature that has evolved on any other planet that might have them, has the same purpose. They all do, whether we're talking about a virus or a fungus or a beautiful orchid or a majestic orca, it doesn't matter. They're all doing the same thing, and that's bad.
- 1.15** It can't be much of a purpose if all of those creatures share it, right? And the reason that this is a bitter pill is that you share it, too. Now, at the very end of this series, we'll talk just a little bit about whether or not you might be in a position to swap out that not-very-good purpose for one that's worthy of you. But for the moment, just realize we are all doing this thing, even though we don't know it. Now let's talk about what that purpose is, just briefly.
- 1.16** The purpose is not the one that we typically talk about in evolutionary biology. Typically evolutionists would say something like, "The purpose is reproduction, reproductive success." We would say that's fitness and that's not right. It's not entirely wrong, but it's not right. The actual purpose is to lodge one's genes as deeply into the future as you can get them, right? Now you don't have a ton of control over what happens 1000 years from now. You have a lot of control over what happens 5 minutes from now. And

so we're very much biased towards the present. But the purpose is to lodge one's genetic spellings, even though we're the only creature that even knows we have genetic spellings, but we behave in ways to lodge those spellings as deeply into the future as we can get them.

- 1.17** Now, the reason that biologists and especially evolutionists will often talk about reproduction as if it is the point, is that reproducing is often a very good way to get your genes at least a little bit into the future. And so if you treat that as the goal, you'll usually be right about creatures. But the problem is, you will occasionally be perfectly stumped because some creature that should be reproducing because that's the purpose isn't. And the reason it's not is, in general, that there are times when not reproducing is a better way to get your genes into the future than reproducing. And so, if you want to understand the sum total of biology, you have to realize what the goal actually is, not this subsidiary goal of reproduction.
- 1.18** So, let's start with a story that you already know, so that you can see how this logic of striving makes something intuitive. We're just going to recreate a story you know. Here we have, obviously, a bee approaching a flower. Now, when I say that biology is intuitive, the only reason I could take that picture is because I knew exactly what was going to happen. I didn't need to see the bee; I just needed to see the flower to know that if I waited there, that this very thing was going to happen. Now, it might have been a different insect. But the point is, that flower told me what it was up to, right? You don't need to know terribly much to get it.
- 1.19** That flower is serving its own interests. It is trying to lodge its genes as deeply into the future as it can get them. How is it doing that? Well, in this case, it's hiring an insect. It is providing nectar and pollen to an insect as an enticement for that insect to come and visit. Now, that flower doesn't know anything. It doesn't have cognition, but it effectively knows that if that bee does visit this flower, that it is going to get covered in pollen. Now, some of that pollen will end up delivered to baby bees, but other parts of that pollen load are going to get delivered to the next flower that this bee visits. And if that next flower is from the same species, that will accomplish a reproductive event across a great distance.
- 1.20** Now, this explains the features of that flower. These petals, for example —they don't really serve a purpose of their own. They don't photosynthesize, they don't collect water. They are not purposeful in that sense. But the reason that the plant has invested in them is that they function like a giant billboard, right? They alert this bee from far off that there is a reward here, right? That is a for-hire sign that this plant has put out.
- 1.21** And the bee, of course, you can explain its structures. It doesn't really care about the flower. It needs to feed its own young, and it needs to get the energy to fly. And so it's serving its own interests, and that explains its wings. It explains its halteres, the reduced wings in the back that help it navigate and steer. It explains the perceptual apparatus that it has. All of those things are means to the same end. You get it? The

petals of the flower are means to an end. The eyes of the bee are a means to an end. And it's the same end—lodging genes as deeply into the future as you can get them.

- 1.22** OK, so now we have a creature that's probably familiar to you, but less so. This is the proboscis monkey. And I put the proboscis monkey here because I want to give you a tool for your toolkit so that you can begin to think about how evolution works. And I should probably say up front that when we say evolution, when you read evolution in the newspaper in general, it's probably actually the wrong word. There's a number of different ways that evolution can happen. Most of them don't matter. They're not interesting. They don't produce patterns. The one that matters is adaptation. Usually when we say evolution, we'd be better off to say adaptation.
- 1.23** Now, what is adaptation? Adaptation is a combination of two things. It's a combination of selection, which is a filter—a filter that lets certain things through. But if I give you an example that just contains selection, you'll see why it's not adaptation. If you have, for example, a grazer—deer, for example—if deer are eating plants in a field and they are preferentially eating the plants that are higher off the ground and they're not getting the ones that are really low, then after the deer have moved on, there will be a bias in that field towards the low plants, right?
- 1.24** But there's no adaptation. There was selection. The deer were selecting against the taller plants, but there was no adaptation. Now, if we add to that story, the second component, in addition to selection, we add heredity, where the low plants that were not eaten by the deer are able to outcompete the taller plants that have now suffered this great loss because they've been chopped off by these grazers, now we begin to get the stuff of adaptation. It is the cumulative coming together of traits that are successfully allowing creatures to lodge their genes into the future. That is adaptation. This is the force that causes things to better and better match the opportunities in their environment.
- 1.25** OK, so here we have a proboscis monkey. Now I'm going to use the proboscis monkey to give you a test for adaptation. Now, we could spend a lot of time talking about the details of the test, the limits of the test, the strains of the test. But for the moment, I just want to give you the test so that you'll have it, and then you can start to play with it, and we can use it on the proboscis monkey here.
- 1.26** The test says that we do not have to worry too much when we go into nature and we see a creature that has a structure on it or has an enzyme or is engaged in some kind of behavior, if the structure or the enzyme or the behavior passes a relatively simple test. And the test is, "Is the thing complex?" Right? Complexity is produced by this process of adaptation. So, in and of itself, complexity should already have us in the mindset that what we are looking at was produced by adaptive evolution.
- 1.27** The second thing is, "Does it have some kind of a cost or a risk that is variable?" That the amount that different individuals have of this expense is visible, right? In other

words, if we see an animal, even if it's a fossil, right? If we see a fossil pelycosaur, a reptile with a large sail on its back, a sail that we do not know the purpose of, we can't and we can't test it because the creatures are extinct. But if we look at that thing and we know that different individuals had different sized sails, then we know they were investing differently in this object, right?

- 1.28** Now, the third piece of the test is, "Does it persist over evolutionary time?" Now, if you've got something that is complex, that has some sort of a cost or a risk associated with it, different individuals have different levels of cost or risk, and yet it sticks around over evolutionary time, well, you know, it must be serving a function, right? Because if it didn't serve a function, then the creatures that were spending less or experiencing less risk from this object or this behavior would outcompete the ones that were spending more. And the fact that they don't outcompete the ones that are spending more, which we know because the structure remains over thousands or millions of years, that tells us that we are very likely looking at an adaptation. And, in fact, what we should think is that the burden of proof is on anyone who says it isn't adaptive.
- 1.29** Now, you will notice, if you look at creatures, that almost everything passes this test. Almost everything you could name about creatures passes this test. On the last day, we will talk about some exceptions, but for the moment, when you look at an animal, a plant, a fungus, you are looking at a collection of adaptations, right? You are not looking at a wasteful process, which leaves you with a puzzle here. You're looking at a proboscis monkey with a dramatically enlarged nose. Do you know what that nose is for?
- 1.30** Smelling?
- 1.31** Well, yes, but what if I told you that the female also does smelling and her nose doesn't look like that? That's a hint about what it's probably about. But the purpose here is just to say, look, you know that that nose is complex, right? The tissues involved are highly organized. The developmental program that produces it is very complex. It obviously costs something. It's made out of the same materials that the rest of the animal is made of. And those materials could be repurposed to something else.
- 1.32** They could be repurposed—the energy in them could be repurposed to trying to gain mating opportunities. The materials in it could be repurposed into muscle. So we know that these animals are spending on the structure, and we know that it is a long-standing expense because it's distributed across the entire species, and the species is not a new one. So we can presume that that strange-looking nose is there for some adaptive reason. That's the easy part.
- 1.33** The harder part is the question of, OK, if I know it's there for a reason, can I figure out what the reason is? And we will come back to some ways in which you might approach that question shortly. Oh, I should also say, before I leave this slide, that the reason that this test works is because trade-offs underlie all of biology. That is to say, we know

that a creature spending on a structure like that is doing it for a reason, because there are competing interests that could equally well do that thing. So trade-off should be on your mind.

1.34 OK, now we can play the same game here. This is an eyelash viper, a very interesting creature. Now, my claim is that these strangely enlarged scales, which give the animal its name—the scales over the eye of the eyelash viper —are presumably an adaptation. Now, we know that because, again, the developmental process that produces those scales is complex. The scales themselves have a highly ordered nature. We know that they cost something to build. We know that animals have different size eyelashes, and we know, again, that they have persisted over evolutionary time. So in some way, this is an adaptation.

1.35 As far as I know, nobody knows what it's for. Now, I have a hypothesis, which does not mean that I believe it to be true. It means that I have an explanation that could conceivably be true and is in need of a test. So, one thing that is very interesting about this particular creature is that, A, I should tell you that picture does not give you a sense for how small this snake is. It's a very tiny little snake. And what they do is they climb up trees and they find a place to sit. They are ambush predators, and they will sit, for example, on a flower. And they will wait for creatures that are coming to the flower—vertebrates, in fact, sometimes birds—and they will strike at them when the animal approaches.

1.36 Now, in order to do that, they need to be camouflaged. But the crazy thing about these snakes is that even within a brood—that is to say, siblings—they can be very different colors. Now, the hypothesis—and again, I'm not saying I think this is true; I'm saying I'm sure it could be true; there's a way in which this could be true—that that animal may be using those scales that hang above its eye to look for places that it could go in which it would be camouflaged. In other words, if it can see those scales as it looks for objects where it might go sit, then it can compare its own color to the color of some object, which would be very important if you were trying to hide from something like a bird, which is likely to have very good color acuity. So again, that's a hypothesis I hope will someday be tested.

1.37 All right, now, here we have a spider. I don't know what spider it is. This is a spider that struck me as I was crossing a stream one day, and it was sitting there majestically on this rock. And the reason I've put it here is because I want you to realize you don't need to know what animal it is to see the striving, right? You can see the striving in the structures here. You see these conspicuous hairs that are sticking off these strangely elongated legs. There is some explanation for all of it.

1.38 Now, I don't know what it is. It might take you a PhD's worth of work to figure it out. As far as I know, nobody studied this animal, like so many of the animals in the rainforest. But the point is, you, were you crossing the stream, could look at that spider and not feel like, "Well, I don't know what animal that is," because you don't need to know

what animal that is. What you need to know is that it is striving to do something you understand very well, because you're doing it. And every creature you do understand is doing it, right? It's the same thing.

- 1.39** OK, so here's an animal that's a little more familiar. This is a wooly monkey. And you can see how this striving, how the adaptive way of understanding biology, applies to this animal. You can see it has this beautifully thick coat, right? Beautifully thick coat, which might strike you as a strange thing for a creature that lives in an uncomfortably warm forest, shall we say, but realize this, too. This is a warm-blooded animal. And warm-blooded animals spend a huge amount just keeping their bodies at the right temperature. For most of them, it's something like 85% of the calories they take in, they spend just idling, right? That's before they move a muscle just to keep warm enough. So a coat like that insulates the animal from the breezes and other things that would rob them of energy. It makes sense.
- 1.40** You can also see these beautifully dexterous hands that it has, and you can see exactly what way they are helping this animal strive to lodge its genes deeply into the future, right? All the fuel that the animal needs is contained in things like these fruits, right? And in order to get these fruits, it uses these dexterous little hands, even choosing between the fruits. You want a ripe fruit, you don't want one that's unripe. You want one that doesn't take an extraordinary amount of effort to get to. The easy fruits are better than the more difficult fruits, right? All of these nuances of the structures of this animal serve this striving purpose.
- 1.41** And you could also see it in this tail, right? This is one of the monkeys that has a prehensile tail. And actually, I'll tell a brief story. Many years ago, first time Heather and I went to Latin America, we were in Copan, which is a gorgeous Mayan ruin at the southern end of the Mayan empire. And for whatever reason, spider monkeys, which also have a prehensile tail, were going extinct in Copan. In fact, we were there when the last spider monkey in Copan was still there, but there were no others, and he was lonely. And at one point, Heather sat down, and the animal put its tail over Heather's shoulder, which is a remarkable experience. This is a large monkey. And it was making a gesture that really did seem like a gesture of friendship.
- 1.42** But the other thing was, this gave us an opportunity to do something that one rarely gets the opportunity to do, which is to look at this prehensile tail. Now, the prehensile tail is not what you think. It's not just one big, furry appendage. At the end of that tail, there's a long pad, and on that long pad, there are fingerprints. Why are there fingerprints on a prehensile tail? Oh, same reason they're on your fingers, to give you better grip, right? This animal needs really good grip. Maybe not in this case, because that tree is dry, but a lot of the time, trees aren't dry, and the animals might be moving at a rapid pace. Or they might be under threat.
- 1.43** In fact, another time, Heather and I were watching some wooly monkeys, and a fight broke out, which is always interesting. And so we watched the fight and, lo and behold,

one of these monkeys came plunging out of this tree. Must have fallen 30, 40 meters. Boom, hit the forest floor, got up stunned, ran back up the tree. So anyway, grip matters and they have this enhancement at the end of their tails to make their grip better. The same thing that you've got on your fingers, undoubtedly borrowed from the same developmental mechanism that put it on that monkey's hands and put it on your hands.

- 1.44** But all right, now let's look at this tree. The tree has fruits on it. Now, these fruits are what we would call cauliflorous, or maybe we should call it caulifruitus. But what that means is that the fruits, instead of growing at the ends of the branches, are actually growing on the thick, woody parts. Now, in some plants, this makes total sense, right? If you're looking at a giant jackfruit, you can't put that at the end of some fragile little branch. The thing will break off, right? So you put it on the trunk and, when you see a jackfruit, this is very obvious.
- 1.45** In this case, I'm not really sure why those are on these woody parts of the plant here. Those fruits are small enough that they could be hung just about anywhere. But I do wonder, again—this is a hypothesis—I wonder if the idea is the fruits aren't too big to be suspended on the ends of those branches. But that's a big monkey, right? Now, if that monkey is the intended distributor of these fruits, then placing it where that monkey has good access and isn't going to break off the branches of the tree would make a lot of sense. So if you were going to test that hypothesis, you would want to see whether or not this tree was best distributed by monkeys large enough to break the other branches. That would at least be a supportive piece of evidence for that hypothesis.
- 1.46** OK, here we have a wood duck. Now, hopefully you can see that this fantastically adorned animal has iridescence on its head. Maybe you're lucky enough that you've spent some time looking at wood ducks up close and you've seen that beautiful, shimmery, that shimmery, glittery thing on its head, or maybe you've seen it on hummingbirds, or maybe you've seen it on butterflies because it's evolved more than once.
- 1.47** What is that? Well, here's the interesting thing about it. There's no green pigment there, right? A pigment is a chemical that absorbs certain wavelengths of light and reflects others. And that's why you see the color that you do. You see the ones that are bouncing off of it, right? That's not what's going on here. What's going on here is there are microscopic little structures on the feathers on that bird's head, and those microscopic little structures have little shelves on them. And those shelves are just tall enough so that a green wavelength of light bounces off the top of the shelf and the bottom of the shelf and reinforces, whereas the other colors cancel. There's no green there, right? This is a trick of physics.
- 1.48** Now, it's a trick of physics that requires incredible developmental precision to produce it, right? Imagine the precision it takes to get a shelf on a feather just the right size to

reflect green and only green light. I mean, that's amazing, right? This is what I mean by miracles, except for the fact that we can understand them, right? It's a miracle, except we can understand it.

- 1.49** But again, is it an adaptation? Well, sure, it's definitely an adaptation, or at least presumptively. It's presumptively an adaptation because all of the machinery that goes into producing those little shelves and the shelves themselves, they're all made out of material. That material has a cost. That cost, the expense, could be repurposed to something else, and yet it isn't. Over the course of millions of years, this animal has continued to have an iridescent head. So that tells you you're looking at an adaptation of some kind, right? This is an animal striving to lodge its genes deeply into the future, and it's doing it by playing tricks with wavelengths of light.
- 1.50** OK, that's pretty cool. All right, here's another one. Here we have a cat delivering a prey item. Now, this is not a structure. This is a behavior. But this behavior also passes the adaptive test, right? This animal is spending to deliver a prey item very often to its owner. I, initially, when I started thinking about this behavior, thought that it might be delivering it to its parent, but I'm no longer convinced that that's what's going on. But in any case, the effort of delivering a prey item so that it can be inspected is substantial. The animal could just eat the item and get the energy from it. Instead, it's going to go to the extra effort of delivering it and possibly losing it to a competitor. Who knows? The behavior is complex.
- 1.51** We've all seen it. So it's widely distributed. That means it's persisted over evolutionary time enough to spread across an entire population. That's an adaptation of some kind. This animal is somehow striving to lodge its genes deeply into the future by doing this strange behavior that doesn't have an obvious explanation. Now, I had an idea a long time ago. My hypothesis was that the animal was delivering these prey items to its parent because its parent knew what was worth hunting in the local area, because its parent had been there hunting. And there might be things that you could perfectly well eat, but the expense of catching them was so high that it wasn't worth it. You should direct your efforts somewhere else. So I sort of thought probably what the animal is doing is delivering sample prey items so that the parent could say, "Yeah, those are good. Not so much on those."
- 1.52** Student of mine came up with a much better hypothesis, I think. And the hypothesis is that this has to do with the domestication of cats. Now, cats are a really interesting domesticated animal because they're kind of only partially domesticated, right? The way they became domesticated is hanging out with people who were trying to protect grain stores. And so the fact that they were hunting the rodents and other things that threaten the grain made them not only tolerable, but worth supporting. Right? It made sense for people to feed cats who were hanging around when there wasn't enough prey in order that the cats would be there when the grain stores were actually threatened.

- 1.53 And so my student Tyler hypothesized that maybe what's going on here is that the animal is demonstrating that it's a good mouser, right? That it's demonstrating that to people, because people don't always support the cats around them. Sometimes they kill them. And so an animal that is advertising, "Hey, I'm actually worth it," is maybe more likely to get its domesticated genes lodged more deeply into the future. And that's how they got here. That's how we end up seeing the behavior. Again, I don't know that this is true, but it's a valid hypothesis. It deserves to be on the table, and it deserves a test.
- 1.54 OK, here we have an owl fly. This owl fly is here roosting on a rootlet that has descended from the canopy. And it has, I hope you can see, two absolutely extraordinary antennae that it has not fanned out into the world. It has them pointed directly up along the object that it is roosting on. Now, hypothesis here. First of all, does it pass the adaptive test? Sure. I don't know how well you can see, but those are some highly structured antennae with some very sensitive little pads on the ends, segmented so beautifully, and all of the architecture involved in being able to move them, right? This is a very complex structure, and it obviously has an expense, and that expense could obviously be reduced by making them shorter. And yet here is this animal with these incredibly long antennae, which have to make them more vulnerable. It's just got to.
- 1.55 So why? Well, here's what I noticed. And again, I don't know the answer, but I know what the animal is striving to do. And in order for this animal to manage to lodge its genes into the future, it has to not get eaten by things. Now, it's not very conspicuous, right? You have to get really close up to this thing to see that there's anything but a rootlet hanging in front of you. But it is vulnerable to something coming down the branch, right? Something that crawled down it could catch it off guard and boom. But when you have antennae that long, that thing can't get very close before this thing gets an early warning. So the hypothesis is the reason this animal is spending on that elaborate, vulnerable structure is that it's an early warning device, and that that explains why it's sitting here with those things aligned right along the path that some predator would have to take to get to it.
- 1.56 OK, next reason this slide is here is to remind me to tell you that all of this striving to get your genes deeply into the future comes in two general flavors, OK? One general flavor is an ecological striving, right? Striving about things in the environment. How do I find more food? How do I not attract the attention of a predator? How do I fend off the pathogens that might attack my physiology, right? These are the things that we typically refer to as natural selection, right?
- 1.57 The other flavor of striving to get your genes deeply into the future, the other cause of adaptation, is what I will call a kind of social selection, which is generally sexual in nature. Doesn't have to be, but it is generally sexual in nature. So here we have two red deer in rutting season. They are adorned with antlers, which are bones that grow

seasonally and then drop off at the end of the season and then grow back a little bigger next year. Right? So it has these very expensive objects on its head. It's got an amazingly complex program that allows these bones to grow out of the top of its skull and then drop off at the right moment. Right? No doubt it's complex. The expense of the bone is obvious, especially with things like calcium being limiting in the environment.

1.58 And obviously, it persists over evolutionary time, enough that it isn't just this species either. I mean, all deer have antlers. So this one, though, clearly suggests to us that it is the result of this sexual striving. And the piece of evidence that makes this so conspicuous is that it is the males that have it and not the females. And this applies to several of the things we've looked at, right? We've just sort of generally talked about the striving to lodge your genes deeply in the future. But in the case of the proboscis monkey, that elaborate nose is a feature of only the males. That strongly implies that sexual selection is involved. The iridescence on the head of that wood duck—also only the males, strongly implying that the type of striving that's being done is a striving for mates.

1.59 OK, now within that second flavor of striving—you've got ecological striving and you've got social striving, which is usually sexual—within that sexual striving, there are two general flavors again. In this case, we have male versus male competition, right? What happens in these deer species is that males fight each other for access to females who have grouped together, largely for predator avoidance reasons. And so the males exclude each other, and whichever male wins these battles mates with the females. And that obviously contributes to his ability to get his genes into the future. However, that is not the sum total of sexual striving here.

1.60 What we have is a frigate bird, and this frigate bird has this elaborate, and I must say, spectacularly vulnerable, inflated pouch. And that is about a different kind of striving. This is about female choice. Now, I'm a little hesitant to divide these things cleanly, because in the case of the deer, we don't know for sure whether or not the females are choosing. The females might be choosing, and they might be allowing the male fight to decide who they like better, or it might be that they're completely indifferent and they're just letting the males do the thing that they do. It's very hard to distinguish between those two things.

1.61 But in the case of this elaborate red pouch on the frigate bird, we know that it is female preference that is causing the elaboration of this structure. And we won't get deeply into it today, but why it is that females would prefer that their mates have this giant, conspicuous, vulnerable sack is a bit of a paradox. It is especially paradoxical in light of the fact that by choosing males that have that ridiculous thing, they are inflicting it on their own male offspring. Right? That's a strange thing for females to do. Now, I do think females have a very good reason for doing it. I do not believe that this is some

kind of frivolous thing. I believe that this is decidedly important, but we will perhaps return to that later.

- 1.62** Here you can see the evidence for what I'm talking about because here's the female frigate bird, and she is absolutely unadorned and drab, right? Ecologically speaking, she is much better adapted. For some reason, it is worth it for this male to step away from being ecologically well adapted in order to appeal to mates. OK, again, all creatures are striving. They're all striving to do the same thing, and it explains all of the expensive, complex features that we see on them.
- 1.63** Now, here's one that is probably not terribly familiar to you. I talked to you a little bit at the beginning. I told you a story about my first encounter with tent bats. Well, here you are looking at the beast in question. These are tiny little animals, and just to orient you, I'm actually lying on the forest floor to take this picture, pointing up. And these bats are none too happy that I'm there. But they're also probably not sure that I see them, because what you can't tell is that they're in the dark, right? The flash on my camera lets you see them clearly, but they don't know that they've been spotted. Probably all day long, animals are walking under their tents and don't know that they're there. And so the bats are being very quiet.
- 1.64** But if you look carefully, you will see that each of these animals has a spear-shaped appendage on its nose. It's called a nose leaf. It's a diagnostic feature of this family of bats. These bats that you're looking at are fruit bats. Not all of the animals that have these nose leaves are fruit bats. Now, if you're beginning to get the theme here, you know that that nose leaf is the product of adaptation. Somehow these animals are striving to get their genes lodged deeply into the future and this appendage on their noses is helping, which is odd. It isn't really clear why that would help, is it? Now, if I tell you that you have a mixed group here of males and females, then you know it probably isn't sexual display. So it's probably about some kind of ecological striving, but again not really obvious why it would be that way.
- 1.65** Now, I'm not going to tell you. In this case, I believe we do know the answer, and it's a glorious one. But I want you to think a little bit about what it might be. And I'm going to tell you a couple of things just so you'll have some idea how to puzzle through it. Like I said, the whole family are the leaf nose bats. It's not the only family that has a leaf nose. There are other bats that have separately evolved a similar structure. There are also bats that have lost this structure, bats whose ancestors had it, and then selection has removed it from them, right? Remember, if you're not using something, if it's not contributing to your lodging your genes deeply in the future, then selection should economize. It should recover those resources and take off this vulnerable structure, which could get caught on something, or it could get an infection. And so we've seen that several species have lost these nose leaves.
- 1.66** I'll give you an example. Vampire bats. Vampire bats came from an ancestor that had a nose leaf like that one. And then they started feeding on blood, probably initially they

started feeding on blood when they would land on an animal that had been injured, and they would lap up the blood from the wound, and then presumably they would start augmenting that process. Maybe they would land on a wounded animal that had a scab, and they would pick at the scab to make it bleed so that they could get their food. And then that process got elaborated even more. And vampire bats now have razor sharp teeth in which they can put a slice in an animal and they can cause it to bleed even without a wound. But if you do that, you don't have a nose leaf.

1.67 These fruit bats, they have a nose leaf. So I want you to think a little bit about what that might be about. Oh, I should also probably tell you that, if you had this animal in your hand, you would know immediately that that nose leaf isn't the cartilaginous structure you're expecting. It's very supple, and it jitters about. If you have the animal in your hand, it's presumably absolutely terrified, probably thinks it's about to be consumed, and that thing is going wild like this. OK? So I promise we will return to that story. It's a very, very good one.

1.68 OK, so now I want to talk about some strategies, and I want you to begin to see how the more you understand about this adaptive striving, the simpler the world gets. OK, so on the left, I believe these are petunias. And on the right, you have —so I'm using these to demonstrate two strategies that you've heard of. On the left, you have an annual, right? Now people who have gardens love annuals. Why? Because an annual flowers itself to death. If you're trying to get flowers, you can't really beat an annual because it just keeps flowering until it's gone, and then you got to redo it next year. You got to put in some new seeds to get more of it. But nonetheless, it's a very showy entity.

1.69 Whereas this perennial, this lupin, plays a different game, right? Now the lupin has to somehow survive the winter because it's not going to come back from a seed. It's going to come back from some sort of organ that has retained integrity, and that's a harder process. So it's not going to flower itself to death. It's going to flower for a time to get some pollen exchange, and then it's going to cut it out and it's going to get down to business and it's going to collect some resources and it's going to store them in some way that, when spring comes, it can go back to work, right? So it's also a cool thing to have in your garden, but not because it's just going to flower and flower and flower. It's a cool thing to have in your garden because it'll be there on its own next year, right? You don't have to do anything about. It's a perennial.

1.70 OK. Those are familiar stories, more or less. These ones won't be. On the left, we have the nest of a bald-faced hornet. Now, a bald-faced hornet is actually a wasp, and it has a life cycle. This nest began when a queen woke up, having been fertilized in the fall. She woke up on the forest floor where she had spent the winter. She climbed up into some tree and began making this paper nest and producing offspring. And when those offspring emerged, they contributed to the building of the nest. And here you have the nest. And then come fall, there will be a reproductive generation produced. Males and

females that are capable of reproduction will leave the nest. Males will try to find females. They will either succeed and then die, or fail and then die. And females who have been fertilized will then fall asleep on the forest floor and go dormant and wake up in the spring, climb a tree, and begin making a nest.

- 1.71** That's an annual. We don't call them annuals because they're animals and we don't use those terms. But there's really no reason we shouldn't, right? That's an annual. It is overwintering as a dormant queen. It's like a seed, right? It's like a seed sitting there on the forest floor. And it has to start the whole process anew. OK. On the right, we have the nest of a thatch ant. Now, ants, bees, and wasps are all in one evolutionary group, the Hymenoptera. They're not that far apart, right? But this animal is using a perennial strategy. The thatch ant collects needles from the forest floor and it accumulates them in these large structures. Sometimes these structures can be 5, 6 feet tall. They're really impressive, especially when the ants are active on the outside, and the thing is just encrusted. It's a living mound of stuff.
- 1.72** When those bits of thatch decay, they generate heat, and that keeps the colony warm, which is part of how it gets through the winter. The other way it gets through the winter is that the foragers, who are largely out feeding on arthropods, primarily insects, bring back the forage and they feed it to the first generation of new ants who don't leave the nest. And those new ants accumulate that resource. They take in all of this food that's been collected during the warm season, and they regurgitate it as a kind of a living pantry for the colony in the winter. That's how they do it. The colony doesn't have to start anew every year. The reason that these things get 4, 5, 6 feet tall is that they're actually very old structures, right? They found a place in the habitat that they can do their thing, they do it effectively, and they just keep building, right? That's a perennial.
- 1.73** So here we're learning something, which is that selection is repeating itself. We call that convergent evolution. And convergent evolution is great when you're an evolutionist, because that's how you know what things are really about, right? When selection repeats itself, you can say, "Well, why did these two different groups do the same thing? Were they solving the same puzzle?" Very often they were. And in this case, we do see them solving the same puzzle. The puzzle is, "What the hell are you going to do about winter?" Right?
- 1.74** If you are in the far north or the far south, where winter is a very harsh phenomenon, you really have to take one of three approaches. You either got to leave, right? You could migrate out, but that's not going to work for an ant or a wasp. Works for some birds, works for the occasional butterfly. But if you're a terrestrial arthropod like that, you're not going to be able to migrate far enough to get away from winter. You could hunker down in something seed-like or very hardy queen that can sleep on the forest floor, right? You can try to do very little over the winter and just persist, have enough of your stuff, persist over the winter, that you have a start on the process next year.

- 1.75 Or you can come up with some really whizbang strategy for actually doing OK during the winter, by making hay while the sun shines and collecting enough extra and storing it in some way that you can access it when you need it, when there's nothing out there to forage on, right? Those are your choices. So it isn't surprising, given that both plants and animals have the same problem, that they would come up with solutions that are similar. And the problem is really on our front. Why do we name them differently? Why don't we say, "That's an annual and that's a perennial"? We probably should. It would simplify things. You don't need multiple terms. You've already got the term.
- 1.76 OK, now I want to throw you a curveball. OK, we've talked about annuals. We've talked about perennials. Now, this is neither. This is a biennial. Now, a biennial is a weird strategy, but it makes a ton of sense once you know where to stand. OK, so this is foxglove. Foxglove is a beautiful plant. It is the source of digitalis, the heart drug. It is the source of that heart drug because it makes a deadly toxin. You do not want to eat the leaves of this plant, and animals in general do not eat the leaves of this plant because they will die if they do. But here's the funny thing about this. This is second-year foxglove, a beautiful flowering plant. Kind of looks like an annual, right? Just vigorous, vigorous flowering, right? In fact, flowering to the point of death.
- 1.77 Here we have it in its first year. No flowers. What the hell? Why a two-year strategy, right? One-year strategy makes sense. You don't have to deal with winter. Permanent strategy, perennial, makes sense. You figure out how to deal with winter and you do it well. Why a two-year strategy? Now you have to deal with winter, but you don't even keep doing it. That's odd. All right. But think about the advantages to this plant. The first year, a seed, one of these flowers gets fertilized by usually a bee, brings it some pollen, turns into a seed, drops off the plant when the plant dies, falls to the ground, manages to embed itself, and it grows, OK?
- 1.78 The first thing it's going to do when it pokes through the surface of the ground is it's going to deploy a solar panel. That solar panel—a leaf, two leaves—is going to allow that creature to start accumulating resources. And what it will do is actually a totally stunning miracle in its own right. We don't even think about it. But this plant is going to take sunlight and is going to use it to turn a component of air into sugar and cellulose, the structural stuff. Right? Air, that's what sugar and cellulose are made out of. They're made out of air, made with sunlight.
- 1.79 That's pretty cool in its own right, but it's a hard job, right? How much of a leaf can you get out of a tiny little seed? Not much, right? It's not a good moment for that plant. It's a very dangerous moment at the point it's deployed its cotyledons. But if it succeeds and it gets like that, it can collect a fair amount of stuff, right? Now, here's what it gets to do for its biennial strategy. It gets to take the stuff that it accumulates in its first successful year and store it.
- 1.80 Now, what happens next is, the following spring, all of these leaves have died back, but it stored the energy and the materials underground. Now it can leap out of the ground

because it's got all of this headstart. It doesn't have to wait for the sun to be shining down on it. It doesn't have to turn the stuff of the air with the sunlight into the materials. It's got the materials on board so it can jump up. And jumping up, from the point of view of a plant that is making hay while the sunshine is close to the ground, is a huge advantage because it can overtop all of those annual plants that are starting from a seed, right? So the advantage of this biennial strategy comes in the second year when it doesn't have to wait. It is not held back by the slow dawning of spring. It can get to competing right away.

1.81 Now, the really interesting thing, once you see the value of that strategy as a component of striving to lodge foxglove genes deep into the future, is that lots of plants have come up with this. And actually, they're some of our favorites. So here's an example. These are radishes. Radishes do this very same thing. Now, we harvest the radishes at the end of their first year, right? So this is us repurposing their strategy for our strategy, for our striving to lodge our genes deeply into the future. The idea is the plant grows and it collects as much resource as it can in its first year, and then it drives it into a storage organ that's pretty darn delicious because of the chemicals that it loads in there to protect it from things like fungi and predators, right? It's got that spicy flavor to it, right?

1.82 We like this plant because we can predict that, at the end of its first year, what it's going to do is it's going to take everything that it can manage to recover from all of its structures, and it's going to put it into a structure that we can utilize, right? Radishes do this, carrots do this, and we harvest them at the end of the first year as part of our striving.

1.83 OK, so I think we've arrived at where I was hoping to get you for this first stage. I hope it has been clear. And what I want you to do—homework, as it were—is to take this idea of striving, to take the test of adaptation, and to start thinking about the creatures that you know and whether or not you can understand more about them with these tools. If nothing comes to mind, you can think about the leaf nose on those tent bats. Why might selection have placed a fleshy leaf on the nose of fruit bats? What could that possibly be for? And why would it no longer be useful in a bat that feeds on blood? Or you could think of these blue-footed boobies and their amazing display and dance. Whatever it takes, think about some creatures and see if you can't find the adaptive logic. Thanks.

Lecture 2: Macroevolution

At a glance — Macroevolution — large-scale evolutionary change.

- 2.1 All right. Good to see everyone. For the second part of this course on Evolutionary Inference, we're going to do a quick whirlwind tour of macroevolution.
- 2.2 Bret was talking about microevolution. The micro and macro refer to time and, to some degree, the number of individuals, therefore, that are involved. Microevolution is about change at the level of populations and smaller, and macroevolution is the reconstruction of deep history. So this is probably going to be the most medicinal-feeling part of the course. There's a fair bit of technical terms, and it takes time, just as it takes time to do what we're going to be doing for the rest of the course —learning how to think evolutionarily, adaptively, seeing traits, and trying to figure out what they're an answer to.
- 2.3 The tree thinking that we're going to be learning here today will take time. You won't have it by the end of the day, if you don't have it yet. But the more exposure you have to evolutionary trees, the more capable you'll get. It turns out that it's a tool that is a lot more useful than you might imagine, even if you don't think that you're going to be wondering, for instance, how birds and crocs are related to each other compared to squamates.
- 2.4 So I'm starting here with a simplest possible evolutionary tree. This is a three-taxon statement. There's only three taxa on the tree. And of course, if you only have two taxa, you can't draw a relationship between them because they're related. They're on Earth, they're related. So you have three taxa. And what this tree says, and we'll be walking through this a few more times this hour, is that birds and crocs, Aves and Crocodylia, including alligators and gharials and caiman and such, are each other's closest relatives because they share that node. And the squamates, the lizards and snakes, technically the lizards, which includes snakes, are the outgroup. They are more distantly related to both birds and crocodiles than birds are to them or than crocodiles are to them.
- 2.5 And this is surprising to people, right? Because you think, "Well, crocodiles kind of look like lizards. So aren't they going to be each other's closest relatives if you're just looking at these three species?" But one of the main points in thinking about the deep history, the phylogenetic history of life on Earth, is that what we have in terms of relationships and the trees that we draw are about relationship and not about similarity.
- 2.6 All else being equal, if evolution proceeds at a uniform pace across all time periods, then you will have evolutionary trees that accurately reflect history that also show similar organisms as their closest relatives. But evolution isn't uniform. Lots of interesting stuff happens. And so you will end up with a truth, which we really believe this tree to be quite robust at this point, that birds and crocodiles are much more

closely related to one another than either are to lizards, even though crocodiles look kind of lizardy.

- 2.7** OK, so what we're going to do in this hour is evolutionary trees, our hypotheses of relationship, how to read them. Then I'm going to talk a little bit about what evidence we use to build evolutionary trees. Some of what Bret was talking about in the first hour—some of those characters, perennial versus annual, leaf noses, the shape, all of these things—this is the kind of evidence that we use to build trees, but we'll go into a little bit more depth on that. So basically, how is it that we're making claims of evolutionary relationship with trees?
- 2.8** Third, why are evolutionary biologists so annoying? By that, I mean that, we say things like, "We're fish. Humans are fish." Get used to it. And "birds are dinosaurs." And it's true, and it seems like we're just being pains in the asses. But it's actually important if you're going to understand things evolutionarily. And finally, ways to be similar. And that brings us full circle back to the convergence that is going to be a lot of what we're talking about for the rest of the course and that Bret was talking about in the first hour. OK, let's get started.
- 2.9** This is the first known evolutionary tree from Darwin's 1837 journal. This is a journal that he started writing after, about a year after he returned home from his 5-year voyage on the Beagle. That voyage was, of course, life-changing for him and world-changing for all of us. Because just a year later, he was saying, "I think this is what's going on. I think that there's relationship between these species." And even so, he then spent more than a couple of decades ruminating before he put it out into the world, in *The Origin of Species* in 1859. So this is early in his thinking, but after he'd already had all of the global experience on the Beagle that he would have that helped him generate his ideas and what was then a hypothesis, but is now a theory of evolution.
- 2.10** And still, he wasn't certain, and he remained uncertain. He got more and more certain over time. But this is what we do. In science in general and in evolution also, no more so, no less so than the rest of science, is that we always retain a level of skepticism and say, "I think this is what's going on, but I'm going to constantly look for evidence to the contrary." So that's what Darwin was doing.
- 2.11** Here we have a non-standard rendering of individual lives on the left and a very colorful, again, non-standard phylogenetic tree on the right. I show this because I think it's useful to remind us of how we get from the micro to the macro. How do we get from individual lives lived to evolutionary trees that reflect whole clades like birds and crocodilians and squamates? If you take the vertical bars on the left as individual lifelines, individual lives, where they're starting at the bottom, that's birth at the bottom of each of these lines and death at the top. The horizontal bars between them are mating events. Then sometimes the mating events yield children, and sometimes they don't, and you get change over time, displayed here just in terms of color and dashes.

- 2.12** So maybe you start out with a red crest, and then it becomes bolder over time because a mutation happened, and then it spread, but it didn't spread. It disappeared because yellow took over instead. So some traits can spread for a little bit and then disappear, and some traits continue on into the future, and they become spotted or striped. And those individual lives are absolutely real, even if the particular character that you're tracking disappears in the larger evolutionary tree that you're looking at. If they have left descendants at all, that still doesn't mean that the particular character that you may be looking at has left traces in modernity or at the point that you're looking at it.
- 2.13** Here's just some language. Once again, this is an evolutionary tree. This is the great apes. Not every one is on here. A tree does not have to have every single possible form of life to be accurate. The more things you put on it, the more information you have. But what we're looking for is a tree, given a set of taxa, what is the relationship between them? So on this tree, before I show you what we're calling the various things, when it's oriented like this, what would be the vertical axis is meaningless, OK?
- 2.14** There is no meaning to be had in the fact that orangutans are at the top of this tree and *Homo erectus* is at the bottom. That doesn't mean anything. The only direction in which you can interpret this tree, the only information in this tree, is along the left-to-right axis. I don't really want to call it an X-axis or a Y-axis because this isn't a Cartesian coordinate system. It's an evolutionary tree. And what we know from looking at this is that —here, actually, I will do the definitions now.
- 2.15** We have the root of the tree, and you can orient it any way you want. All of the trees I'm going to be showing you are going to be oriented this way. You have the root of the tree, where there's some most recent common ancestor. Sorry. There are branches of the tree and there are terminal taxa, which are what are at the tips of the branches on a tree, and their extant forms. That is to say, they're alive today with us, unless indicated by a cross, in which case they're extinct. And we have clades. We're going to be returning to this question of what is a clade when we talk about why evolutionary biologists can be so annoying about what to call things. But what a clade is for now is a most recent common ancestor and everything descending from it.
- 2.16** So, well, we'll come back here. But if I were to say, you know what? Humans is something different. Humans is an act of special creation, so we're going to take this out of here and we're going to look at all the rest of these. You no longer have a group that's informative historically, and it makes it very difficult to talk about what it is that we are and what it is that we're trying to do when we try to interpret evolutionary history.
- 2.17** Then these nodes, which are the branching points, are for every single node on the tree. Again, we have a node here between Pan, the chimps and the bonobos. We have a node here between humans, modern humans and extinct humans. I happened to have picked *Homo erectus*, but it could have been anything because we only got two there. And then we have a node here. And that node points to the fact that, as far as

we know, as far as we understand it currently, and there's a lot of good evidence to support this hypothesis relationship, Pan, chimps and bonobos together, are together the closest relatives of all of our closer relatives, Homonini or hominins, including Homo and also Australopithecus, and many others as well, many other extinct forms.

- 2.18** We are not more closely related to chimps or bonobos, because we share a node right here with both of them. We are equally related to chimps and bonobos, and together, chimps, bonobos, and all of the humans, all of the extinct humans and the modern humans, are closely related, are sister to gorillas. And humans plus chimps and bonobos and gorillas are sister to orangutans. And the thing that gets confusing for people, back to this tree, is that you can swap these nodes as much as you want, and that doesn't change the meaning of the tree.
- 2.19** So let me take a step back for a moment and say first, we've got three taxa again, right? Lizards, birds, and crocs —the simplest possible tree you could make because with two taxa you don't have relationship. There are two other possible hypotheses for relationship that I could draw for these three. This hypothesis, which we are highly certain of, but still not completely certain of, is that birds and crocs are more closely related to one another than they are to lizards. An alternative hypothesis with these three taxa would be that birds and lizards are each other's closest relatives, and crocs are the outgroup.
- 2.20** The final possible hypothesis for three taxa, for these three taxa, would be that birds — that crocodiles and squamates and lizards are each other's closest relatives, and birds are the outgroup, which is the thing that most people would probably think if they weren't being walked through an argument to the contrary. So there's three possible hypotheses. This is the one that we have strong evidence is actually true. And again, that doesn't suggest that there are other taxa that we could add.
- 2.21** Like, what about dinosaurs? Where do dinosaurs show up? Well, birds are actually dinosaurs. So if I tell you we have a lot of evidence from fossils, and even from some behavioral evidence, fossil nests, that birds are dinosaurs, then we're going to put dinosaurs right there on the tree. This doesn't render the previous tree any less true or any more true. We've just added information. It's more complicated because now that we've got four taxa, there are a lot more possible hypotheses of relationship. So we're going to stick with three, just to keep it simple. But that's where we would put dinosaurs, given what we currently understand about deep history of this clade.
- 2.22** This tree, this hypothesis again—birds and crocs are each other's closest relatives, and squamates are on the outs—is exactly the same hypothesis as this one, where I've just swapped a node. That bottom node, the crocs and birds node, I've just swapped. It's also exactly the same hypothesis relationship as this tree. Now I've swapped the bottom node. All I did was flip the node, and that didn't change anything about the hypothesis being demonstrated on the tree. But many people, because of how English works, because of how most languages work, we look at trees and we go, "Oh, well,

this is telling me something that's important. I'm going to read from top to bottom." But the reason to play around with swapping nodes here, and I encourage you to do it if you're interested in this thinking. There's some good open-source software wherein you can make trees and then swap nodes around.

- 2.23** All of these trees that I've shown you here, all 4 of these trees, are the same in terms of what they mean about evolutionary history. Birds and crocs share a node and therefore are sister to one another, and squamates, lizards, and snakes are on the outs. Still, it's surprising. I haven't said that much about why we think that's true. We're soon going to get into the question of what is the nature of the evidence? Not for this tree, particularly, but what kinds of evidence do we use? But the thing that requires some thinking, but really some just exploration with trees, is swapping nodes around and assuring yourself this tree and this tree mean the same thing. Those are the same evolutionary hypotheses of relationship.
- 2.24** All right. Second of 4 broad parts in terms of our whirlwind tour through deep history, through macroevolution. How do we assess evolutionary relatedness? Again, this is in the background. Faded out is Darwin's first tree. Trees, also known somewhat imprecisely as cladograms, are constructed using characters. Most people who aren't evolutionary biologists would say characteristics. We call it characters. I'm not sure why. Probably just to be confusing. We use characters of the organisms—morphological, molecular, developmental, behavioral, or biochemical characters, and we'll get back to that—that are shared and derived, that are synapomorphies.
- 2.25** Shared and derived characters are what we use to build hypotheses of relationship. So morphological characters—things like bone, including fossils; things like cardiovascular changes, four-chambered heart versus three-chambered heart versus two-chambered heart; muscle, both size and placement, origin and insertion points. Anything that you can observe with a scalpel and maybe a microscope is going to be a morphological truth about an organism. You can then code it and say, "OK, I'm looking at these 3, 10, 200, 8000 taxa. What is the state of the origin point of this muscle that I think is the same thing across all of these taxa?"
- 2.26** So molecular characters—things like DNA, RNA, proteins, enzymes. So what molecules are being made by particular organisms and aren't being made by others? Developmental characters—embryogenesis, metamorphosis. What is the timing and order of particular developmental stages? It doesn't work very well if you're looking across vast gaps. How do you use a developmental character to distinguish between a human and a cypress tree and an algae? It's going to be a little tough, even though algae and cypress are more closely related to either than are to humans. But development isn't going to work as well there because development changes pretty rapidly.
- 2.27** Similarly, behavioral characters. I mentioned nest building, and that'll come up again. Nest building, culture, territoriality, tendency to lock horns or lock antlers. Was that a

red deer that you showed in the first hour? To the degree of sexual dimorphism, which is a morphological character, but then you will see evidence of that in the behavior of the organisms. You can use some characters to predict the behavior, to predict other characters.

- 2.28** And biochemical characters—things like venom types between the two broad clades of poisonous snakes, for instance. We have Elapids, which is like cobras and kraits and sea snakes, which make neurotoxins, neurovenoms, and the vipers, like the eyelash viper that Bret showed, which make a muscle toxin. Those toxins, which are highly derived, quite complex, variable in extent, and persistent, therefore adaptive, they are adaptations, are so different between these two clades of poisonous snake, of venomous snake, that that is one part of the evidence that we use to draw evolutionary trees that have those two kinds of snakes, even though they both have the potential to be deadly as not all that closely related to one another.
- 2.29** Synapomorphies—long, unfortunately technical-sounding word, but basically what this means is shared, sin, apo, new, morphy, morph, shape or size. The word just proceeds the molecular evolution. So characters —synapomorphies—might be molecular. We still call them synapomorphies. These are shared, derived traits between organisms. So what are some?
- 2.30** We have in the Salmonids, which is salmon and trout and whitefish and char and grayling, among other things—they're all tetraploid. What that means is that while our gametes are haploid—we have one copy of each of our chromosomes in each of our sex cells, sperm or eggs, depending. Humans are diploid. All mammals are diploid. We have two copies of each of our chromosomes, one from mom, one from dad, in each of our cells—there are some clades out there that have gone triploid, tetraploid, pentaploid, octoploid. It goes crazy, especially in frogs. The frogs do this a lot, and it's interesting to consider why.
- 2.31** But all the Salmonids are tetraploid. That is one of the characters that is shared within that group and derived relative to their ancestors that then maps onto —you can put into a data set, and that character then reveals, along with a bunch of other characters, a relationship between the salmon, trout, whitefish, char, etc. And as such, we then say, "You know what? This looks like a clade. We're going to call it Salmonids."
- 2.32** OK, we have amniotes, which is reptiles, including birds, plus mammals. Amniotes have eggs with 4 layers of protection. We have the amnion, the chorion, the allantois, and the yolk sac. What amniotes are emerging from is non-amniote tetrapods, which to most of us means amphibians. So what the amniotic egg does, which is—you know, this is a group named for this signature synapomorphy, this signature shared derived trait that is unique to and present in, to some degree modified in many species, all of the amniotes—it allows the amniotes to become fully free of water. So once you have an amniotic egg, you don't have to keep returning to the water to reproduce as the amphibians do.

- 2.33** And we're amniotes, even though we don't seem to have these 4 layers in the same way that a chicken egg does. It's just modified. So we have these 4 layers. We are amniotes, and we have modifications that have added to the original amniotic egg.
- 2.34** Archosaurs. Again, archosaurs are birds, crocs, and dinosaurs. They all build nests. There are a lot of pieces of evidence. There are many characters. There are many synapomorphies that describe this group that we now call archosaurs, and that we use then to define Archosauria as a good group, as a monophyletic group. And nest building is one of them. We have in mammals non-nucleated red blood cells, these torus-looking red blood cells. There is at least one other group that has non-nucleated red blood cells.
- 2.35** If you were building a tree and you had a data set and you said, "Well, non-nucleated red blood cells is the only thing I'm going to use to figure out who's related to whom," you're going to get confused. So you need to use a lot of characters. You can't just use nest building and say, "That's what makes archosaurs archosaurs." You can't just say, "Tetraploidy—that's what makes salmonids salmonids." I'm picking a single synapomorphy, a single shared derived characteristic, character, for each of these clades, or in the case of mammals, a few, but three isn't enough.
- 2.36** Mammals have hair. Yes, true, even though we're this supposedly hairless ape, and whales, cetaceans, have mostly lost their hair in all of this. But if it shows up at the origin of the group, even if some members of the group then lose it, it's still diagnostic of the group. For instance, tetrapods, four legs. Tetrapods are the vertebrates who came out onto land and then made a go of it. Snakes are tetrapods, even though they've lost their legs. They're still tetrapods. They're still members of the group, even if one of the most signature diagnostic characteristics, one of the synapomorphies that is the thing that we think of when we think of the group tetrapods—they've got four legs, they can walk on land—even if that disappears, it doesn't change your history. You can't change your history. You're still a member of the Tetrapoda, even if you've lost some characters that were the thing that enabled you to do the thing that you're doing now.
- 2.37** Oh, and a diaphragm. We have a transverse septum that separates our perineal cavity from our pericardial and our pleural cavities, our heart and our lungs and all the rest of our viscera. And in non-mammal amniotes at least—I think it's actually maybe all of Osteichthyes, maybe all of bony fishes—but in non-mammal tetrapods at least, this is just a septum. It's just a membrane that separates between the cavities. And in humans, it becomes muscularized. We give it a special name because we do that, because we're humans, and we call it a diaphragm.
- 2.38** That allows us to breathe effectively, and one of the additional synapomorphies of mammals, and also interestingly of birds, even though they're not closely related to one another, is endothermy, is warm-bloodedness. And so the diaphragm is part of what mammals use to be endothermic and to be able to breathe fast enough to do the

job of endothermy. And birds accomplish this a totally different way. They don't have a diaphragm. They're endothermic, but they've solved the problem of how to pull that off via different means.

- 2.39** Primates is another good group. It's a monophyletic group. They have opposable thumbs. We have opposable thumbs. We have bony protection around the eyes. You feel this here on the side of your temple. That is not necessarily present in other mammals. Primates all have this bony protection around our eyes. What you see through the evolution of primates is also an increasing reliance on the visual. Whereas mammals, in general, are mostly about olfaction. If you take all the senses and you say, "What are mammals really, really good at?", it's about smell. We tend to have big noses, and a lot of our brain, the entire cerebral cortex, is ancestral olfaction brain.
- 2.40** And what you have in primates is a move towards a shrinking of our noses in general, except for that proboscis monkey, and an increase in reliance on our vision. You also have a move towards what's called frontation, where our eyes move from the side of our heads, which is good for closer to a 360 view, but you don't get much depth of field, to a lower field of vision, but what you do have, you can triangulate. And so you can see and you can get a sense of distance. And so that's what frontation is good for. And you tend to see binocular vision in organisms that are highly visually reliant, as we are, and on things that are basically doing, giving chase at high speed. Predators. Predators tend to end up needing frontation of some sort with their eyes, unless they're using other senses, like for instance, some of those vipers, which are also using chemosensory modes by which to find each other.
- 2.41** OK, so this is just a smattering of synapomorphies. Just to remind us, how do we construct trees? Cladograms, evolutionary trees, phylogenetic trees— those are all synonyms. They're constructed using characters of the organisms—again, morphological, molecular, developmental, behavioral, biochemical, more, that are derived and shared. Species develop novelties, characters, over time. Mutations happen, and then selection hits, and we end up with, often, a process by which complexity evolves. Adaptation happens at the end of that process, and those characters that we see are the result of, usually, mutation plus selection.
- 2.42** The most parsimonious, that is to say, efficient distribution of synapomorphies, characters, describes the tree that is most likely to be a true description of evolutionary history. If you have a character set of 100 characters and 3 taxa and you say, "OK, for each of these characters, maybe they're binary characters, maybe they've got 3, 4, 5 states, maybe it's eye color and number of limbs and how many chambers of your heart you have and a number of other things." You can build a tree and say, "What is the most efficient distribution of those characters over these three taxa? Which tree could I build that requires the fewest number of evolutionary changes?"
- 2.43** That tree, the one that has the most efficient distribution of characters, the one that is most parsimonious with regard to the evidence that we have, is most likely to be true.

Not necessarily true, because parsimony doesn't always obtain, but usually it does. And that is an assumption of building evolutionary trees, that once you have your data set—which could be, if you're talking about morphological characters, it might be 50. If you're talking about molecular characters, it could be hundreds of thousands—once you have your character set, you throw that into usually a heuristic algorithm piece of software and say, "Generate for me the most efficient possible topology for a tree given these characters."

- 2.44 And you're going to end up with some ambiguities, and that's fine. And that doesn't mean that history was ambiguous. History happened. It happened one way, only one way. Our current thinking about what happened doesn't change what actually happened. But that doesn't mean that we know what was true.
- 2.45 OK, so just to go back to this simple 3-taxa statement. One of many characters that support this hypothesis relationship, again, is nest building. And so this is how, if we were in a classroom and able to spend time in a computer lab generating trees and learning how to do this over the course of weeks, I would be suggesting that you figure out, generate a bunch of characters, generate a data set, and then map them yourself, rather than letting the computer do it for you, map them onto tree and see where things show up. See where things get surprising. Between, for instance, birds and mammals, things get very surprising. There are a lot of shared synapomorphies between birds and mammals, even though we are pretty certain, like, really about as certain as you can get in evolutionary biology, that birds and mammals are not each other's closest relatives.
- 2.46 So that's an example of non-parsimony in evolution. You've got endothermy evolving twice. You've got the four-chambered heart evolving twice. You've got the ability to run fast and have high-burst speed. And yes, it's flight for birds and doesn't tend to be flight for mammals, although a quarter of them are bats, so for some of them it is. You've got these things evolving separately in these two distinct clades. And that doesn't mean that they are each other's closest relatives. And to that point, we'll come back to that for the fourth part of this whirlwind tour through evolutionary biology. How can you be similar? What are the different ways to be similar? And how do we know what we're looking at?
- 2.47 All right. Why are evolutionary biologists so annoying? Always going on about things like, "We're fish. It's important that you know this." And most people are like, "It's really not. I just want to buy a fish. I don't care. I'm not interested in thinking about the fact that I'm a fish, or birds are dinosaurs." OK, that's cute. It makes for a nice party trick, but why does it matter? Why could this possibly be relevant? And the short version of why it's relevant is that names that evolutionary biologists give to groups need to be reflective of history.
- 2.48 Now, names do a lot of different jobs in the world, and that conversation is one for another time, a class that is perhaps more broadly focused on macroevolution. And

some of what names are doing is certainly allowing humans to communicate with one another. But for evolutionary biologists, if the names don't reflect evolutionary history, they become meaningless and they become worse than useless. They become actually a hazard to communication. So, us going on about we're fish and birds are dinosaurs does sometimes make it more difficult to communicate with people who aren't evolutionary biologists, but it's necessary for us to actually do the work that we're doing.

2.49 How do we name the clades that we discover when we build evolutionary trees? Again, this gets a bit technical, but hang out with me here. A group that includes all the descendants of a most recent common ancestor —again, that node on evolutionary trees. The most recent common ancestor is the node and everything downstream, everything on the branches and the terminal taxa on the branches—if you have a clade that includes all of that, that's monophyletic. Single evolution, monophyletic.

2.50 Whereas, and I'm going to go through some examples here, if you exclude some members and you say, "No, we're not fish," well, that's paraphyletic. Then you're saying, "Well, there's somehow two different things going on, even though there's only one most recent common ancestor. How are we going to keep track?" That's not a good approach if you're trying to understand history. Sometimes we borrow from two totally different parts of history in order to name things. That's also not evolutionarily informative or accurate.

2.51 Here are some examples. This is a slightly bigger tree than the one I showed you before. This includes two members of the so-called lesser apes. We're great apes—you could tell who named the clades. We're also Eutherians. We're the true mammals. The marsupials and the echidnas and the duck-billed platypus—they don't get to be true mammals, even though they're absolutely true mammals. We have these naming things that reveal that we have our arrogance in names. But here we go. This is the apes. It's not all the apes. There are a lot more gibbons than just the one. Siamang is the largest gibbon, and there are a bunch of other gibbons. There are, of course, many, many extinct forms that aren't on this tree. But as far as we know at this point, all the evidence —again, the morphological, the molecular, the behavioral, the biochemical, the developmental evidence, the characters that have been compiled by many different systematists, evolutionary biologists, whose job it is to infer deep history—suggests that this is an accurate hypothesis of relationship for the apes.

2.52 If you clip this tree in one place and you say, "OK, these are great apes. Great. That's great apes," there are going to be people, though, who say, "No, but we're not apes. That's not cool. I don't want to be an ape. I'm not interested." That's a different question. Does it make you feel uncomfortable? Do you not want to be grouped with them? But evolutionarily, this is where we are. If you clip us off—once again, so a group that includes all the descendants of a most recent common ancestor and that common ancestor is monophyletic —if you clip it again, though, if you clip that tree a second

time and say, "I'm actually going to exclude the humans," that makes the great apes paraphyletic.

- 2.53** Paraphyletic groups aren't evolutionarily informative, and they aren't what evolutionary biologists traffic in in terms of trying to make sense of what it is that history has brought us. You still have us and our relatives being monophyletic if you've clipped that twice. But because you've excluded us from the rest of the apes, you've lost meaning. And so we don't want to have paraphyletic groups in evolutionary biology.
- 2.54** Another example of paraphyly. Here is a tree with a bunch a fish on it. Evolutionary tree with a bunch of fish on it. Cephalochordates, I think, probably don't get called fish, so there's the outgroup. But starting here at this node, right there, we have fish, lampreys and hagfish sister to one another, maybe. That's actually in contention. But they're definitely down near the base of the vertebrate tree. We've got Elasmobranchii, which is sharks, skates, and rays. That's before the evolution of bone. Sharks don't have any bone at all. They have mineralized tissue. They've got sharp teeth, but they don't have any bone.
- 2.55** Then at the level of this node here, we have the evolution of bone. So that's the Osteichthyes, that's the bony fishes. That includes things like sturgeon and gars, and Teleosts, which is almost any fish that I haven't listed here on this slide. If you think of it, it's a teleost. So, I don't know. Name some fish.
- 2.56** Salmon.
- 2.57** Salmon, yes.
- 2.58** Bass.
- 2.59** Bass, yes. Teleost. What else? Once again?
- 2.60** Tilapia.
- 2.61** Tilapia. Yes, teleost.
- 2.62** Beta.
- 2.63** Once again?
- 2.64** Beta.
- 2.65** Beta. Yes, beta is a teleost.
- 2.66** Goldfish.
- 2.67** Goldfish. Yes. I think goldfish may actually be a couple different species. I'm not sure that goldfish is a clade, but I think all the goldfish are teleosts. Yes. Basically, almost every fishy fish that you can think of is a teleost. That's where the vast majority of fish diversity lives. What this group is here —the sturgeon and the gars and the teleosts, and there's a couple more that I just didn't include on the tree here—those are the so-

called ray-finned fishes. They've got a lot of rays in their fins, a lot of skeletal elements, and they tend to have webbing between them, and they use them as fins.

- 2.68** And then where Osteichthyes, where bony fish, branches into ray-finned fishes, it also branches into lobe-finned fishes. We have fewer skeletal elements at the distal ends of our limbs. And in some of us, that becomes lungfish, and in some, it becomes coelacanths, and in some of us, it becomes tetrapods. So tetrapods are lobe-finned fishes, are bony fishes, are jawed fishes. We've got the jawed fishes here-ish, depending on where hagfish goes. So we got the evolution of jaws in fishes. We got the evolution of bone in fishes. We have the evolution of lobe fins in fishes. And then we have the evolution of being on land and being awesome at it, here, but that doesn't make us any less fishy.
- 2.69** So mostly when you ask people if tetrapods—if they know what a tetrapod is—if you say, "Oh, is a snake a fish?" "No." "Is a whale a fish?" "No, of course not." "Is a human a fish?" "No." Well, actually, the answer to all of those questions is yes. Snakes are fish, whales are fish, which is confusing because that's the opposite of what you learned in elementary school, right? And humans are fish. We're all fish, but not in the way you think it means, because the problem is that evolutionary biologists are using the term, and also everyone else is using the term, and that's where it gets tricky.
- 2.70** You probably don't care that much when we're talking about terms that only we talk about, and we start informing you that actually we are Sarcopterygians. Cool. I'm good with that, right? Well, if I tell you what that translates to is lobe-finned fishes, then that's when it gets a little tricky. But this is why. This is why we're fish, because we're embedded within all the other fishiness that evolved on Earth. We are, again, lobe-finned fishes and bony fishes and jawed fishes. And yes, we're doing something very untraditionally fish-like, but that doesn't make us any less fish, because we can't change history. What we can change, and what evolution changes, is the form that moves forward into the future.
- 2.71** OK, so here's an example of polyphyly. Salmon. When you think of salmon, you may think of the life history of Pacific salmon, being born in streams and going out to sea and spending many years at sea and then coming back to their natal streams and breeding once and then dying. It's a pretty great story, and it's true. Or you may be thinking of the Atlantic salmon, or you may be thinking of the thing you like to order for dinner, and you think, "Well, it's got to have pink flesh." That's definitely a diagnostic feature of salmon, is the pink flesh. And maybe there's some other things you think about when you think about salmon.
- 2.72** It turns out that Atlantic salmon, *Salmo*, and Pacific salmon, which is all of these guys down here—all the things we call salmon are not each other's closest relatives. We have in the middle, in the intermediate place between the *Salmo* and the *Oncorhynchus*, the Atlantic salmon and the Pacific salmon, we have some trout and some char. What that means is that salmon is useful as a culinary diagnostic, I think. It's

a fairly firm, pink-fleshed, firm-fleshed, pink-fleshed fish, but it's not useful as an evolutionary descriptor.

- 2.73** Now, this, again, this is not complete, but this is tree of Salmonids, the family that includes all the salmon-like things. And so Salmonid is a monophyletic group, but salmon itself is not. It is polyphyletic because you have one evolution here and you've got another evolution down here, and they're not each other's closest relatives. That makes things confusing for evolutionary biologists to think through things. And it's why we can talk about Salmonids, monophyletic group, but not salmon, because it's polyphyletic.
- 2.74** All right. This is, I think, the fun part. How do we end up being similar? How do organisms end up being similar to one another, and what does it mean when we are? We can be similar through common descent, through homology. Again, some more technical terms. Or we can be similar by having independently arrived at common solutions to shared problems. The technical term for that is homoplasy, but usually it will come about in forms like convergence. Two different distantly related species converged upon a similar solution to a similar problem, and now they look similar.
- 2.75** And maybe it's impossible to tell them apart just on the basis of that one character. My God, they both fly. If I'm just looking at flight, dragonflies and beetles and bats and swans are each other's closest relatives, right? If I'm only looking at flight as the character by which I draw an evolutionary tree. This is one reason why we need to use lots and lots of characters to build evolutionary trees. One character will tend to be confusing, not necessarily, but it can be confusing because evolution is not always parsimonious. Because while in the aggregate, we expect that the most efficient distribution of those synapomorphies, those shared derived characters, onto an evolutionary tree will yield an accurate representation of history, that's not necessarily the case, and it's definitely not necessarily the case for individual characters.
- 2.76** We can be similar through homology, like here. This is just from a basic biology textbook. Here we have the 4 limbs of some tetrapods. As it turns out, it's all mammals and then a generic bird. Color-coded. You've got the humerus in dark brown, and it looks like the radius is in white—what your thumb radiates around. Your tibia. I mean, your tibia—the fore limb, not the hind limb, your radius and your ulna, and then you got the metacarpals and the carpals, the digits.
- 2.77** So humans, you will recognize that. Hopefully, you haven't seen that in yourself, but you will recognize it from other presentations or through dissection. Look like cats. We're pretty different from cats, but compared to any of these 6 images here, it's clear that we and cats are pretty similar and, color-coded, we know that the humerus is the humerus from other means. And the humerus of a human, the humerus of a cat, the radius and the ulna, the carpals, the metacarpals—they're all pretty similar.

- 2.78** But what happens when you get to a horse? Things change a lot. Or bat, the Latin for which is Chiroptera, meaning hand-wing. The bat has basically got an extension of its hand bones, not mostly its arm bones, to turn it into a wing, whereas the bird has used its whole arm to turn that into a wing. So the flight in birds and bats is not only distinct from one another, because the most recent common ancestor, birds and bats, didn't fly. That's how we know in part. But also they've accomplished the jobs in very different ways. The bat has actually just used his hand. The proto-bat, the most recent common ancestor of all bats, started to modify their hand into a wing, whereas the proto-bird, the most recent common ancestor of all birds—which was, again, a dinosaur —turned its entire forelimb into a wing.
- 2.79** And then, of course, in whales, where we have a secondary return to the sea, itself fascinating —we have several secondary returns to the sea in tetrapods. Having emerged onto land, having gone through all of the changes that are necessary for that to happen, then you have some lineages going, "You know what? Go back." They never go back fully. Whales are fully aquatic, but they have to be there. You have all of these secondary returns to the sea in tetrapods, but none of them actually redeveloped gills. They all retain the necessity of doing their respiration through lungs. So the modification of the humerus, radius, ulna, carpals, metacarpals in the whale, compared to all the rest of these, points out that these are the same bones, but they've been modified. These bones are homologous to one another, even though they are doing very different tasks in each of these organisms.
- 2.80** OK, here's some more. We've got chimps and bats, and I think I will not be talking to you guys about these guys in this course, but these are my frogs, these are Mantella, the Madagascar poison frogs. Birds—gosh, is that a kinkajou? Yeah, it's a kinkajou. Beautiful. Frugivorous carnivoran. OK, what does that mean? That's actually relevant here. Carnivoran refers to a piece of history. That's a descriptor of the clade that it's in. And other things in the Carnivora are all the cats and all the dogs and bears and seals and lots of other things as well, hyenas. But just because you're in the Carnivora, and presumably the proto-carnivoran started eating meat and becoming modified to be really good at eating meat and only meat, doesn't mean that once you're in that group, you might not find a niche doing something different.
- 2.81** So this animal is firmly embedded within the Carnivora. His closest relatives are carnivorans, but he doesn't eat meat. He eats fruit. He's a fruit specialist. He's a frugivore. The first thing, frugivore, is an ecological descriptor of what he does. The second thing, carnivoran, member of the Carnivora, is a phylogenetic description of what piece of history he belongs to. This is the tension that we also find, especially in this, how is it that you're the same? The forelimbs of that guy are homologous to the forelimbs of the chimp. The teeth of a kinkajou are going to be somewhat modified from his closest relatives, but they are still the teeth that he inherited from his carnivoran ancestors and his mammalian ancestors and farther and farther back. And in

fact, dentition in mammals is another synapomorphy of mammals, the particular way that we have a very different dentition.

2.82 So all of these four limbs are homologous to one another. They emerge from the same basic framework, but they've all been modified to do something pretty different. Some more homologies. Scales, feathers, and fur. Scales, which evolved before feathers or fur did, appear to have given rise to both feathers and fur. So most people are not going to mistake feathers for fur or fur for scales or scales for feathers, mostly. But what it means to say that they're homologous to one another is feathers are modified scales. Fur, hair, is modified scales. So you could be really annoying as an evolutionary biologist to say, "Well, then, technically we have scales." Not very useful, right? It's not very useful to say that, but it is useful to know, "OK, the fur that we have, the hair that we have, is modified from and inherited from a scaly ancestor, and it's those scales that gave rise to the fur that we have."

2.83 This is a hyrax. We have the elongate incisors. Those aren't canines. If you have pointy teeth in mammals, usually it's the canines that are pointy. These are actually incisors that are elongate in hyraxes and dugongs and elephants. Hyraxes, dugongs, and elephants are remarkably—dugongs are another secondary return to the sea, like manatees, in mammals. Elephants, hyraxes, and dugongs are actually a monophyletic clade. They're actually a good group. They're a monophyletic group that are each other's closest relatives within mammals, even though the three of them don't really look much alike. But one of the characters, one of the diagnostic characters, that is shared between them, one of the synapomorphies of this group is these elongate incisors, which in hyraxes just shows up looking cute and fierce because these are tiny. These are tiny little animals. They're not going to give you any trouble, although this guy is looking like he's trying to.

2.84 Here's another homology. Jaws from a gill arch. Our early vertebrate ancestors had no jaws. Lampreys have no jaws. Hagfish have no jaws. Then you get the transformation of either the anterior-most or the second back, depending on who you ask, gill arch, which is made of all these skeletal elements into jaws over time. And other gill arches change in other ways. In mammals, some of the posterior gill arches in early vertebrates have transformed into middle ear bones. And we can attribute some of our acute hearing as mammals to the transformation of early vertebrate gill arches. Those middle ear bones are homologous to gill arches of early vertebrates. But that doesn't tell you very much about what their function is. So function is over in a microevolution space. And "what is it homologous to" lives in this "where did it come from and who's it related to" space.

2.85 OK, so that's homology. There are, as I said, broadly two ways to be similar. We've got structures or mechanisms or behaviors being similar between two organisms, due to shared history. That's homology. For instance, kingfishers, like this beauty, and ducks both have wings. Kingfishers and ducks both have wings because their most recent

common ancestor had wings. Pretend that you don't know the term bird at the moment or bat or anything. Just say, "OK, I've got a kingfisher and I've got a duck. How is it they both have wings?" Well, we have evidence that their most recent common ancestor had wings. Therefore, their wings are homologous to one another.

- 2.86** In contrast, structures that can be similar—structures can be similar to each other due to convergence, such as kingfishers and fruit bats. Those Phyllostomids that Bret will be talking more about—the tent bats—have wings, of course. And so do kingfishers. But do kingfishers and fruit bats both have wings because their most recent common ancestor had wings? No. No, they do not. Birds and mammals are not each other's closest relatives. Birds are dinosaurs, and dinosaurs and birds are Archosaurs, and they're reptiles, and mammals split off from the reptiles a very, very, very long time ago.
- 2.87** So we've got at least a couple hundred millions of years, I think, of history since we had a most recent common ancestor with birds, and that most recent common ancestor most definitely did not have wings. Therefore, the reason, the way that kingfishers and fruit bats are the same with regard to their wingedness is due to convergence, not due to shared history. It's due to homoplasy, not homology.
- 2.88** Here's some more examples. We have tuna and lamnid sharks. Lamnid sharks are like the great whites and the makos and a few others, but those are the familiar ones. Here's a reminder of this tree that I showed you earlier of the fish, with elasmobranchs—shark, skates, and rays—in red on the top red line, and tuna are one of many species of teleosts.
- 2.89** On this tree, it doesn't look like there's that much distance between them. But actually, and this is telling, because these trees do not accurately tell you anything about how much time has passed. They don't try to. They're just telling you relatively who is most closely related to one another. But as it turns out, there's about 400 million years of evolution between tuna and sharks. That is to say, that most recent common ancestor, that node right there, is about 40 million years ago. So to the degree that tuna and lamnid sharks have really high levels of similarity, that's going to be convergence. That's not due to share history. That's not due to homology.
- 2.90** So specifically, we have—body form is broadly the same between tuna and lamnid sharks. Muscle dynamics at even the micro-muscular level is broadly the same between tuna and sharks and lamnid sharks. And the overall swimming mechanics, and this actually is totally fascinating. They're doing similar jobs in the world. Both of these are effectively apex predators who need to move fast and in open water. And there are some other sharks that do that, presumably, and there are some other teleosts—again, some fishy fish that do that. But tuna and lamnid sharks are uniquely similar in terms of, for instance, not swimming by undulating their whole body.

- 2.91** Imagine, if you will, and some of you will have more familiarity with this than others, but looking down at a fish as it moves through water and you will see the entire body undulate laterally. Not so for these guys. Not so for tuna and not so for lamnate sharks. They actually keep most of their body stable. And they use their caudal half and really, like their back end, their back third. Almost all of the propulsion and the lateral movement is in that back end. And as a result of that shared arrival at a similar solution to a similar problem, they then also have shared, again, muscle dynamics and a number of other physiological adaptations that look remarkably similar, even though, again, it's been 400 million years since these guys had a most recent common ancestor.
- 2.92** Now, within the lamnid sharks, they're each other's closest relatives, so they do it the same way as each other, due to homology. But lamnid sharks and tuna are anatomically, physiologically, and behaviorally similar due to homoplasy, due to convergence. OK, a few more examples of homoplasy. Let's just say convergence and reversals.
- 2.93** You've got nectarivory. This is a glossophagine bat, and this is a photograph of Bret's with his head in a flower going after nectar. That's what nectarivory means. You've got honey creepers and hummingbirds—three very different evolutions—and there are more than that. There are many evolutions of highly adapted nectarivory in birds that aren't each other's closest relatives. And so you have often a very long curved beak. You sometimes have a long tongue. You have the ability to hover. Bats obviously don't have the beak, but they have the ability to hover, and they have the sensory ability, across all of these guys, to find the nectar when it is available in the flowers. So that's highly convergent.
- 2.94** You have the skulls of extinct saber-tooth carnivorans. And these look really similar, but they also are highly distantly related. You have the marsupial, *Thylacosmilus*, on the left, I believe on the left, and the placental mammal—that's like us—*Smilodon*, which is a saber-tooth tiger of North America. So the saber-tooth tiger of North America and the marsupial tiger, effectively, of Australia, converged on very similar solutions to "how do I break into the skulls, basically, the things I want to eat?" And they've got these skulls that look remarkably alike.
- 2.95** But marsupial mammals and placental mammals are —do not have at the base of their tree, their most recent common ancestor, didn't look like this. The last time we had a most recent common ancestor with *Thylacosmilus* and *Smilodon*, we didn't look like this. We looked like a pretty nondescript, generic, fuzzy little mammal. So these two sets of characteristics have converged upon one another. And that is another example of homoplasy, another way to be the same, other than homology.
- 2.96** I think, finally, my last example of homoplasy here is multiple evolutions of leglessness in lizards. So on the right, we have a vine snake. Snakes are lizards. That's modifylied talking again. That's evolutionary biologists being annoying again. But snakes are lizards. But not only are snakes lizards, but there are other lineages of lizards that have

lost their legs, and they're not snakes. That seems confusing. Why would we do that, right? Well, because there are at least 3, and maybe as many as 6, I believe, different evolutions of leglessness in lizards.

- 2.97** One of them, the one that became really remarkably diverse and has spread all over the world—not so much into the Arctic; squamates don't do so well in the Arctic—but the one that's become very diverse and that we have names for, everyone in here could name a snake—that's snakes. But then we have *lialis*. That's an Australian legless lizard that, unless you're from Australia and like lizards, you probably have never heard of before. But it's not a snake. And in fact, if you look at it, you think, "Yeah, it doesn't really look like a snake. It looks like a lizard that lost its legs." Like, it's not quite as far down the road of really mastering the "I'm a lizard and I don't have legs" bit.
- 2.98** Snakes have mastered it. Snakes have managed to master it, and they tree-climb, and they swim, and they burrow, and they go across land, and they do all of these things in many different ways, mostly not within the same species. But they're remarkable. But a number of different lizard lineages have also evolved leglessness in other places at other times. And not all of them are snakes, but all of the legless lizards are lizards. So homoplasy, again.
- 2.99** And why? Why would you lose your legs? There are a number of possibilities. One of them is, if you are in a landscape like this *lialis* is, the legless lizard on the left here, with a lot of sand and it's hot, and either your prey is underground or your predators are overground, or you need to moderate your temperature because you're an ectotherm and you can't do so internally, and so you want to burrow—and legs get in the way of burrowing. And so you will often have in legless lizards a burrowing snout on the front end, and that tells you something about what likely drove the evolution of leglessness.
- 2.100** But at this point, across snakes, we have—this is a vine snake. These guys hang out in lowland tropical, neotropical forests in Central and South America, and they look like vines unless you know what you're looking for. And if you're sliding down a hill trying to catch yourself and you want to grab a vine, you probably shouldn't because these guys are Colubrids. They're rear-fanged, not really venomous snakes. They couldn't hurt you. They can't even get their gait wide enough to hurt you, and they don't really have toxins to speak of. But you could definitely grab one thinking it was a vine. They're that good. They're that good at being cryptic. So they're a long ways from the first moment that their lineage went legless, but they're still part of that lineage.
- 2.101** So phylogenetic systematics, the reconstruction of deep history. Here is a summary of the basic assumptions that we have. And in all of science, what you're trying to do is reduce the number of assumptions you go in with to know what your assumptions are and to keep on checking yourself that you haven't added some assumption to your set of criteria. Because if you have, then you're kind of cheating and you're more likely to confuse yourself and be wrong about what your conclusion is. So you want to minimize your assumptions in all cases and know what they are.

- 2.102** So the basic assumptions in reconstructing deep history are groups of organisms are related by descent from a common ancestor. Change in characteristics occurs in lineages over time. And there's a bifurcating pattern of cladogenesis. I'm going to explain all of these. But first, I will say the first two can be summarized as descent with modification. That's the phrase. That's almost all you need. What do you need to understand the reconstruction of deep history? Descent with modification happened. That's the assumption, and you're off to the races. That's a combination of groups of organisms are related by descent from a common ancestor, and change in characteristics occurs in lineages over time. Descent with modification.
- 2.103** And the third one, bifurcating, just means species split into 2. They don't split into 3. Darwin's tree—he's drawn stuff splitting into 3, 4, 5, right? But statistically speaking, what are the chances that one lineage simultaneously becomes 3? Really, really, really low, right? It probably, actually, never happens. Even if you have a split, a bifurcating pattern, that then very quickly becomes another bifurcating pattern, so that what we can infer from history, it looks like that was a 3-way split. It was probably a bifurcation and then a second bifurcation. So the assumptions of deep history reconstruction and evolution are descent with modification and species split into 2, not 3 or more.
- 2.104** All right. Coming tomorrow, we're going to talk about the hostile forces of nature, and then we're going to play a little bit of evolutionary jeopardy, which is to say, again, what is the ecological question, or the social question, that goes with the evolutionary answer that you can see?
- 2.105** So not all tomorrow, but we're going to be talking about gliders. Why glide? I'll be talking about hoatzin, which are crazy Amazonian birds. I'll talk about leafcutter ants. And this is not a complete list. And we're going to talk about people on the final day. The final section, I'll be talking about people. This happens to be our two sons many, many years ago. All right. I think we're good.

Lecture 3: Evolutionary Ingenuity

At a glance — Evolutionary ingenuity, building on micro- and macroevolution.

- 3.1 We've already introduced you to the basics of microevolution and macroevolution. We're going to walk through some of the hostile forces of nature today and then embark on what we already introduced in the abstract, what we're calling evolutionary jeopardy, where we present to you organisms, characters, and say, "What ecological question or social question is that an evolutionary answer to?"
- 3.2 I'm going to try to get you guys to maybe play along a little bit, even though the people watching at home are not going to be able to. But hopefully that will provide a bit of the experience of—this is not actually going to work best as me, sage on a stage, delivering content unto you. It's going to work much better if you can actually be engaging material in real-time as we're going.
- 3.3 So let's just start by me asking a question of you, and this one doesn't necessarily have to be one that you respond to in real-time, although if anyone has something to say, go ahead and raise your hand. So the "hostile forces of nature"—it's a term coined by the great evolutionary biologist Richard D. Alexander. He was actually Bret's major advisor in graduate school, and he was one of my advisors as well. And it was an idea of Darwin's, although Darwin didn't actually use the term.
- 3.4 And it basically addresses, "What is it that gets in the way of organisms doing what they need to do?" If, as Bret was talking about yesterday, if what organisms are trying to do is lodge their genes as far in the future as possible, what gets in their way? And so broadly speaking, there are abiotic—non-living—and biotic problems that organisms run into. I just would ask you first, say, imagine you're a mouse in a forest. "Is a bigger problem for you winter or competitors or predators or disease?" Which of these things do you think is actually going to be the thing most likely to take you out or most likely to limit your success?
- 3.5 And it's going to depend, right, on what the forest is, how old you are, what your experiences are, how much of your future reproductive success you have ahead of you, how much is behind you, who you're still protecting at home, how experienced you are. All of these things are going to change—oh, also, as I said, where you are. But where you are depends specifically, and not just on the type of forest, but where you are on the globe in terms of latitude. So that's a question that is too, probably too big to go into in any detail here, but all of these pertain to which hostile forces are the things that you're going to be dealing with most.
- 3.6 So we have abiotic and two kinds of biotic forces, of hostile forces of nature, that we'll walk through. First, the abiotic. Within abiotic, there are broadly two types, predictable and unpredictable, or stochastic abiotic forces. So what might a predictable abiotic force be? Anyone have any ideas?

- 3.7 Shark.
- 3.8 Shark?
- 3.9 Rain.
- 3.10 Shark is not abiotic, so shark would be a biotic. Rain? Rain when seasonal. Yeah, exactly. Rain, when seasonal, is at least somewhat predictable, right? What else? Hurricanes. To some degree, yes, certainly where we are standing here, you can rely on there being something happening within some number of months each year. You don't necessarily know the extent of it, the strength of it, but they will happen unlike, say, well, tornadoes are also seasonal in the places where they exist, but they're going to be even more difficult to predict. So compare that to unpredictable, or stochastic, forces. What kinds of abiotics, so non-living, forces might be stochastic?
- 3.11 Earthquakes.
- 3.12 Earthquakes, yeah. Once again?
- 3.13 Tsunamis.
- 3.14 Tsunamis, yeah. Earthquakes and tsunamis, they go together, right? And we'll be talking about those. And volcanoes. Floods to some degree. If the rain that comes is seasonal and therefore predictable, and you've got a 100-year flood, well, you don't necessarily know when the 100-year flood is going to come, but at least you have some sense of maybe what was predicting it.
- 3.15 So organisms have to deal, and that includes us, although we've done a very good job, humans, of protecting ourselves from most of Darwin's hostile forces. We've become, actually, again, in the words of Dick Alexander, our own greatest enemy, our own greatest hostile force, because we've eradicated most of the other hostile forces from our lives. So winter is coming. How will I survive? Will next year look like last year? In general, next year looks a lot like last year, but it doesn't always, because those rare winters happen. Sometimes winters are extremely harsh, and they wipe out a lot of organisms that didn't put enough away.
- 3.16 And so there's always a question, "OK, if I'm a mouse, and I'm caching food for the winter, how much do I cache? Do I cache to the mean? Do I cache to what will get me through the average winter?" Well, no. Half the time, I'm going to not make it through the winter. So do I over-cache? Do I put more away than seems necessary and then waste some? Probably that. But that looks wasteful to many ecologists who come through and say, "Well, look at these. Look at these animals. They're just being wasteful. They're doing too much. They're putting too much away." But the organisms that over-plan are more likely to make it through the harsh winter, and the ones that under-plan are less likely to make it through.
- 3.17 But the ones that under-plan, if it's an easy winter, or a couple of easy winters, or a couple of years without a bad hurricane, or whatever it is—however it is that they're

planning, they are more likely to win against their more robust and planning competitors because they have more in reserve that they didn't spend on putting away food for the winter. And they can spend that on looking great or finding mates or doing whatever it is they want to do. So that's a predictable abiotic force of nature. And then also unpredictable—and two of you guys hit exactly the one that I'm going to talk about, earthquakes and tsunamis.

- 3.18** So again, the Boxing Day tsunami of 2004, which was centered—the earthquake was centered off of Sumatra and wiped out people and, to some degree, other organisms, although not as much, around the Indian Ocean. Elephants in Phuket, in Thailand, didn't die. Much as the Andaman Islanders, the Moken people of the Andaman Islands, didn't die because they retained a cultural memory of what to do when the sea recedes and also knew how to look, the elephants didn't die. Why didn't the elephants die?
- 3.19** Elephants didn't die, as it turns out, because they have, probably, there's probably other things, too, but the tsunami travels at 700 km/h. The soundwave that begins at the time that the earthquake starts travels at 1260 km/h. And elephants have an ability to hear very, very low sounds that we cannot hear. And the difference between the rate of the sound and the water, as it was traveling towards the elephants, a lot of elephants in Phuket and in a few other places where it was looked at, to hear the tsunami 38 minutes before it actually arrived. And they also knew what to do. They went to higher ground and they didn't die.
- 3.20** So that's an unpredictable abiotic response, an unpredictable abiotic hostile force, that they have the sensory capabilities on board. Plus, elephants are incredibly intelligent, social, long-lived, generational overlap organisms. They also have the ability to communicate to one another: "OK, I'm hearing that. This is what that means." Who knows to what degree there's any detail in that this is what that means. Maybe it's just, "Now we go. Now we go up to higher land." OK, so that's abiotic hostile forces of nature.
- 3.21** Then we have heterospecifics. So here we have two organisms, more really. Well, two species because there's two deer in the picture. These are two deer grazing on plants. The plants don't want to be eaten. Fruit is actually designed to be eaten, as Bret talked about some yesterday, but leaves never want to be eaten. So how is it that plants are going to defend themselves against grazers? There are a number of ways. In general, "how do I avoid becoming someone else's lunch?" is one of the questions that if we're talking about the hostile forces of nature—so heterospecific just means individuals, biotic life of something that is not your species, hetero-specific, as compared to conspecifics, individuals of your own species. So all we humans sitting in the room here today are conspecifics, and there's probably some heterospecifics around, as it's not a purely sanitized space, right?
- 3.22** So deer on plants is a heterospecific hostile force of nature on the plants. And the answer to "how do we avoid becoming someone else's lunch?" is going to be different

if you're a plant or an animal. If you're a plant, you can't move. There are a few plants that can move a little bit. They're the exceptions, and they still don't move very far, and they don't move very fast. So the plants are going to tend to defend themselves with chemical defenses. They are going to tend to put toxins in their leaves, or with physical defenses, with spines and such, to make it difficult to get to them. Whereas if you're an animal, you will tend to defend yourself actively with the armor that you have on you or by fleeing. And neither of those are the same thing as what plants will tend to do.

- 3.23** Other hostile forces of nature that have to do with heterospecifics are, "How do I feed myself?" Like, how is it? How does this deer know that those leaves are going to be good for him? Him in this case, for sure, right? He's got antlers. How do you find—if you're a carnivore, how do you find your food? How do you outwit it? How do you stay cryptic enough until you're close enough to launch yourself at it that it doesn't run away?
- 3.24** If you're an herbivore, again how do you know what to eat and how do you choose? What kinds of senses are going on there? Are you tasting a little bit and then going away and coming back and seeing if you feel OK and then eating more? Are you preferentially choosing young leaves? Probably, because the young across all species are less well-defended. So that's part of why the young are more defenseless and both, you know, babies are defenseless. Human babies are defenseless. Baby leaves are also defenseless. They don't have their toxins on board yet, and so they tend to be more likely to taste delicious.
- 3.25** How do I avoid infection? Parasitism. One way is the long-distance migration of birds. We could have played this as evolutionary jeopardy, but I'm just going to insert it in here. Many of the shorebirds that migrate through North America along various corridors—I'm most familiar with the ones on the West Coast, but they also come with the East Coast, and there are a few corridors that come up through the middle of the continent as well—over winter in Mexico and have breeding grounds in the Arctic or the subarctic. Or the boreal zones. And that's a lot of work. That's a lot of energy to spend. So why do it? Why migrate?
- 3.26** There are a lot of reasons to migrate. One of them is, if you have nests —and again, young or defenseless—you have nests and you've got eggs, which can't move at all, and then you've got hatchlings, chicks, which don't move very well, they are going to be very attractive to predators. And so you are going to be likely to concentrate. And so we have these shorebird migrations that end up in these vast concentrations of nests in their summer nesting grounds as a way to basically avoid being eaten by predators. But what that brings in, once you're concentrated in space, you can maybe better defend yourself. You can send up alarms about when predators show up.
- 3.27** But once you are found by a parasite, like the ectoparasites that tend to lunch on these birds, it's really hard to get rid of them. They will spread across a nesting colony, and they're not going to go away. And so by leaving, once your kids are volant, once your

kids are able to fly, by flying out of there and abandoning your nests, you abandon all of your ectoparasites, too. And the few that may be clinging to your feathers as you fly away will presumably fall off and not be able to persist all the way back down to your wintering grounds. And by the time you get back—whatever it's going to be, 8, 10 months later—they're gone. Those ectoparasites also didn't survive winter. So that is one way to deal with one of the hostile forces of nature that comes in the form of heterospecifics.

- 3.28** Also, how do I avoid being stolen from? I think we will have time later in the course to talk a little bit about poison frogs. And I'll just give you a little teaser here that the poison frogs in Madagascar brood in these tree holes or bamboo wells, basically little pools of water—phytotelmata—that aren't open to the rest of the water system. So they don't have predators, they don't have fish. They do actually have predators, but it's not in streams or ponds. And as a result, these little tree holes are highly valuable. And so the poison frogs that I worked on find them, defend them fiercely. They're territorial about them. They keep everyone else away that they can.
- 3.29** But there's other frogs that like these tree holes, too. And with regard to competition and fighting, you're much more likely to be good at fighting your own kind than you are some other kind. We've probably all seen these videos of a guy boxing a kangaroo, or trying to go after something else that isn't a human. And it goes silly, it goes wrong, it goes weird, but mostly what it goes is unpredictable. It's just really hard to know what someone else is going to throw at you if they're not your species.
- 3.30** And also when you're talking about frogs, they're going to be a wildly different size. So "how do I avoid being stolen from?" is going to be a problem when you can't use the same kinds of defenses that you would against your own kind. And that brings us to your own kind, which is conspecifics. So again, hostile forces of nature come broadly in abiotic versus biotic. And within the biotic types, we have anyone that's not you, what kinds of problems do they present to you? And the problems might be, "How do I eat it? How do I get around its defenses?"
- 3.31** But conspecifics pose problems, too, right? "How do I win against the guy who wants the same things I do?" And the same things I do may be that piece of food, that territory, that mate, and related, but not exactly the same question, and in some species very much not the same question, "How do I find and attract and keep a mate?" Right, so these are hostile forces of nature, which we think of like, "How do I find and attract and keep a mate?" Why are we calling this hostile? Well, it's one of the things that if you can't do it successfully, it's going to get in your road, and you're not going to be able to lodge those genes of yours very far into the future.
- 3.32** And again, "How do I avoid being stolen from?" So back to the frogs. If the frogs are dealing with competitors who come in who are of the same species, they can use techniques that they are well-skilled in. And if it's other species, they're more likely to either flee or hide, as opposed to fight. OK, so just to recap, hostile forces of nature.

Winter is coming. How will I survive? What will next year look like? That's a predictable abiotic force.

3.33 A non-predictable, a stochastic force, might be a tornado, a volcano, an earthquake. How do I avoid becoming someone else's lunch? What are the heterospecific organisms out there who I either want a piece of, or they want a piece of me, and I need to do everything I can to get what I want because this is about how does an individual do its best in the world? And then how do I win against the guy who wants the same things I do? Conspecifics. OK, so that's hostile forces of nature.

3.34 Now we're going to do this evolutionary jeopardy. What is the question to that answer? I've got 3 organisms or traits to walk you through in this hour today. Let's start with hoatzin. We're going to do hoatzin today and gliders and kelp, a strange, a strange group. Hoatzin, *Opisthocomus hoazin*—I think is how you pronounce that specific epithet, although I don't know why it's missing a T. That's actually how it's spelled. So taxonomists just name things unusually. These are remarkable birds. They are in the Amazon. This is not either of our pictures, but we have seen these birds.

3.35 They often live in oxbow lakes. Oxbow lakes are lakes that, in a big meandering river like the Amazon, or a tributary of the Amazon, like the Tiputini or the Shiripuno, where we have spent time, as water levels fall and rise, as they do a lot, because the heavy rains that come to the Amazon are not particularly predictable—they will come; you can predict that, but when they will come is not predictable—and so as water rises and falls, other things may be happening with regard to the land, and sometimes you get a piece of a river separated, and that's an oxbow lake. What was a river, what was a piece of a river, the river is now rerouted, and now you've got a sliver of a lake. And they often don't last for a tremendously long time, but long enough for a whole ecosystem to grow up around them, a distinct ecosystem from the rest of the Amazonian ecosystem around them.

3.36 These guys are neotropical, meaning they're the New World tropics, which is the Americas, and the tropics, meaning between 23.5 degrees north and south latitude. They're arboreal, meaning they're entirely in the trees. You don't find them on the ground. They're really wacky-looking, which is not a technical term, but just look at them. They're very strange-looking. Probably they find each other beautiful. We tend to, conspecifically, we tend to find each other beautiful when we are. And it's hard to imagine what they're seeing in each other, but there it is.

3.37 They're smelly. It's strange. Birds don't tend to smell, but they're smelly. They're folivorous, meaning they eat leaves. They fly badly. They fly really badly. They live in these, where we have seen them and most of the other places where they live—they live on oxbow lakes in these spiny palm islands that grow up in the middle of these oxbow lakes. And they make their nests in these spiny palms. And the spiny palms are very, very spiny. That is how the palms are defending themselves from being eaten. And these hoatzins, which are large birds, crash into the spiny palms and then

somehow grab on. It does not look safe or elegant or like it's what was intended, but it's pretty much what they do every time they land.

- 3.38** Except sometimes they'll find something. This is a picture taken. I don't actually know where this was. It's from Getty, but this is a picture taken with them not on a piece of spiny branch, and so maybe it was easier to land there. They are communally roosting, meaning they make these nests in big clusters and that you never find a hoatzin nest all by itself. And they have helpers at the nest, which means that, very often, last year's young will stay non-reproductive, even though they are capable of being reproductive, and stay and hang out with mom and dad and help them raise the next breed.
- 3.39** Helpers at the nest is a pretty widespread, well-known strategy that a lot of birds employ, wherein long-term, maybe the helpers at the nest get to inherit a piece of mom or mom and dad's territory. Often mom and dad—birds have a very high rate of monogamy, unlike almost everything else that has evolved on Earth. So maybe they get to pick up a piece of mom and dad's territory. Maybe they get to learn how to be a hoatzin in this case, more than they would have by just staying there for the early season. But so, that's what hoatzin are. Neo-tropical, arboreal, wacky-looking, smelly, folivorous, badly flying, communal-roosting birds with helpers at the nest.
- 3.40** There's a lot of questions we could ask here. There's a lot that we could ask. So I'm going to start with this. How do hoatzin survive on a diet of leaves? And I'm going to come back to that and ask you to actually see if you have any ideas. But first, let's talk about why eat leaves at all. Let's give you a little background on folivory in general. So why eat leaves? Leaves are, as we've already talked about, not intended to be eaten. There's very little in the world that is eaten that's intended to be eaten, right? Nectar and fruit and milk, that's pretty much it. And most of the world is not surviving on nectar, fruit, and milk. And of course, most of the milk that humans eat is not the milk that was intended to be eaten by them, by the same species.
- 3.41** So leaves, like taproots, like trunks—trunks are hard to eat, not much eats it—but the rest of the plant, the part of the plant that's not fruit or nectar, never wants to get eaten, and so it's going to defend itself. Furthermore, leaves are little solar panels, and they're well-defended chemically. Sometimes they're defended physically as well. They have little spines or thorns or something. But mostly, they're just not very high quality. They don't have much in them. They don't have a lot of nutrient load. So eating leaves, you have to eat a lot of them. You have to eat a lot of them. And you also have to eat from a lot of different species in order not to poison yourself. Because leaves are so well-protected chemically that if you eat all from one species, you are likely to end up poisoned by whatever that species has put into it.
- 3.42** So given all that, why would you eat leaves? Anyone think of—given all of the disadvantages I just walked through, what are the advantages of eating leaves? It's easy. Yeah, you could find them. Totally.

- 3.43 Other people eat them or animals?
- 3.44 Yeah. So it probably was not the first choice, right? So it may have been the empty niche. And at this point, everything is being eaten. We'll talk about that a little bit more when we talk about leafcutter ants. But yeah. Say again?
- 3.45 Availability.
- 3.46 Availability, yeah. They're available and they're easy to find. They don't run away.
- 3.47 They can get water at the same time?
- 3.48 Oh, good. Yeah, they can get water at the same time because many, many animals that drink take an additional risk when they have to go to water to drink. If you can combine, that's good. That's not one that I thought of. If you can combine the food with the water, and take the same risk at the same time, you limit your risk from predation, potentially. Yeah.
- 3.49 A particular nutritional need that the leaves satiate for the m?
- 3.50 Probably so, yeah. So even though they are low in energy, they have a lot of elements and micronutrients and such. So that is probably, that is undoubtedly so. If they're eating it, they're clearly getting something from it that they need. It may be even more relevant if we ask a question about why does some carnivore also occasionally snack on grass, for instance, as most of us have noticed doing, our pet dogs or cats doing. That may be in part about getting just a little bit of stuff that they need that they're not getting from their all-meat diet.
- 3.51 Great. Easy to find, abundant, don't run away. I guess maybe abundant. I think all of that was included in what you guys came up with. You're never going to have to spend time sitting and waiting for them to walk by because they don't walk. They don't run away, and they're all over the place, especially in a forest. It's going to be hard to be a folivore in the desert, but if you're in the right ecosystem, this makes sense. And then the disadvantages, I've already walked through many of them. They tend to be well-defended with secondary compounds—secondary compound just being the term of art for the molecules that plants put on board in order to, in order to keep herbivores away, and they have low nutritional value.
- 3.52 So flight and a low nutritional value diet don't tend to go well together. It's a problem, right? It's much easier to fly if you eat meat or if you eat fruit, something with high protein and fat or high sugar that will give you a burst of energy and allow you to fly, or that will allow you to put on the right muscle so that you can fly. Leaves take—not only are they low nutritional value, but it takes a lot to process them. Think about cows. Think about what cows have to do in order to make what they're eating into actual food. They're fermenting their food in their stomachs, and then they're actually bringing it back up, and they're fermenting their food in their foregut, in their stomachs, and having it go through a digestive process that is basically repeated a few

times in order to get both all the nutritional value they can out of the really low-quality forage they eat, which is grass in the wild, and also in order to detoxify what they're eating.

- 3.53** So in general, if you eat leaves, you don't fly because you're not getting enough. You're not getting enough energy at any point, at any given point. And you have to spend so much of your time eating leaves and finding leaves in order to get enough that it's just not going to work. There are, for instance, no herbivorous bats. A quarter of known mammals are bats, something over a thousand species of bats. None of them are herbivorous. We think that there were no herbivorous pterosaurs. They're extinct now. We don't know for sure. But what fossil evidence reveals, we think that there were no herbivorous pterosaurs.
- 3.54** There are lots of—well, there are a few folivorous birds, but they're generally lost flight. They're non-volant. So they're non-flying herbivorous birds, or they are terrible fliers, like these guys. And then there are some insectivorous insects, but almost to a species, the insectivorous insects are insectivorous when they're in their larval stage, when they're not flying. And then once they go through metamorphosis, they move into a different diet. They're either eating nectar or insects or something else with a higher energy load.
- 3.55** OK, so all these reasons to eat leaves, and yet this constraint, if you're a flying organism, against eating leaves. So how do they do it? What do you think? How are they surviving on leaves?
- 3.56** Would it be possible that—you know how sometimes people eat something that's a little bit poisonous to go about being immune to it? Does that happen in the wild with these birds?
- 3.57** I don't think so, but I don't think that we know. So the question was, in case that didn't get picked up, was, could it be basically a kind of titrating of toxins, so that getting used to it, so that they can continue to eat leaves. They may well. No, I guess I'm going to amend my answer. I think they may well do that. But if they do that, that still doesn't solve the problem of how are they managing to basically make bank on. Like how are they managing to get everything they need from the leaves, even if they're detoxifying it more efficiently than we think they should be able to because they actually have some immunity to it. How are they managing to basically fly?
- 3.58** Do they eat their feces?
- 3.59** No. Interesting question. Do they eat their feces? We don't think so. They're nesting in these spiny palms above oxbow lakes, so they probably have no access to it. It just disappears into the water.
- 3.60** Is there another food that they eat that maybe neutralizes the poisons?

- 3.61** Great question. We, I don't work on these guys, but the people who work on them don't think so. Do you have something back there?
- 3.62** You said they're eating down the trees? Is that because they are constantly eating to get the right amount of nutrition?
- 3.63** They do seem to be eating a lot, yes. They eat a lot more of their time. Their time budget is spent more on folivory than the time budget of, say, a frugivorous, a fruit-eating bird would be. Although a fruit-eating bird would then be spending more of its time looking for the fruit and not as much time actually eating the fruit.
- 3.64** Do they sleep disproportionately compared to other birds?
- 3.65** I don't think so. But why might you think that they would sleep disproportionately compared to other birds?
- 3.66** Because they wouldn't have to consume as much in that state.
- 3.67** Interesting. Maybe they're just running at a lower metabolic temperature. Probably not, being endotherms. Sleep budgets do vary across both birds and mammals, the two differently evolving groups of endotherms. But the fact is they still have to fly, or they do fly. Maybe they don't have to, but they do fly. And so even if they sleep more, they still have to be pulling the nutrients out of the leaves somehow. Yeah.
- 3.68** Have they evolved a ruminated cow-like stomach?
- 3.69** Why, yes, they have.
- 3.70** Cows actually have the same thing. They have multiple stomachs.
- 3.71** They're actually foregut fermenters, just like cows. It's crazy, and it's alimentary, my dear hoatzin. Yeah, I know. That's his fault, actually. So they actually do the same thing that cows do, and not exactly the same way, and it's convergent. This is a homoplasy. This is a convergent evolution of foregut fermentation. The trade-offs of this are it takes a lot of time and space. Just like it takes a lot of time to find enough leaves to eat, it takes a lot of time to actually ferment. And so you got to spend a lot of time in rest and maybe in sleep. Maybe they are spending more time in sleep, but they're at least spending a lot more time basically doing that digestive stuff.
- 3.72** But it also takes a lot of space. Think about a cow. Cows have big stomachs. Well, these guys, they also have really big stomachs. And what that means in part, that those big guts that they've got, which is where they're doing all that foregut fermentation in order to get the nutritional value out of the leaves that they're eating, is that they don't fly very well. They've got, there's basically not very much room for flight muscles on what's left of where the flight muscles would usually attach up here. And so even though they're these big-bodied birds, light compared to a similar-sized mammal, because birds always are because they fly, again—but they fly really badly, in part

because they're foregut fermenters, which means they've got big guts, which means they don't have room for flight muscles to attach to.

- 3.73** So let's go back to this list that I gave at the beginning. And say, OK, they're neotropical, arboreal, wacky-looking, smelly, folivorous, badly flying, communal-roosting birds with helpers at the nest. Are there any clues in there that a naturalist who was walking along or on a canoe on an oxbow lake, which is where you're most likely to see these guys —if you noticed all of those things, you might not see the helpers at the nest. You have to spend some time watching them before you'd know that was going on. Which of these things might cause you to think, "Huh, I wonder if?"
- 3.74** The smelliness might be attributed to the fermentation that their foregut undergoes in their system.
- 3.75** Yeah, the smelliness might be attributed to the foregut fermentation, and indeed it is. And so I said smelly here. I didn't say that early naturalists in the 19th century apparently described them, had no idea what they were doing. They saw them eating leaves, knew that they were folivorous, never attributed to foregut fermentation. But the early naturalists said, "God, they smell like fresh cow manure. What is that? Why does a bird smell like fresh cow manure?" As it turns out, it's because they're doing a very similar job that fresh cows are doing. When they produce the manure, they're making very much the same smell. Now, because the poop basically is going into the water and it's not with them, you don't have as overwhelming a smell, but the whole bird smells a bit. Just like if you get right up close to a cow, it smells a bit.
- 3.76** And then the badly flying thing here, too, on this list is maybe a slight clue to, OK, why would a bird be badly flying? We know there are some birds that don't fly. You could be a penguin or a kiwi or an ostrich, or you could be a swallow or a peregrine falcon or a duck. And you can either fly really well or not fly at all, but why be badly flying? And it does seem that these guys make a go of flying badly between these spiny palms and a few other trees in the adjacent forest. And they don't ever go long distances. You're never going to mistake them for something that really has flying down. There are toucans and parrots and all sorts of really beautifully flying birds in these forests, and you're never going to mistake a hoatzin for any of them, also because they're patently ridiculous-looking.
- 3.77** OK, one more bonus question about hoatzin before we move on to gliders. Baby hoatzin wing claws. OK, so baby hoatzins have on their wings claws. No other bird species has this, and no adult bird species have wing claws. Adult hoatzins lose their claws. Babies have them. What do you think they might be using them for? What could that be about?
- 3.78** Is it because they fly so badly that when they crash in the thing, they need something that'll latch on?

- 3.79** I like that. This is actually, I didn't tell you, this is pre-volant. This is before they can fly at all, they've got the wing claws. Once they can fly, basically, the flight, the volancy—I don't think that's a word—but the ability to fly is developing at the same time as they're losing their wing claws.
- 3.80** Since they're going to get their food essentially from the trees, they can climb to eat and get their leaves?
- 3.81** So, great. How about if I tell you, which I didn't tell you before, that at the point that they've got the wing claws, they're still being fed by their parents? Otherwise, fantastic answer.
- 3.82** To cling to their parents?
- 3.83** Their parents are not carrying them around, so it's not like primates. I don't think that happens, but interesting possibility.
- 3.84** Is it actually the spines in the nest? That spiny wood?
- 3.85** What would it have to do with that?
- 3.86** Maneuvering around it somehow. I don't know if they evolved growing in those types of environments, specifically with that style of wood.
- 3.87** OK. A few of you are centering, getting down into it, but we aren't quite there yet.
- 3.88** Is it for a better reach of leaves, for a better ability to feed themselves?
- 3.89** Yeah, they're mostly being fed by their parents at this point, at the point that they have the wing claws. So they're early in development. They're not flying. They're not feeding themselves. But yeah.
- 3.90** Could it be to protect themselves from predators because they can't fly?
- 3.91** It is that, but they're not directly protecting themselves from predators this way.
- 3.92** Is it to maneuver away from predators?
- 3.93** Again, so now I think we have triangulating all three of these answers, and several from over here as well. I think we basically arrived at it. What it is, is that predators will come at them aerially, mostly, because there will be some big snakes, but because they're on these spiny palm islands, they're in the middle of these oxbow lakes, there's not that many terrestrial predators that will come up at them. So the predators are mostly going to be coming in on the wing, things like a harpy eagle, possibly, or maybe, maybe monkeys. Although, again, these spiny palm islands are pretty isolated from all the other trees that the monkeys would use to get to. You'd have to be a very dedicated, interested-in-eating-hoatzin monkey to swim across the lake to do it.
- 3.94** So if the predators are tending to come in through the air—we did this, here's just a natural history illustration of their wing claws—they respond to predators by hurling themselves out of their nests into the water. As predators come in, as predators on the

wing come in at them, they just drop out of the tree into the water. And an aerial predator is unlikely to follow a baby bird that has just disappeared into the water. Most aerial predators are not going to do a good job of being able to get into the water and then get back out again.

- 3.95** OK, but that doesn't explain the claws yet. They use the claws then to—they don't use the claws to swim. They swim back to their natal tree, and then they use the claws to climb back up to their nest. And that is the only example that we have in birds, and birds are pretty well studied, of any baby birds having claws at all. And it's a pretty specific adaptation, isn't it? Yeah, well done, hoatzin. Exactly.
- 3.96** All right. OK, now we're going to talk about gliders. Gliders is not just one species, it's a number of species. This is a colugo, also known as a flying lemur, because they don't fly and they're not lemurs. I don't know why they're called flying lemurs. I actually have some idea, but it's a silly name. This one, this is not either of our pictures, is almost certainly carrying a baby on her belly because she just has a really big—it's hard to see in this picture, but if you look really carefully, it looks like there's another animal grabbing on and attaching herself. They're not like hoatzin. They don't normally have these big guts.
- 3.97** But what problem—this is only one of many gliders. We'll talk about a few. What problem is solved by gliding? Some predators are faster than me on land, and in the trees, but if I can escape through the air, then I'm in good shape. But that raises the question of, OK, why not fly? Isn't flying better? Didn't we just talk about flying being better? Yeah.
- 3.98** Conservation of energy.
- 3.99** Yeah. You want to say a little bit more about that?
- 3.100** Flying uses up energy.
- 3.101** Flying is more metabolically expensive, and it's also just more difficult to obtain.
- 3.102** I'm just going to say the same thing. It's like those glider jets that people use, and they can go hundreds of miles on no gas, basically, versus jet fuel and trying to move a bunch of parts. So they can get through greater distances with low energy cost.
- 3.103** Right. Basically, powered flight has this gigantic adaptive valley, which is to say, if you're here, if you're gliding and you're trying to get to flight, you got to go through some stuff that is not as good as what you've got before you get to flight. Climbing is easier. Basically, you can preload your fuel effectively by climbing and then launching yourself from on high and using what you already preloaded into you in order to glide. It's obviously not as effective at doing all of the things that flying does. And unless you hit a thermal, you're unlikely to be able to go up. And usually, gliders are not in a space where they're going to be able to rely on thermals.

- 3.104** But it has evolved many, many times. And that's the question to think about here is, flight is amazing and powerful and useful and almost ubiquitous in birds, ubiquitous in bats, lots of insects, pterosaurs. We've got lots of extinct clades that are fliers, many extant clades. So it's happened a lot, but so too has gliding happened a lot. So once again, flying lemurs don't fly, and they aren't lemurs.
- 3.105** There is some question as to what they are, like what they are phylogenetically, what group they belong to. And some systematists are still saying, "Maybe they are primates," but almost basically, the molecular evidence and most of the morphological evidence at this point says, "Yeah, they're not." They might be sister to primates. They have some primate-y characteristics, but they don't seem to be primates, and they don't fly. They glide. So they're gliding non-lemurs, which doesn't make for as good a name. They're colugos. And these guys live in Southeast Asia. Interestingly, actually, a lot of the gliders live in Southeast Asia. What else? Can you think of other things that glide as an anti-predator defense?
- 3.106** When you look from above, they look like something else, like leaves? When you see from above, a bird or anything, they might look like a leaf, if they camouflage?
- 3.107** Yeah, so that's perhaps one reason that you might glide. I'm looking now for other organisms that do glide.
- 3.108** Was going to say that they're quieter?
- 3.109** Oh, actually, that's another reason to glide. Yes. That it is potentially quieter, although fliers can glide, gliders can't fly. So that goes one direction and not the other. You might not be able to as effectively, like you might not be able to glide as effectively as a flier, especially depending on what you're maximized for. If you're maximized for short turns and aerobatic, like acrobatic flight, you probably don't glide very well. But if you sometimes fly over long distances in open spaces, you can probably glide OK, too. What other things glide?
- 3.110** Flying squirrels.
- 3.111** Flying squirrels, yes. We'll talk about those a little bit. Yes.
- 3.112** Snakes and geckos.
- 3.113** Snakes? Yes. Actually, yes. Hopefully, you've now scared a number of people because, yeah, there's some gliding snakes, which is bizarre. I don't actually know that there are—I guess there are these parachuting geckos is the distinction that many herpetologists make. And so the geckos don't show up here because they have the webbing between the toes and some of the flanges on the body, and they can sort of parachute down, but it's more vertical. They don't have much lateral movement, the geckos that do this.
- 3.114** Also, these beauties. This is *Agalychnis callidryas*. This is the red-eyed tree frog. She—I'm imagining, I don't know why—was very nice and let me take her picture. This is a Hylid. This is a New World tree frog. There are also Rhacophorids in the Old World.

Different evolutions of arboreal frogness, different evolutions of tree frogness. They have somewhat different characteristics. The tree frogs, the true tree frogs like this one have—for instance, you can see the expanded toe pads and also additional digits, so they can actually wrap their hands more around trees. They have more digits. But that's about climbing.

3.115 And a few of them, like this red-eyed tree frog, actually do a good job of gliding as well. And what characteristics of a frog—I've just talked a little bit about what characteristics of a frog might make you a good climber, but what characteristics might make you a good glider? What do you expect to see in a glider?

3.116 I think very light and having a lot of elasticity in the skin.

3.117 Yeah, light for sure, and that doesn't show up here, but yes. Why elasticity in the skin?

3.118 Because if they were, say, scaly and rigid, they wouldn't be able to stretch out these parts and go from climbing to gliding.

3.119 Yeah, OK. So, and maybe that translates to, in just the language of herpetology, like having skin folds, like having extra skin, and also smooth. So smooth, which amphibians have, and also having extra folds. Yes. Anything else?

3.120 Aerodynamic?

3.121 Yeah. So what will make them aerodynamic?

3.122 The ability to fold themselves in a way that reduces the resistance to air.

3.123 Yeah. So in part, the aerodynamism will come from having skin that maybe often lies flat but can be spread out. It's also going to have webbing between the fingers and toes. And basically, a lot of places where you might not expect non-gliding congeners, non-gliding relatives of these guys, that they will have extra flanges and such, which can be used as control surfaces during flight, so that they can basically steer a little bit, even though they don't have power. So in a study of the Old World tree frogs, they found full webbing on hands and feet, enlarged hands and feet as well. So that's again control surfaces and ability to steer and additional flaps of skin on the arms and legs, exactly as you were saying.

3.124 OK, how about these guys? This is *Draco volans*, the flying dragon from Indonesia. It's a tiny little guy. It's called a flying dragon, but it's not a big animal. These are 70 to 80 millimeters, snout-vent length, so that's head to butt, and their tail is longer. Like many herps, like many lizards, like many reptiles and amphibians, they are sexually dimorphic, but the males are smaller. So unlike what we expect in mammals when they're sexually dimorphic, the males are smaller than the females, but they're still like, the males are like 70 and the females are like 80 millimeters, snout-vent length, really small little guys.

- 3.125** And they have color. They're convergent with the New World anoles, which you may actually have in Miami, but certainly in the Caribbean, where the males have a dewlap that they extend, and it's a different color by species, and they use it to display, and both to warn other males to stay away and to attract females. So these guys also have a dewlap that they use to display, but they also have patagia that they extend when they're flying. And both sexes have it. They're sexually dimorphic in a lot of regards, but both sexes have these patagia, but they're differently colored. And the tips of the patagia are blue in one sex and yellow in the other. I don't remember which. I have not been fortunate enough to be in the habitat these guys are. Haven't seen them. But they are, again, gliding to escape from predators.
- 3.126** OK, and then we have, as has already been mentioned, the southern flying squirrel. There's also northern flying squirrels, *Glaucomys volans*. This is a composite photo, of course, not one of ours, from launch to gliding to landing. So often we have seen—the specific epithet in the scientific name is *volans*, which means flying, even though they don't actually fly. And you don't tend to get *volans* in the scientific name when it's an actual flier. You only get it when it's sort of adjacent to flying.
- 3.127** Anyway, the southern flying squirrel is in fact a squirrel, but does not fly. They nest in, sometimes in leaves made of—I mean, in nests made of leaves, but very often in tree hole cavities. They take over old woodpecker holes, and they launch themselves again to get away from predators. Then finally, among the gliders that we'll talk about here today, all of these guys have control surfaces. We talked specifically about what makes you more likely to be a really good glider if you're a frog, for instance—having a lot of webbing, having a lot of extra skin, having bigger hands and feet so that you can use those as control surfaces.
- 3.128** But snakes don't have control surfaces. And yet this one genus, *Chrysopelea*, glides. And there's a lot of interesting work on this. If you are snake-sensitive, maybe look away. But here's just a couple of videos of—this is done in a controlled situation. This is part of research. This isn't in the wild. Here we go.
- 3.129** Yeah, you don't want to run into that if you weren't expecting it. And here's another.
- 3.130** Yeah. So most snakes can't do that, I promise. Most snakes have no ability to do that. You could see, at least in the second video, and maybe a little bit in the first, too, but they dorso-laterally flatten when they are launching themselves and as they are flying—even I do it—as they're gliding through the air. And they use the usual muscle, like snake muscle techniques, of coiling themselves and using the traction, the friction, on the surface they're on to launch themselves up. But then once they're in the air, that's a trick that other snakes do not do. This is the only snake that we know this from. And gosh, I've forgotten. I think this also is in Southeast Asia, *Chrysopelea*. Definitely not New World.

- 3.131** And they do it without the control surfaces. They obviously don't have webbing between their hands and feet because they don't have hands and feet. And they don't have extra skin flaps because they're snakes. Most of what they're doing all throughout most of their lives is doing snake stuff. And snake stuff is mostly not about gliding. But in this one clade, and it is a couple of species, in this one clade, they have also, like the colugos and like the frogs, the tree frogs in the New World and the Old World, and the flying dragons and the flying squirrels, have managed to learn, even without control surfaces, how to glide.
- 3.132** So just one quote from the guy who did the research. He was at University of Chicago at the time he did this research. "Although many cylindrical animals swim through water, flying snakes of the genus *Chrysopelea* are the only limbless animals that glide through air." So that's just in there again as a —if that, if that freaked you out, as it does a lot of people, you're very unlikely to run into it. It's only the one genus.
- 3.133** All right. So segue to talking about kelp, the final evolutionary jeopardy organism that I will be talking to you about today. Kelp. Probably everyone has seen it. You've been walking on the beach and you see there are a lot of species, of course. They're in the water. Otters like to hang out with them. If you're in a boat, you might get tangled in them if you weren't expecting it. There are a lot of marine organisms that live in kelp forests, especially. I'm, again, most familiar with the ones in the Pacific Rim. We'll talk about that just briefly here for a moment, in a moment. But the question, one of the many questions I could ask, about what is it that they're doing is what problem are they solving with the floats?
- 3.134** One, they get a free ride from the current, so they can spread out with no energy cost. Then also, they're staying closer to the sun versus, let's say, plants that are underwater and aren't getting as much light.
- 3.135** Yeah. So with regard to the first answer, they're getting a free ride. Mostly, they are rooted. Mostly, they are not able to just move completely freely. But yeah, they're closer to the sun. And that's going to be completely relevant to being a successful kelp. Yeah.
- 3.136** It's a mechanism to lodge their genes as far into the future as possible.
- 3.137** Well, it surely is. But that's so, what you said, in case it wasn't picked up, was a mechanism by which to lodge their genes as far into the future as possible. Yes, but not very precise. Not a very precise answer. So let's talk first. It's going to largely be about sun. But let's take a step back first and say, OK, we've got trees, kelp, and vines in the world. There's lots of other kinds of plant forms, but those are three. Kelp is not technically a plant, but even most of the researchers who work on kelp will say, "You know what? I'm just going to call it a plant to make it easier to talk about." So they're not actually technically plants, they're big alga, but we're going to pretend that they are.

- 3.138** What they are for sure is autotrophs. OK. They are making energy from the sun. They don't need any other organisms to make their energy. We, of course, are heterotrophs. We need to steal someone else's energy that they have gotten from the sun, or we steal someone else's energy that they have gotten from someone else that they have stolen from someone who's made it from the sun, right? So heterotrophs are all dependent on autotrophs, to some degree, and autotrophs are all dependent on the sun. And what autotrophs need, therefore, like plants, including, for purposes of this discussion, kelp, is to have access to the sun with which to photosynthesize and convert that sunlight into sugar.
- 3.139** How do you do it, though? So there are a lot of ways that plants exist, and most of them aren't actually phylogenetically constrained. Like, trees don't show up once or even a few times in history and they're like, "Oh, that's the tree clade." No, trees show up almost everywhere in every family that plants exist. Vines show up almost everywhere. Shrubs. All of these forms are really variable within plants, unlike animals. Animals tend to develop a form, and then we kind of stick with it. We're constrained. We're much more constrained by the developmental pattern that we inherit than plants are. But trees are all over the place. Why are trees tall?
- 3.140** To avoid being eaten by animals on the ground, and also again to reach a little closer to their food source.
- 3.141** Yeah. So two answers—to avoid being eaten by organisms on the ground and to get closer to their food source. To avoid being eaten by things on the ground, they wouldn't have to be that tall. I mean, depending on what the hostile forces of nature are for that particular tree. Do they have climbers that can get up into the upper branches? Then no matter how tall you are, it doesn't matter. Do they just have browsers who could maybe stand on their hind legs like the deer? Then all you have to do is maybe be 10 feet tall and put your leaves above that, and then maybe you're clear. But yeah, avoiding terrestrial herbivory is going to be one concern, but there will also be avoiding aerial herbivory or climbing herbivory. There's going to be something that wants to eat you no matter what.
- 3.142** So probably, mostly, it's about competition between other trees in order to get access to the light. A prediction of that is that, if you have a species of tree alone in a prairie that can survive well there, that isn't outside of the center of its really successful zone to live, and compare it to the same species of tree in a similar habitat, control for everything that you can in the middle of a forest, you would predict that the tree in the middle of the prairie wouldn't have grown as tall. That's a prediction of the hypothesis that trees are tall because they're competing for sunlight.
- 3.143** Vines managed to get to sun by a couple of different means. There's vines and there's lianas. Some vines grow up from seed and reach the height without ever having to put down any woody stems, and they basically parasitize the work of tree trunks, and they wind around them, and they get to the heights. Some basically start out at the top, and

then they grow down, and they become maybe rooted on the forest floor, and maybe not. But all of them are needing to make bank, to basically make sugar from the sun, which means they need sun.

- 3.144** So differences between them—they are reaching for the light, but they have opposite constraints. Working against gravity with wood versus staying buoyant. And so kelp has these floats, but it wouldn't work at all if what was attached to those floats was hard, was rigid, because the currents that they're in would break those, break those kelp trunks that don't exist into many pieces. So they have two sort of competing constraints, which are, "How do I stay pliable in the current?" And trees have to withstand winds. Different trees have to withstand different kinds of winds, depending on where they are. But trees have to withstand winds, which is not nearly as consistent or, usually in most species, in most situations, as chaotic as the currents that kelp has to deal with. And so they've got floats to keep them on the surface and completely flexible bottom parts.
- 3.145** And what's right below the floats are photosynthesizing because, unlike air, if you have a light gap, it doesn't matter if you're on the forest floor. If the sun hits it, you get just as good a light as you do at the top of the canopy, as long as the light hits it. In water, of course, the sunlight degrades very, very quickly. And so you are only able to photosynthesize within a few feet of the surface of the water.
- 3.146** OK, one last bonus here. First, so the problem solved is "how do you photosynthesize when you're underwater and a trunk would snap, but you need to stay close to the surface?" So that's some of the backstory on the floats that you see when you walk on the beach and see kelp floats. So one bonus here about kelp is that the peopleing of the Americas may have been made possible by kelp. The so-called "kelp highway" hypothesis is interesting and compelling and has considerable evidence for it. And it is that when people came over from Asia, overland through Beringia and down into the Americas, maybe as was proposed and considered to be the likely thing for a very long time, maybe they took an inland route through a newly ice-free corridor and had to contend with all that the great North American landmass had to offer in terms of hostile forces of nature.
- 3.147** Maybe they took a coastal route on land and basically hopped between refugia on land, between the glaciers before the glaciers retreated entirely. Or maybe they were in boats. Maybe, the kelp highway hypothesis proposes, the people came over in boats, and this has a number of things going for it. It allows you to be at sea level the entire time. And so if you're on land, you have to deal with a lot of changes in elevation and other difficulties that you may not be equipped to deal with. Kelp forests are incredibly diverse. They're these three-dimensional landscapes where, if you're in a boat on the water, you can access a ton of marine resources relatively easy once you know how to do it.

- 3.148** Across the northern Pacific Rim, basically from Japan all the way down to Baja, on both sides of the Pacific, the species in the kelp along the Pacific Rim, but especially in kelp forests, are either identical or very, very similar. They're either closely related species or doing very similar things or they're identical species. And so if you're a fisherman, if you're a Beringian who is traveling the coastal highway in your boats and you know how to fish up north, it's going to change a little bit as you start heading south, but you're basically going to be using the same techniques in the same boats, eating the same stuff, cooking it the same way as you did up north.
- 3.149** So as the climate changes, as all of the vagaries of difficulty of moving into a new continent for the first time happen to you, at least maybe your food and the way that you accrue it and how you prepare it is more or less stable. Then finally, just like otters use kelp as holdfasts while they are eating their food or while they're sleeping or they're just hanging out, so too can people in boats use kelp as holdfasts. And the kelp are basically diminishing near-shore wave energy tremendously. And people can put their boats between the kelp and the coast. They can hold onto the kelp. It's much more protective than being in open water. And so that's a hypothesis on the table at the moment in terms of how the Americas were peopled, which to me seems quite remarkable.
- 3.150** All right. Thank you, guys. Otters up next.
- 3.151** Thank you.
- 3.152** Yeah.

Lecture 4: Adaptive Evolution

At a glance — Adaptive evolution, illustrated with sea otters.

- 4.1 All right, guys. We are going to talk about sea otters. And remember, the framework that we are using is that organisms are evolutionary answers to ecological questions. So you have the answer in front of you. And the thing that we're after is, well, what question does that answer? You can also play this with the traits of organisms. Sometimes you can play it with a clade, where a clade, a whole group of organisms, answers a set of related questions. Sometimes you can't because a clade can divide into two things.
- 4.2 So for example, the set Artiodactyls involve a lot of things like deer and giraffes, which may be answering similar questions. But a branch of the set Artiodactyls became whales, and they're obviously answering a whole different set of questions. So it doesn't necessarily apply at the level of clade, but at the level of traits and the level of creatures, it's a pretty good way of thinking evolutionarily and adaptively.
- 4.3 So here, of course, we have a sea otter, and we're going to get to the question, but I do just want to point out, it's kind of an odd creature to find where it is. It looks a little bit like a terrestrial carnivore, and there's a very good reason for that. If you think about the cladistic context, the phylogeny that it is embedded in, its closest relatives are the other otters. Now, the other otters spend a lot of their time in water. They do spend some time in saltwater and places where they encounter it, a lot of time in freshwater, but they breed and they sleep and they burrow terrestrially. Sea otters never come on land. They actually breed and sleep in the water. So they've become quite extreme in this regard.
- 4.4 This is their sister taxa. That means the next rung out and sister taxa to the otters, including the sea otters, are the mink and its relatives. Now, minks are also associated with water. They're primarily terrestrial, but they are associated with shorelines. They do swim, and they swim well, but most of the time they spend on the shore meandering up and down, making a profit, in that intermediate zone. And then outside of that group, sister to the mink and all of the otters, we have the other weasels.
- 4.5 This is a wolverine, which is perhaps an extreme example. A wolverine is, in some sense, the answer to the question, "How can you make a profit from carrion of animals that die, especially during the winter, when it's very hard to find them?" So wolverines have extremely large ranges, and they're very good at finding fallen animals, animals that are very likely to die in the winter because it's harsh for everybody.
- 4.6 OK. So before we get to the ecological question to which sea otters might be the answer, I wanted to point out an anomaly that suggests just how weird this evolutionary game is. Up in the San Juan Islands, where Heather and I live, and in the Puget Sound, you would expect sea otters. It's right in the middle of their range as you

go along the coast. But you don't typically find sea otters in the Sound or in the San Juan Islands. We do have river otters. We see them all the time, and they spend a lot of time in the sea. They are, in effect, in the sea otter's niche, but somehow competitively superior there.

4.7 Now, that's odd at first, and I spent a lot of time thinking about why that might be. Never came up with anything really solid until I realized that sea otters have a problem. Because they don't go on land, they are exposed to predation from the sea. Now, there aren't a lot of things that will eat them, but there are a few, and on that list are sharks, like a great white shark and orcas. Now, we don't have a very high density of great white sharks, but we do have orcas in the Sound and in the San Juan Islands.

4.8 And so one possibility is that the density of orcas in the San Juan Islands makes it very hard to be a sea otter that has to sleep close into the shore because the likelihood of an encounter with an orca is very high. And therefore, river otters might have an advantage in the sense that they spend the night on land in their burrows, safe, where there are no terrestrial predators large enough to eat them. So again, it's a hypothesis, but it would at least explain why in the sea where you have rocky habitat and kelp, you don't have many sea otters.

4.9 OK. Now, in addressing the question of what it is that explains sea otterness, what the question is to which they are the answer, it is important to realize that their current range is not a good representation of their ancestral range. Here you can see on this map the ancestral range for sea otters, and it stretches from Hokkaido, Japan, all the way down to southern Baja. They existed in that entire range. And it is, in fact, one of their specializations, one of the the things that makes them an entirely unique species, that caused them to be nearly exterminated throughout that range in the 18th and 19th century. It is their coat.

4.10 Sea otters have an incredibly dense coat. In fact, it's the densest fur of any of the nearly 5,000 mammal species. It is so dense, in fact, that when the animal goes in the water, it does not get wet. The fur excludes the water, which is a partial answer to how you can take a small endotherm like this one and have it survive in waters throughout much of that range, which are very cold. That's surprising that they can do that because endotherms are already, even under excellent circumstances, spending a huge fraction of their caloric budget just to stay warm enough for their enzymes and other molecular structures to function properly. They spend something like 85% of their budget on just staying warm when they're outside of the water.

4.11 You put them in water, and they have a much bigger problem because the specific heat of water is incredibly high. So as you know, if you've jumped into cold water, it robs the body of heat really, really quickly. So that problem is even further exacerbated if you're a small animal because your surface-to-volume ratio is high. You have a lot of surface area with which to radiate heat. So the evolution of an incredibly dense pelt allows sea

otters to neutralize not all, but much of the problem that comes from a small endothermic animal entering cold water and staying there full-time.

4.12 Another thing here shows that is important to realize is a key to the answer that they provide. You can see in this picture, these paws are very hand-like. This is an animal that has incredible dexterity. Now, when we talk about the niche that these animals are in, you might think, well, it would be more natural for something else to be exploiting the food sources that they are eating, right? Wouldn't it be natural to have something like a seal, which is larger and has less of a problem with radiating heat, for example, is better able to defend itself against larger predators? Wouldn't that be a good choice for a platform on which to build something that feeds in their niche?

4.13 Well, you can't build something that needs dexterity out of a seal. A seal is actually evolved from ancestral bears, but it has lost the dexterity in its hands because they've become flippers. So this animal has retained its dexterity, and that's going to be key to understanding how it is that they exploit a niche that other creatures have found difficult. I also want you to see here—this is a hard shot to get because they don't usually show it to you—but this is the hind foot of the otter, and you can see how beautifully webbed it is. So this animal can propel itself along at surprising speed. It also has a flattened tail, and it can dive down very effectively. It's an efficient creature with respect to diving relatively small distances. They can go, I believe, up to 100 feet is the maximum they've been observed diving. But they're good divers, and they're efficient swimmers, given that they're built out of that terrestrial carnivore platform.

4.14 All right. Now, here we're going to get to what question it is that sea otters uniquely answer. Here you see this sea otter doing something that they have become quite famous for. This animal is going to bang that clam with the rock that it has picked off the floor of the shallow sea here, and it's going to break it. Boom. And then it's going to extract the contents. So what you have in the shallow zones where these animals are living are a tremendous number, a huge diversity, of invertebrates, all of which have defenses so that they don't get eaten. This one has a hard shell. It's very difficult to get into it, right? But the animal it's dealing with, not only does it have the dexterity to pick up a rock and break that shell open, but it also has the strength and the intelligence to pull it off, right?

4.15 This is an animal that's floating on the surface, using its own bouyancy as a anvil, and then using a rock to break open an animal that has otherwise a sufficient defense to keep the animal out. You will recognize that other creatures have solved this in different ways. You may have observed gulls taking mollusks high into the air and dropping them onto rocks, using a force they cannot generate other than by lifting it up and allowing gravity to do the work. But you've got an intelligent creature who has solved a problem with tools available in the environment and has extracted a resource that, frankly, a seal couldn't get to. That's a pretty remarkable thing.

- 4.16 And so underneath all of these issues of, well, "what question does that animal answer?", we are haunted by competition. If you can figure out how to get at some resource that some other creature can't figure out how to get to, well, then you've solved a problem in doing so. So if we're talking about hoatzin, the hoatzin have figured out how to make use of a resource that to almost every other creature of their kind in the forest is totally useless. It's too well defended. You can't get through the cellulose and the leaves to extract the stuff that you could actually use. Well, the hoatzin has solved it.
- 4.17 And here you've got an otter that has solved the problem of hard-shelled mollusks. Here you can see it's picked up crab. I unfortunately did not get a picture of them eating sea urchin, which is a big part of their diet. In fact, it's such a big part of their diet that it has a very positive impact on the kelp forest that they depend on. They eat so much sea urchin that they free the kelp from the effective herbivory that the urchins otherwise exert. So they have an important ecological effect.
- 4.18 Here we have another invertebrate, and this is an invertebrate that has to be dug out from between the rocks. So you've got a highly intelligent creature, capable of eating a wide diversity of things and solving whichever problem it is that accompanies each food item. And I must tell you, I spent my whole life waiting to get the opportunity to photograph sea otters up close, and I finally got it. But inside of 2 hours, I must have seen it eat 7 or 8 different species, solved in different ways. I saw it not only use a stone to break open a shell on its belly, but I saw it do the reverse. I saw it take the shell and bang it on a stone on its belly. It's a very dynamic mechanism for feeding.
- 4.19 And the impression that it leaves when you watch it is that there's a lot of stuff to eat down there. It's just all wrapped up in puzzles that you've got to solve in order to do it. And so no other creature is solving them. Some of the creatures are intelligent enough but don't have dexterity. Some of them don't have the intelligence or the dexterity. And this terrestrial carnivore has the basic mechanisms to do it all. Now I want to talk about the final little piece of this puzzle, and it's going to provide us a tool that we're going to use for further discussions down the road in this lecture series.
- 4.20 Here what you have is an otter feeding its offspring a prey item that she has captured from the sea floor. Now, probably the most important thing, the most subtle and yet significant fact of the question for which sea otters are an answer, is that they don't have to learn it all themselves by trial and error. What we have here is a dual inheritance system. It is true, as we talked to you about yesterday, that the goal of all evolved creatures is to lodge their genes as deeply into the future as they can. But some creatures have dual inheritance.
- 4.21 That is to say, they have culture in addition to genes. Now culture, for those creatures, is not the point. Genes are still the point. Culture is a means to an end. But what culture does is it allows individuals who have solved a problem to pass on the solution to that problem to, in general, offspring, sometimes to other conspecifics or related

individuals. But the point is, when you look at that range, what you are looking at is a hugely idiosyncratic number of opportunities. The beaches and the rocky outcrops in Baja don't look anything like they do on Hokkaido. They're very different. What's excellent about something like a sea otter is that it can pick up local knowledge in a way that would be impossible if there was no generational overlap and cultural transmission.

- 4.22 If this creature were one that hatched out of an egg and never met its parent and operated some program to go extract resources from the environment, it might be very effective in some place. But if you took that program and you pointed it at a different environment, it would likely fail because that program, even if it was close to the right program, probably wouldn't generate enough resources to allow survival. So you would have to have many different species around this range in order for all of the different opportunities to be appropriately exploited.
- 4.23 But when you have something like a sea otter, a mammal, where the babies always meet at least one parent, and spend time with that parent as the parent is foraging and feeding the offspring, the parent is training the offspring. What's good to eat here and, what's more, how do you get it? At first, the baby is just consuming resources. Later on, the baby is learning. There is, at the very least, information contained in there about what it is the mother thinks is useful to forage for here. And that may be very specific. It may be that that mother prefers this food in that spot, but that elsewhere in her range, she prefers something different. And that's information that that baby can potentially pick up.
- 4.24 So when we look at something like a sea otter, we should think of it differently than we think of creatures that don't meet their parents or don't meaningfully exchange information with others of their kind, because what happens is, a software program is built for the offspring after they are born. And that software package allows it to be tailored to the specific piece of environment where that offspring is going to live. Now, maybe not the exact place, because at some point, that offspring is going to leave its mother. The mother is not going to turn over the territory to the offspring unless she dies, and there's no expectation that she will here.
- 4.25 So the baby is going to move on, and it's going to have to innovate something. It's going to have to find some place on the coast that the program it picked up from its mother is good enough. And more than likely, it's going to find a habitat as similar to the one that it grew up in as it can, because that's the one for which it has the specific skills that are going to allow it to survive well. So the answer is, idiosyncratic coastline is well exploited by a dexterous creature that can make a small profit off of a bunch of different species that combine to one large profit, where each of those species represents a distinct problem. That's the problem to which sea otters appear to be the amazing answer.

- 4.26 OK, now we are going to return to the question that I left you with yesterday. We were talking about the nose-leaf structures in bats. And I showed you this picture. And you will recall that these bats have this extension rising off of their nose above the nostrils. And I told you that, if you were to hold one of these animals in your hand, you would watch that thing twitch back and forth wildly, right? So the question is, what is this for? And it isn't just the species. There are hundreds of species that have a nose-leaf. What's even more interesting is that it has evolved multiple times, at least three. So I want to show you a little bit about the diversity of nose-leaves, and then we're going to talk about what question it is that they might be the answer to.
- 4.27 First, we're going to talk about Merlin Tuttle, who took all of the photographs I'm about to show you. He is, without a doubt, the greatest bat photographer who ever lived. Also a wonderful human being and a great conservationist. All right, here we have a leaf-nosed bat. This is *Macrotus* in Jamaica with a katydid. This is not a frugivore. The bat I showed you before, under the leaf, those are frugivores. This guy eats arthropods that he gleans off the surface of trees.
- 4.28 And I will say, when I did my graduate work, I studied fruit bats, but I learned to radio-track bats from a woman who was studying katydid specialists. And so you would go to two different parts of the forest with a radio transmitter on the bat, and you would use an antenna to figure out where the bat was, and you could triangulate its position. But you learn something about these gleaning bats. First of all, you learn that katydids don't fly very well. Katydid usually land with a crash. So you hear the crash, and then you detect the bat moving from some perch where it's been waiting to hear that crash, and it flies, and it picks off the insect and then flies off somewhere to eat it. It's got a nose-leaf.
- 4.29 Here we've got *Trachops*, the frog-eating bat. It has a nose-leaf. *Trachops*, again, the frog-eating bat, not listening to the fact that it's a frog-eating bat about to capture this lizard. This is *Vampyrus spectrum*. This is a hunter of vertebrates, including birds, other bats, rodents. This is a large animal. It can have a 3-foot wingspan. It's truly ferocious.
- 4.30 It also points out something interesting about the leaping to conclusions that sometimes goes on in science, because, like so many of the New World bats, it has an allusion to vampire in its name. This is *Vampyrus spectrum*. It's not a vampire. There are vampires in the New World tropics, vampire bats, but somehow scientists were so frightened by the fact that they knew there were animals out there that actually drank blood that they went and they named a bunch of fruit bats and other bats as if they were vampires, and only later found out that it had nothing to do with it.
- 4.31 OK, this is a pallid bat. This is a pallid bat in the desert, and it's carrying a centipede, obviously. It's got a nose-leaf. Here's the same species carrying a scorpion. It's got a nose-leaf. And then here's the most interesting thing. Here we have a horseshoe bat. All the bats I've shown you so far with the nose-leaf, those are all New World bats. They

are in the Americas, and they're all part of the same family, the Phyllostomidae. This is a whole different family. This is a separate evolution of nose-leaves. And look at that. It's got a big insect in its mouth. That's interesting. Nose -leaf, big insect in its mouth. What the heck is going on?

- 4.32 All right. And then this is another evolution. This is Megaderma of the Megadermatids, and it is in Thailand. And here, wouldn't you know it, it's carrying a large cricket. It's got a nose-leaf. So what we've got is almost the perfect case. This is now a place where we have a bunch of evidence about what this nose-leaf thing is. But it's a weird story, right? It's evolved multiple different times on different continents. You've got a bunch of different food sources. What does eating fruit have to do with eating katydids? Why would this nose-leaf thing have evolved? What's it for? Anybody got it? Speak loud.
- 4.33 Is it supposed to resemble an insect?
- 4.34 No, it doesn't resemble an insect.
- 4.35 I think it has something to do with echolocation.
- 4.36 Has something to do with echolocation. Do you have it?
- 4.37 It's a receptor?
- 4.38 It's a receptor. No. Good try.
- 4.39 I was going to say sensory, but it's an olfactory sense?
- 4.40 Olfactory sense. An excellent hypothesis, but wrong. Nobody else? All right. Here is the marvelous answer. I got to tell you, I love this one. All of the bats that have that nose-leaf are echolocators. They echolocate. But it's hard to echolocate when you're carrying stuff in your mouth, right? If you've got a big katydid in your mouth and you're sending out sounds that have to bounce off stuff and come back to your ears, it's going to be all muffled. It's not going to work. It's like driving while looking through binoculars or something, right? It's not a good idea.
- 4.41 So what's happened is these guys have been modified to echolocate out their noses. And that's not a receptor. That's a transmitter. That's a focuser of sound, right? That nose-leaf modifies the sound, puts it over the object that the animals are carrying, so that they can hear clearly and know where they're going. You got to admit that's a great answer, right?
- 4.42 And here's the last piece of the puzzle that really suggests that it's right. That animal is *Desmodus rotundus*. That is a vampire bat. It evolved from an ancestor with a nose-leaf. It has lost that noseleaf as it picked up the drinking of blood because it doesn't carry anything. Now, its nose is still funny-looking, but there's no appendage there. It doesn't need it. And so, of course, evolution economized, and it repurposed the materials, and it removed the risk of having that fleshy, vulnerable appendage right on its nose because, of course, that's what selection would do.

- 4.43 All right. We are going to move now into a discussion of tent bats, and you will be thrilled, no doubt, to find out that tent bats are all fruit bats, so we're not leaving the subject completely. This is not a tent bat. I put this here to remind me to tell you that there's a wide diversity of ways that bats roost. This is one of my favorites. This is a sucker-footed bat. This animal literally has suction cups on its wrists and ankles that it can operate with little muscles. And this leaf here is very smooth and shiny. That is a young leaf on a plant that deploys that leaf as a rolled-up structure and then unfurls it slowly. And there's a period of 3 or 4 days where there's a tube that's just the right size for these bats to hide in it.
- 4.44 And it's the kind of thing, if you don't know it's there, you walk by these things all the time and you don't realize that there's something absolutely magical in those tubes. But if you do know it, it's very hard not to check because every so often you get lucky and you see something marvelous like that. You can find them, my favorite way, the way that does not typically scare them into leaving all of a sudden, is you take a headlamp and you shine it and you look for the bats through the leaf. That's the best way to do it. And that's how I got that picture, because I knew that bat was there before I'd scared it off.
- 4.45 But anyway, this is not a frugivore. It's not a tent bat. It doesn't modify that leaf, but it does take advantage of the fact that in the forest where this animal lives, there are always leaves in the process of unfurling, and they have this glorious smooth surface on which suction cups work, and it has modified them radically. This is one of the only bats that roost head-up. OK, now to tents.
- 4.46 This is a bat tent. You will know from the story I told you yesterday that it is much easier to understand that it might be a bat tent looking at it in a picture than it is to find these things in the forest. They are devilishly difficult to find until you train your mind to see them, and then they're all over the place. But anyway, what you can see here is, you've got—I think this is a cyclanth leaf, and it would rise up like a fish tail. And what's happened is the animal has cut in ways that were until recently mysterious, has cut twice along this leaf in such a way that the fish tail collapses, making a structure that hides the animals from all sides.
- 4.47 Here's another one. This is a little dilapidated. You can see it's been a while since this one was built. It's beginning to die from the back. That is, incidentally, no guarantee that you won't find bats in it. Sometimes you do. In general, I would say I probably find 30 empty tents for every one I find that has bats in it. It's one of the conspicuous facts of these things. There's a huge number of these objects for the number of bats that are actually out there. Person on a good day might find a couple hundred of these if you did 6 or 8 hours of walking.
- 4.48 They are typically pretty low to the ground. I've seen them as low as a meter. More typically, they might be 2-3 meters. Sometimes they could be 4 or 5. They often occur in clusters. Sometimes if you find one of these, you'll find another one right there. Or if

you look around in that area, there might be a few within a number of meters. These are all clues to what might be going on here. You actually can see two different different structures here. This leaf, this is a *Philodendron*. This *Philodendron* has been cut twice at the back here, and it's been cut once along the rachis. This one has not been cut there. There's a lot of variation in the structures. And in fact, there are 8 recognized architectures that are distinct.

- 4.49 Here you see one of the radically different architectures. This is the only multi-leaf structure that is known. And this particular version is interesting, too, because if you look down here, that's an old tent. A new tent has been built above it. This animal or some other animal came here, returned to a plant that had already been modified once, and modified it again. Typically, if you go and you look at one of these tents, there'll be nobody home. But every so often you see them.
- 4.50 Now, here you can see what the modification has done. That 's the modified tent. That's what the plant looks like, unmodified. So the animals are cutting all of those leaf rachises and creating a cylindrical structure that hides them from all sides. So it is certainly true. You've heard me say it twice now. These structures do hide the bats, but there's more going on. So this is the animals that were inside that last tent I showed you. That tent. These bats. That's *Uroderma bilobatum*. And notice the pronounced nose-leaf. These are definitely frugivores.
- 4.51 All right, here's another one. You can see a palm leaf. You can see what it would look like if it were unmodified. And then what's inside? Oh, look at that. This is *Artibeus watsoni*, and that is a female, and that's her offspring. Now, that offspring is probably old enough to fly, but you can tell it's not fully mature because it's the wrong color. It will turn brown when it's ready. It's a bit gray until then. Oftentimes in the forest where I worked, you would find pairs, a male and a female, and often offspring, living together. In this case, it's just the mother. That could be because the male didn't stick around, or it could be because the male got eaten while he was foraging one day. But in any case, he's not present.
- 4.52 OK, here's another case. You can see this is a palmate leaf. Looks like that when it's not modified. All the rachises have been cut. And there you go. *Artibeus phaeotis*. So again, a frugivore, pronounced nose-leaf. So what the heck is going on? You've got all these nose-leaf bats. None of the other nose-leaf bats make tents. All the tent-makers are frugivores. What's that about? What's more, there are a lot of frugivorous bats in the world. All the tent-makers are small. It's small frugivores that make tents. And tents have evolved at least 3 times, probably 4, but at least 3 times separately, always in small fruit bats.
- 4.53 All right, here's yet another structure. See that? It's been cut down the middle, and I can tell you there's also been some work along that seam. And there you go. Another *Artibeus*. You can also see that they've been using this leaf for a long time. They've left scars. I call them roosting scars. Basically, as they hang, they put their claws into the

roof of the tent, and it causes some damage, and eventually fungus will begin to eat them. You can also see that there's wear. Underside of these leaves are often chalky white, but you can see that the bats flying in and out have worn that coating off the surface, and they've left it shiny.

- 4.54 All right. Here's yet another structure. I call this a boat tent. And there you go. *Artibeus watsoni*. All right. So any idea what the hell is going on? Why would this be small, frugivorous bats that have evolved at least 3 times to modify leaves in order to roost? Anybody see it? How about if I tell you one more thing? The expense of flying per gram of bat goes up the smaller the animal is. That is to say, small bats are inherently less efficient at flying than large bats. And it's the small bats that have evolved 3 times this pattern of making their own roost structures on the forest floor.
- 4.55 So that means they have to conserve more energy and spend way more time sleeping, which makes them more vulnerable to getting picked off. So they need more protection.
- 4.56 Well, you're on the right track. The efficiency thing is there, but they don't get any more time to sleep. They do the same amount of sleeping.
- 4.57 Close to their food source?
- 4.58 Close to their food source. Beautiful. This is the answer. This is the hypothesis I tested when I was doing my dissertation work, was that what's really going on here is that they are lowering their commuting costs by putting tents near fruiting trees. Imagine for a second that you're looking down on a tropical forest. Where 's the fruit? Well, a majority of those canopy trees make fruit. A lot of those species make fruit that's edible by small bats. But the point is, where the fruit is now isn't where it's going to be a month from now. You might have a tree that's in fruit for 2 weeks. That's a great food source. But then a month later, that part of the forest is bereft of anything that they can consume.
- 4.59 And if they had a permanent roost, what they would have to do is commute from wherever it was to wherever the fruit happened to be. And that's a lot of expense. Now, the surprising thing is I now know, because no one had seen a tent made, I did finally, finally manage to observe a tent in the process of being made. What I discovered was that at least the more elaborate tents are really expensive. It takes an individual several days to complete one of these things. So it's not energetically cheap, but apparently it is cheaper than flying those long distances to all of those places where the trees are in fruit.
- 4.60 Now, I should probably tell you a little bit more about these fruit bats. They do not—there are a couple of exceptions to this, but the animals that you're seeing, the tent bats, do not eat the fruit in the tree where it's fruiting. This is very different from something like monkeys. Monkeys famously sit in the tree, eat fruit, throw away the parts they don't want. It just rains down in the forest next to them. These animals

don't. And the interpretation that has been given to that is that it's too dangerous, that predators learn to recognize these fruiting trees, and they sit there and they wait for bats to show up. And so you don't want to stick around in such a tree.

- 4.61 What the animals do instead is they grab a fruit, sometimes a surprisingly large fruit, given how tiny these guys are, and they fly off to a favorite spot, and they hang, and then they eat bites of fruit, and they drop the remains below them. They roll the fruit on the roof of their mouth to squeeze the juice out, and they drop the remains. And if you know what you're looking for, you can find these little piles of chewed -up fruit pulp. And if you look up, you will see a place where a bat roosted, like I think I mentioned to you yesterday. Now, that tells you something else that links up to a story we've just heard. These animals are limiting their costs yet again. The problem is all of the value of the fruit is in the sugar. It's in the juice. It's not in the pulp. But if they eat the pulp, then as they fly, they're heavier. And so every kilometer they fly is that much more expensive. So they go way out of their way to extract the value so that they can fly lighter.
- 4.62 Now, this is sort of the inverse of the story of the hoatzin. The hoatzin have accepted the cost of eating leaves, which means eating a lot of material that takes a long time to break down all of that cellulose and get out all the good stuff, right? So they accept it, and they accept the crappy flight that comes along with it. These guys are very good fliers, and they're very good at figuring out how to make themselves even that much more efficient. And they have lots of excess capacity. Mothers will fly with offspring attached to them, and they can be up to 25% the weight of the mother. They do eventually start leaving them in the tent, even before the animal can fly itself. But at the beginning, they fly with them attached.
- 4.63 Here you can see the characteristic J-shaped cuts that they make in the bottoms of the leaves. I think I forgot to mention. It was long hypothesized that they made these things by biting, and it's not how they do it. What they do is they take the claw—remember Heather was talking about the fact that bats are the Chiroptera, hand-wing. That wing is made from the fingers of the animal that have been elongated and membranes—patagia—have been stretched between those elongated bones. The thumb sticks off the top of the wing, and the thumb retains its claw.
- 4.64 So what these animals do when they make these tents is they dig those claws in and they pull, they tear, using the strongest muscles in their body, which are their flight muscles attached to their wings. And they use those claws to rip these things until they eventually collapse in just the right way. Now, I should tell you, I've seen thousands of tents. I've seen tents that make no sense. I've seen ones that didn't work. I don't know what it's about. I don't know how these animals learn it. I don't know if they learn it by watching their parents. I don't know if they learn it by looking at tents and inferring how you have to do it. I don't know how much trial and error there is, but there has got to be an incredible story there. How these animals learn these 8 different architectures.

- 4.65 It seems, from the pattern I've observed in the forest, that animals tend to stick to a tent type. There are enough leaves that if you're a Philodendron tent-builder, you'll stick with Philodendrons. And in general, that'll be characteristic for your species. And some other species will use one of these other different plants. But this changes as you move across tropical forest. The species turnover is incredible. Tropical forests are highly diverse, but that diversity is rapidly changing. That is to say, very small shifts in condition cause one species to fail and another one to take over. So as you move across tropical space, the number of times that you're seeing very different species erupt is large.
- 4.66 But these bats have fairly large ranges. So how is it that one animal that doesn't have a built-in program, presumably, is able to figure out how to make tents in the particular species that it finds in the particular patch of forest where it lands? That is a good question, and it's reminiscent of the question about the sea otters. The idiosyncratic nature of the forest, the particular species that are available to make tents—can you figure out how to make a tent that works? The change in architecture. In some forests, These cuts are curved like this. In some, they're pronounced Js. In some, they're straight. I don't know where that information is held. It's probably culturally transmitted, but there's bound to be an interesting story there.
- 4.67 All right. And then there's a final piece to this story. So what we've got so far is tent bats are the answer to an ecological question. The question is, how, if you're a tiny animal that spends a lot for every kilometer it flies, can you survive on fruit that is very patchy and where the patches move around the forest based on what month it is? That's the question. And you can see in that question some of the reasons for the patterns I described to you. Why would you have a tent that has an old dilapidated version with a new one built right on top of it?
- 4.68 Different season?
- 4.69 Well, it's the same season. It's back, a year later. So animals that roosted there probably are returning to the same plant because, A, they know the nuances of the forest there. So when I was catching bats to do my work, I paid very close attention when I released them. And it was shocking. You would have a bat in a cloth bag to protect it from harm, and you would hike it out into the forest, always to the exact place that you caught it, and you would release it.
- 4.70 And it was amazing how many times I saw animals fly, as I took them out of the bag, they would fly directly to a tent that they clearly knew was there. So somehow they're able to resolve this image suddenly, even while terrified, enough to be able to go directly where they're going. That means they know that patch of forest, right? The same way if I dropped you in the central square of your town, it wouldn't matter where I put you, you'd know immediately where you were and where you wanted to go. So it's that. An animal that has spent 2 weeks, 3 weeks, or something like this, flying out of that tent to one or more fruiting trees in the exact neighborhood might very strongly

prefer to be right in that exact spot because it knows how to deal with that spot. It knows where the vulnerabilities are. Maybe it knows that that tent was never attacked by anything.

4.71 I showed you that picture yesterday, and I said that I thought those animals were terrified that I was looking up at them, but they weren't sure that I saw them. The fact is, all the pictures are misleading because they're all taken with a flash. Now, there are a couple of species of bats that are very sensitive to things like flash. There are also bats that are very sensitive to certain noises. There's a tape that we use in forest work called marking tape or flagging tape. And as you unroll it, it makes a sound that you don't even know it makes until you realize that it terrifies *Vampyressa pusilla*, the tiny little tent bat. They fly out if you unroll the tape, so you don't unroll the tape anymore. But the point is that sound is very triggering to them for some reason.

4.72 A flash, not for most of these animals. And I don't know why that is. I assume it's because they deal with lightning all the time. So random flashes of light don't have a strong implication for safety. So maybe they take the flash to be that. But the thing that they are sensitive to, strangely so at one level, is any touching of that plant. The slightest tap on the trunk of that plant will send the bats flying. And in fact, tent bats were discovered twice by two different individuals working in Panama on Barro Colorado Island, which also happens to be the island where I did my dissertation work. They both discovered it the same way, accidentally. They were both walking down the trail separately, and they slightly tripped over a vine, the way one often does in these forests. And that vine was connected across the ground to something. And suddenly, bats filled the air. And they thought, "What did I do? Where did the bats come from?" And they went looking, and then they put together the pattern, and they realized, "Oh, there are these tent things."

4.73 So this also tells us why you would find clusters like this, right? You've got some fruiting tree. Maybe it's a *Dipteryx panamensis*, a beautiful canopy tree that flowers purple, that bats eat the fruit, right? Maybe it's suddenly productive. Well, you get a bunch of bats that are feeding from that same tree, and maybe they get an advantage, like schooling of fish or flocking in birds, where they're less likely to be consumed if they're in a group. Just statistically, they're each safer for the fact of being together.

4.74 These are squirrel monkeys. Squirrel monkeys have been observed—they're the only animal that I'm aware of that has been observed to have a specialist method for hunting tent bats. So remember, tent bats are exquisitely sensitive to anything that touches their plant. That's the thing that reliably sends them flying. That rules out most of the things that an animal could do in order to eat them, right? If you imagine a snake trying to climb up that plant, no, those bats are going to be long gone. If you imagine some quadruped underneath trying to reach up to get it, it's going to bang that leaf before it gets to those animals. They're going to be gone.

- 4.75 Squirrel monkeys are a little bit like the otters, right? This is a highly intelligent creature with a cultural mode. The offspring always meet the parents, and the parents have all sorts of opportunity to convey information about what kind of things you can eat in this here patch of forest. And if you've got a high density of fruit bats, well, the question is, how are you going to get your paws on them, right? They're there. They're certainly edible. There's got to be a way.
- 4.76 So what the squirrel monkeys have been observed doing is climbing above the tent and then diving onto it, hitting the tent so it knocks the bats to the ground, and then another animal is waiting and picks up the bats. It doesn't always work. But the point is, that's a very clever solution to a very difficult problem created by tent bats and their amazing roosting ecology.
- 4.77 OK, maybe this is the final point. The final point is, are the tents there to hide the animals, or are the tents there to reduce commuting costs? Obviously, both. So the point is, you could look at those tents and you could say, "Well, what are they for?" And you could even test the hypothesis that they're there as a way of hiding bats, and a proper test would prove that they are, and you might quit at that point. So there must be some motivational force that created the impulse to start modifying leaves.
- 4.78 Almost undoubtedly, it began with animals roosting under leaves that weren't terribly modified. And then maybe they roosted under leaves that had broken in just such a way that they happened to hide the animals beneath. And then there might be a point at which a bat was hanging below some leaf that was broken, but just not quite broken enough, right? And so maybe it just broke it a little more, just to pull the flap down a little and hide it just a little better. And then obviously, it's a pretty gentle slope from there to get to, "Well, that leaf isn't broken at all, but maybe I could break it." Somehow this happened.
- 4.79 And what we don't know, and what I suspect we will never know, is how much of the program is being handed off genetically. Do these animals have an impulse to make these tents that they then fulfill through trial and error as they tear at leaves and they collapse not quite? Don't know. Or is it entirely cultural? And again, if it is entirely cultural, even in that case, the purpose is still to lodge the genes of the animals as deeply into the future as can be accomplished. In fact, it occurred to me yesterday when we introduced that concept that maybe we should talk about the principle, owing to Richard Dawkins, that a chicken is an egg's way of making eating another egg. What Dawkins means by that is that the purpose is to advance the genes into the future, and the chicken is a clever solution to how you do that. It's a means to an end, like all of the rest of what we've talked about. It's a means to an end.
- 4.80 And so you should think about it that way. It's so counterintuitive that the purpose of that system is to produce eggs, not to be chickens. But nonetheless, evolutionarily speaking, it's really at least equally valid to think of it as an egg production scheme than it is to think of it as something about chickenness. And all the rest of these creatures

are that same way. So that leaves us, again, in that funny place where we're looking at an animal. I mean, that animal really is miraculous. If you watch it, I challenge you to think anything less of it. It's an amazing animal, the way it is able to model the 3D space around it, to leap from one branch to the next, risking life and limb, and to do so with grace and agility, sometimes to do so with a baby on its back. This is a remarkable creature that has that same mind-numbing purpose that all creatures have, right? That's the sort of, the bitter pill aspect of the whole thing.

- 4.81 OK. Now we have arrived at the end of the evolutionary jeopardy part of this presentation. But it occurred to me, after we met yesterday—I was asked a question, and I'm going to see if we can figure out who asked me that question, if they can rephrase it. Might have been you. Do you remember the question you asked?
- 4.82 You might ask why make a distinction between the purpose of animals being to toss their genetics in the future versus just reproduction?
- 4.83 Reproduction. Right. That is the question. So let me just phrase it in a way that everybody can hear it and knows what it means. And then I realized, I think the answer I gave was right, but it deserves a better one. So the question is, is there really a difference between saying "lodging your genes deeply in the future" and "reproduction"? Aren't they so close to similar that it's a distinction without a difference? That's the question. Now, what I wanted to do was come up with a couple of examples that would show you just how different they can be. Now, this is going to be a little bit mind-bending enough that it will probably just sound wrong. But nonetheless, after a great deal of thought, I believe it is absolutely correct.
- 4.84 Each of you sitting here is composed of something like 30 trillion cells. Each of those cells could be a single-celled organism. In fact, you contain things that function exactly like single-celled organisms. Your immune system has something in it called a macrophage, which is like an amoeba, and it moves around in your interstitial spaces, and it consumes garbage. It's a garbage-collecting amoeba, but it works for you. Why does it work for you? Because it has your identical genes. And so what that means is that the way that a macrophage gets its genes into the future is to sacrifice its entire life on your behalf. Whatever garbage you need collected, it's there because it can't put its genes into the world any other way, and you have all of them. So the best thing it can do is keep you as healthy as possible, so your chance of producing offspring is as good as it can be.
- 4.85 But then look at this the other way. If you contain single -celled organisms which contain all of your genes—they have your entire genome—then you could be 30 trillion single -celled organisms. You could, if you really wanted to maximize the number of units of reproduction that you had done, you could go from being one individual, who is likely to produce 2 offspring, each of whom is 50% related to you, resulting in you just simply replacing yourself in the next generation. That's what you're probably going to do.

- 4.86 But you could get a return on investment that would take you to 30,000 full clones. Not 30,000. 30 trillion full clones in the next generation if you just simply surrender to the idea of being a bunch of single-celled organisms. So why don't you? Why does your biology choose not to even ask that question? Now, the answer to that is likely to be that the utility that you get out of surrendering all of those reproductive opportunities is higher with respect to the likelihood that your genes will be around 1000 years from now. If you decided to win right now, by making 30 trillion copies of yourself instead of producing 2 offspring who are 50% related to you, if you decided to do that, you would get a big win right now, and you would surrender almost all of the ability to control your circumstances that you currently possess.
- 4.87 So almost all of your cells have just agreed not to do their own reproductive thing and to collaborate in a very arduous, slow process of human reproduction because that's the best way to get those genes into the future. And it is so much the best way that we don't even find animals experimenting with this. We don't see an animal that then gives rise to a bunch of single-celled organisms just to see if something has changed, and maybe it's a good time for that strategy. It doesn't happen because the fact is, what you are, as much as it's counterproductive to the process of leaving a lot of offspring in the next generation, is useful in the long-term from the point of view of the goal that we describe. So that's one way to answer this question, right? This is dramatically different than reproduction when you see it correctly.
- 4.88 Here's the other way. Think about the predicament of a female. Some of you will not find that difficult because you are a female. But a female is in a position, biologically, to produce a clone of herself if she wants. In fact, it's easier in many ways to produce a clone of yourself than it is to sexually reproduce. And the reason for that is, if you've reached reproductive age, then your genome has got to be pretty well coordinated. You wouldn't have gotten there if it wasn't. So you know something about that genome. And when you sexually reproduce, what you do is you take half of that genome, chosen in a kind of random shuffle, and you fuse it to somebody else's half-genome, chosen in a random shuffle. And you build for the first time ever a unique genome that's going to exist exactly once.
- 4.89 And you don't know if that's going to work. Usually does, because the system is pretty good, but it's not guaranteed. And there's nothing that says it's going to be as well coordinated as a genome that got somebody to reproductive age. So the obvious thing to do if you're female is just not divide your genome in half and to produce another one just like you. And that doesn't condemn you to having an identical person running around. That person will have a different developmental environment. They will just have your same physiology. It's a good deal. In fact, it's a hugely good deal from the point of view of fitness.
- 4.90 If reproduction was the point, when a female reproduces through sexual reproduction, she produces an offspring that only counts at 50% because it's 50% somebody else. A

female who produces an offspring that is a clone gets twice that level of credit. So if the point, if selection's point was really to maximize your reproduction in the next generation, the obvious way to do it is for females to dispense with males and clone themselves. It's a winner. And the fact is, it would even work. It would work for generations.

4.91 There's no question that whatever argument we evolutionists make about the value of having genetic diversity, about the way that that might protect you from things like disease, it's going to take generations for the advantage of producing an offspring that's twice as evolutionarily valuable. It's going to take generations for anything to catch up with that level of advantage accruing each and every generation. So the fact that women never do this, the fact that female mammals don't do this, is evidence that every time selection has experimented with this mode, it has discovered that at least in mammals, it's a loser. It's ultimately a loser. It's not proximately a loser. It's not immediately a loser. Yes.

4.92 The reason that we have to distinguish, if we want to understand adaptive evolution between reproductive success—that is to say, how many offspring you leave—and fitness, which is the characteristics that lodge your genes as deeply into the future as possible, is that on some of the most profound questions, they deliver diametrically opposed answers. Do you want to be a bunch of cells stuck together for life, collaborating on a project that will produce tiny numbers of offspring? Or do you want to be trillions of cells, each of which fends for themselves, and some will win and some will lose? Well, it's counterintuitive, but you want to be a collaborative organism because your likelihood of having your genes around 1000 years from now is higher, not lower.

4.93 Same thing goes for, should you clone yourself? Should you dispense with all of the complexity of sexual reproduction and get twice as much evolutionary payoff for each offspring? Nope, it's a loser. It 's not a loser now, but again, it's not "get your genes into the future." It's "get your genes as far into the future as you can get them." And the way you get them far into the future is you see the things that work now and will mess you up later, will mess you up 2 generations, 10 generations, down the road. You anticipate those things and you avoid them now because the whole point of the project is to continue to be around indefinitely.

Lecture 5: Ant Farms and Poison Frogs

At a glance — Ant farms and poison frogs as case studies in adaptation.

- 5.1** All right. Welcome back, everyone. We're going to be talking today, I'm going to be talking today, about leafcutter ants and about poison frogs. This is a view from a boat that we were on in the Amazon, in the Ecuadorian Amazon. This happens to be on the Rio Shiripuno, which is a tributary of the Cononaco, which is a tributary of the Curaray, which is a tributary of the Napo, which is a tributary of the Amazon. That's the way of the Amazon Basin, is that it's very rare that you're actually on the Amazon. If you are, it doesn't look like this. It's massive.
- 5.2** And leafcutter ants and poison frogs, some of them, both show up in the Amazon, although it's not the only place that they show up. So let's start with leafcutter ants. They are in a couple of different genera, *Acromyrmex* and *Atta*. And I'm just going to start with, again, the evolutionary jeopardy framing. They cut leaves. They're called leafcutter ants. They cut leaves. What is the question to that answer? That is an evolutionary answer to some ecological question they are presented with. What might they be answering? Any ideas? This one's a bit more— there's a few more steps of abstraction than what we were talking about yesterday with hoatzin and kelp and such. Yeah, Jennifer.
- 5.3** Leaf litter.
- 5.4** So what about leaf litter?
- 5.5** Well, if they burrow underneath, I'm not sure if that might close their openings.
- 5.6** OK, so what was said was, if they're burrowing, maybe cutting the leaves is maybe covering up openings that they have? Is that the idea?
- 5.7** Or some way to organize.
- 5.8** Some way to organize a nest, perhaps. In a way, that is the answer, but it's not as complex as the answer turns out to be. Yeah. Can anyone add to that?
- 5.9** So they can carry them.
- 5.10** Oh, they cut them so they can carry them. OK, but why did they carry them? Yeah.
- 5.11** So they have food for the future.
- 5.12** So they what?
- 5.13** Have food for the future.
- 5.14** So they have food for the future. So do we think that they're eating them?
- 5.15** Or they ferment them.

- 5.16 They ferment them and then eat them. Not quite, but very close. So what they're actually doing—here, I just have a little video. This is from the Amazon. This is actually from a different —this is from Tiputini, which is a different place in the Ecuadorian Amazon. Just a video from my iPhone as they're going into their nest. And you can see in this not-very-extraordinary video that they are entirely—I'll play it once more—all the leaves are going in one direction. And it's not just leaves, they're carrying some other stuff as well. But all the leaves are going in, which in this case, in this framing, is up-screen. And there are a lot of ants coming the other way, but none of them are carrying leaves.
- 5.17 So they're carrying leaves in, and they're not carrying them out. And they're doing something with them inside, but we haven't yet quite landed on exactly what they're doing. They're not fermenting the leaves, but they are, in fact, growing fungus on the leaves. They are extraordinary farmers, and that's what we're going to be talking about today, farming in ants and the contrast between how ants farm and how humans farm.
- 5.18 So there's 210 species in this tribe, Attini. They're entirely New World. They're in the Amazon and throughout basically the New World tropics, up through Central America. Fungiculture, which is just a form of agriculture—it's just a particular form of agriculture in which you are farming fungus as opposed to farming plants, which is most of what humans do —evolved once and only once in ants, but it has many, many degrees of refinement within those ants. And so there are hundreds of hours of stories to be told about the ways, the degrees of farming, that the Attini, that the ants, that the leafcutter ants, are part of, have learned to farm. The leafcutter ants are sort of the pinnacle.
- 5.19 As some of the researchers who work on these ants have said, "Achromyrmex and Atta —again, those are just the two genera, the two genres, genera, of ants that do the most obligate farming of fungus—are extraordinary farmers," they write, "and the dominant herbivores of the New World tropics." Amazing when you think about how many other things eat plant parts, not just folivory, like we talked about with the hoatzin, and not just frugivores, like we've talked about with bats and nectarivores. All of those are herbivores, and yet it's these ants that are considered to be the dominant herbivores. They are doing that much modifying of their environment to be called dominant within the New World tropics, which, of course, is a vast, vast place.
- 5.20 OK, so why farm if you're an ant? And then we'll talk a little bit about why farm if you're a human? Why should we farm? Why should they farm? Well, their historic food, the historic food of the group of ants that they're in, and of most ants, is arthropods, both live and dead. Most ants, most New World ants, and most ants in general, are capturing live, small, prey or eating newly dead prey. In the tropics where ants live, and in the New World tropics where these live, it's pretty moist. It's very hot, and so rot happens quickly. You have a highly productive environment in terms of growth and also a high rate of rot. And that means that competition for arthropods is high. And once they die,

you got to be on them really fast, or else they're going to decay into something that even the ants don't want. And again, these ants don't want arthropods at all.

5.21 So we think, we, the people who are working on these ants and all of their predecessors, think that they began to farm probably as a way to open up a new diet. There was such intense competition for the arthropods because so many things eat arthropods, arthropods mostly being insects—there's so much competition for arthropods, and all the ants are already eating them, but so are so many other things, eating arthropods, that farming fungus released them from that competition. So they evaded one hostile force, one of Darwin's hostile forces of nature, and have, from competition with predators and scavengers, the predators of live arthropods and the scavengers of dead arthropods, that they're now free from them. But of course, they've opened up competition with herbivores because they're taking the leaves that folivores would eat.

5.22 And so you can't free yourself from competition unless you're literally the first one into a niche. But it's been a long time since anyone was first into a niche on this planet. There's been life happening for so many billions of years, not very many billions, but so many hundreds of millions of years, that empty niches are hard to find. So they moved from a highly competitive environment into maybe a slightly less so, but mostly they changed their competitors. But this also means, if you can be successful at farming fungus, about growing fungus on the right substrate —and that's what these leaves are, they're the right substrate to grow fungus on—then your risk of losing food to rot goes down.

5.23 Because just as your risk of losing food to crop predators in humans goes down if you are actively tending a garden, as opposed to if you are out foraging and you say, "Oh, I saw that thing a week ago. I know it's about to ripen. I'm going to go back and see if it's still there." Well, there's a good chance it's not still there. I mean, this even happens — we'll talk a little bit later today about the poison frogs in Madagascar that I worked on. And I was watching a large fruit ripen on a tree, eager to see if I could get up it and get it at some point. And literally the day that I was planning to go, a group of brown lemurs, a troop of brown lemurs, came in and destroyed the thing and ate it, mostly, but also left little bits on the ground, not appealing to a human who was also interested in it.

5.24 So the theft of wild fruits, of wild foods, of wild potential foods, is rampant. Whereas if you farm, you have control. So you get a greater amount of control over your diet. But of course, it requires vigilance and tending and a whole additional set of skills. And it's not that you've freed yourself from competition, you've just switched over what your competition is. OK, so how does the agriculture of ants compare to the agriculture of humans? Well, humans have been farming for about 10 to 12,000 years, ants for about 50 million. They've got a lot more practice. And it's not to say that they are better at it, but they have a lot more practice.

- 5.25 Ant farming behaviors are presumably entirely genetic. They're much more robot-like than we are, obviously. There's very little robot-like about us. And so whatever they're doing, if conditions change such that what they're doing isn't as useful as it used to be, it's going to be a bit harder for them to switch it up because they are not as labile, they're not as plastic, in terms of what they can do because it's not coming from a cultural level, presumably.
- 5.26 Similarities between human and ant farming, though, include that ants seem to have moved domesticates into ecosystems that they could not otherwise have gotten to or thrived in. So this means that ants have taken their fungal cultivars, the fungus that they actually eat, not the leaves that they're cutting, which is the famous invisible part of what they're doing. The leaves that they're cutting, they do that with whatever leaves they find on which the fungus will grow. But the fungal cultivars that they're actually eating, when in tight symbiosis with these ants, live in places that they never could have gotten to otherwise.
- 5.27 So this raises a question about who's actually driving. This is a tight coevolution between these these fungi that are in obligate relationship with these ants. And because the ants are eating the fungi, we think, well, obviously the ants have parasitized the fungi. They brought them in, this is against the will of the fungi. And of course, that's speaking in this evolutionary shorthand that doesn't actually describe consciousness. But maybe the fungus has not just as much interest in, but maybe was driving the original interest in being cultivated by the ants as a way to get into ecosystems that they couldn't otherwise get to. It's a possibility.
- 5.28 More comparisons between ant agriculture and human agriculture. Humans retain all of our options, even when we farm. We can still hunt and we can still gather. Most of us don't. When we do, it's very lightly. We may pick berries during the summer or some such, or grab a fruit from a tree. But most of us don't rely on hunting and gathering, as all of our ancestors did for a long time, even though the vast majority of us, probably everyone watching this, is effectively an agriculturalist, effectively a post-agriculturalist, getting most of our food from agriculture.
- 5.29 But we retain the flexibility. It's still facultative. We aren't reliant on agriculture to earn our living, whereas ants are. They cannot make a living. At least the leafcutter ants that are the most derived of all of these 200-plus species of ants, they cannot make a living if you take away their fungal cultivars. They die. So that's a big difference between human and ant agriculture.
- 5.30 Both of us, both humans and ants, who are agriculturalists or post-agriculturalists, gain reliability, as we talked about, and predictability when we begin to farm. But ants always farm just a single species, and it's very rare for humans to farm just a single species. Modern mega-agriculture, monocropping operations notwithstanding, those operations are about business interests, not about what a human can survive on. It is very rare for a human to eat one and only one species and be healthy. And certainly

farming, be it animal farming or plant farming or fungus farming, just one and only one species successfully and eating only that is, as far as I know, never been accomplished in a human. Even if you are on a single ingredient diet, there's a very good chance that you're paying other people to do all the raising and the butchering and the slaughtering and the butchering of those animals.

- 5.31** By comparison, ants are in tight symbiosis with a single species. And we already talked about this—competitors change when you start farming. That's true for humans as well. The biggest risks to us in terms of who might be taking our food, before humans were farming, might be other frugivores or other predators. Whereas what happens once we start engaging in pastoralism or agriculture, the risks may become—still, there are risks from predators. Fox in a henhouse is a real problem—but the bigger risk is, just as with the reason for the long-distance migrating birds that I talked about yesterday, when you've got a densely packed resource, then the parasites can find you. And so parasites on crops and parasites on your food in general is much more of a risk when you're farming than when you're hunting and gathering because you've concentrated all your food in one place.
- 5.32** Agriculture thus invites pathogens. There's just no way around it. If you're going to concentrate your food in one place, the pathogens are going to find you. And unless you've got a roving situation, which to some degree, swidden agriculture, colloquially known as slash-and-burn agriculture, accomplishes. Slash-and-burn is actually very effective at low population levels, at low population densities, and usually is described as useful because of the very low fertility in tropical soils, that you raise 2 or 3 crops on a patch of soil that has recently been slashed and had the nutrients from the forest burned into it. And after 2, 3, 4 years, depending on where you are exactly, the soil nutrients are gone, and so you need to go to an adjacent plot. And after 20, 30 years, depending on where you are, sometimes 10, sometimes as few as 10, you can come back to the original plot, and enough has grown up that you can slash and burn onto it again.
- 5.33** But it's also true that by moving plots around, you will leave those plots fallow long enough for whatever soil pathogens were growing in them to also disappear. There are advantages, and of course, in the modern world, we mostly don't have that option. Slash-and-burn is no longer a viable option, really for anyone, because we all live at too high population density to pull it off.
- 5.34** I think I've already mostly gone through this, but the high density of crops facilitates movement of pathogens between hosts, and that increases virulence of those pathogens. And the genetic diversity of crops is lower than in the wild. So we tend to select for, and the ants will tend to select for as well. Anytime you're farming, you will tend to select for what you like best. And in a healthy system, what you like best is what is best for you. That's not necessarily the case in our hyper-novel world, of course. But in the beginning of farming, certainly, and we can hope that we can still find this

truth in our farming, what you are most seeking is your body telling you, "That is what I need to be healthy."

5.35 So that tends to be somewhat narrow within any given species, which means that we tend to select for a reduction in diversity and genetic variability of every given crop. And once you have a reduction in genetic variation in a crop, then the pathogens have an even easier time finding you. So you've got two big issues. You've got the crops all in one place, which makes you easy to find. I guess 3 issues. High density, which means it's a great opportunity, and lower diversity of the crops, which means if some pathogen can take advantage of you, it's really likely to be able to take advantage of the entire crop. So these are 3 reasons that farming tends to, and maybe always does, produce pathogens.

5.36 So diversity in farming, as in so many ways, is a strength. It's actually, it produces an antifragility to collapse. I've taken out all the names here, and really what I want you to look at here is the structure, the topology, of these 3 trees. We've got ants on the left tree, with their fungal cultivars; the food that they're eating, the fungus that they're eating, in the middle tree; and then they've got parasites on the right. And it is tightly coveolved. It is not perfect. These trees aren't exactly identical between them. But I don't know of another example that we know of that's so well cataloged in the natural world, where you have an organism, something that it is cultivating, and then the parasite that has moved in on the cultivar, and see tight coveolution between them. So for each species of ant, there's a cultivar that it uses, and there's a parasite that comes in and attacks that fungal cultivar. Amazingly, amazingly tight coevolution here.

5.37 But it's even more mutualistic than that. So they've evolved mutualistically with their crops. Then these pathogens come in and coveolve with them. The ants don't want them there. Presumably, the fungus doesn't want them there, but the parasites are there. They want themselves there, and they're there. So what happens? What are the ants going to do? If you've got this tight evolution, and ants, unlike swidden agriculturalists, don't move their nests around all the time. They don't move them around frequently. They do it sometimes, but it's not on a tight schedule, as far as we know, like it is with some early forms of agriculture.

5.38 So what are they going to do? How are they going to get rid of their pathogens? And we'll talk in a minute about how human agriculturalists tend to respond to pathogens. But how do the ants do it? Any thoughts? How might the ants deal with this? How might the ants deal with these tightly evolved, these bacteria? No, micro-fungal parasites that are evolved to the fungus that the ants are absolutely, obligately eating and farming? Say again?

5.39 Biological? Maybe another critter?

5.40 The suggestion was it's going to be something biological, maybe another critter. Yes, that's exactly it. So it could have been a physical response maybe, erecting some kind of

barriers, but what are you going to do with a tiny parasite? Maybe a chemical response. There are a lot of chemosensory cues that ants use to communicate with one another. So maybe they could be erecting some kind of chemical barrier, like some human agriculturalists do as well. Plant something around the border of the garden that the insects that would eat the garden don't want to eat.

- 5.41 But no, they're not doing that. They're actually using another organism that acts as an antibiotic. They are carrying around with them bacteria from the group Actinomycetes that suppress the growth of their parasite, which is Escovopsis. They've been doing this for millions of years, at least. That's as far as we can figure. So they have collected, and the Actinomycetes bacteria, I don't know if either they, we don't understand their phylogeny yet, or if it's not as tightly evolved, or if it's one bacteria across all of these pathogens, these pathogens here on the right. But I can't draw—there is no fourth tree here for the mutualistic bacteria that are acting as antibiotics.
- 5.42 But all of the—as far as we can tell, as far as what's been looked at—all of the leafcutter ants that are in tight and obligate coevolutionary relationship with the fungus and that have this tight, and they don't want it, but it's their relationship with these parasites, are also carrying around a fourth species of bacteria that they use as antibiotic against the parasite. Extraordinary.
- 5.43 How do we deal with it? We've already talked about this a little bit. How do we deal with crop pathogens? We can try to exclude them through physical or chemical means. We can eradicate them, often by killing, again sometimes by chemical means. We can protect our crops, again sometimes by physical means. A lot of, for instance, modern bananas, which are their own crazy story because they can't reproduce on their own, but are protected with these blue bags that you see. Any non-organic banana plantation that you walk into is going to have these blue bags over all of the bananas that is protecting them from crop pathogens. And we can also use immunization or resistance. We can apply something to make our crops immune to pests.
- 5.44 And ants do all of these as well. They do a little bit of, obviously, when they see a problem, when they have a bunch of fungus that is clearly being parasitized, they'll remove it. But the main thing that they seem to be doing is this fourth category that humans are doing, immunization and resistance. And in some ways, and I think in the most important way, they're better at it than we are. Because where we might come in with an antibiotic, say for our cows, that antibiotic is a static molecule, and it's going to respond to, well, hopefully, to whatever pathogen it was developed for. But it can't evolve in real time because it's a static molecule that you're giving as a medicine.
- 5.45 Whereas these ants are carrying around wild-type bacteria from, again, the clade Actinomycetes. And those Actinomycetes are being spread by these ants, are thriving because of their use by the ants, and so are presumably not just able to, but have an evolutionary advantage if they do, respond to any changes in the pathogen and continue to do a better and better job of responding to that pathogen. So they've got a

live antibiotic, a live, evolving antibiotic, whereas for the most part, what humans have is something that's dead, that we're taking that's static. And of course, evolution isn't static. That's the big point, right?

5.46 All right, so what is the question to that answer? Why are they cutting leaves? Well, there's a lot of questions, obviously. But one of them, which I've alluded to already, is that it may have to do with the fungus more than the ants, that this tight symbiosis may have begun as a way for the fungus to exert selective pressure on the ants, to use the ants to get the fungus to explore new niches. That the ants have not—as far as we can tell, but this is a tough thing to infer from history—been able to go new places since they've been farming, but the fungus has gone new places since the ants have been farming them. So that suggests maybe that the fungus is actually driving this interaction rather than the ants driving it, which I raise that in part to remind us that, when we see two organisms, two species, or even two individuals in relationship, we can very easily jump to a conclusion about whose idea it was, who's at advantage, who's getting the most from it, and it might be the exact opposite of what we think.

5.47 OK, one more thing, which again itself could be a whole 10, 20 hours of talking here. Why are they so collaborative? Why are these ants able to farm so collaboratively? Some of them are going out for leaves, some of them are back tending the garden, some of them are tending the gardens by getting rid of those that have become parasitized, some of them are tending the babies, and yet they never seem to have a disagreement about whose job is which, about, "Hey, there's some ants that are going to strike out on their own and build their own colony." Why not? Why are they so collaborative?

5.48 There's a hive mind that goes through the queen.

5.49 There's a hive mind that goes through the queen. That is true, but not yet explanatory. Why is there a hive mind for these guys and not for crickets? There are lots of insects that live socially. And there are a few others that aren't included in the clade that ants are in, which is Hymenoptera, which is ants, bees, and wasps, that also live eusocially. Termites, for instance, and there's a different explanation for them, but other than that, basically, insects, even who live socially, will always have some differences of opinion without it being in any way conscious about what their best interests are at any given point in mind.

5.50 So the answer, in short, is that the Hymenoptera—again, the ants, bees, and wasps—are haplodiploid. And in that one word lives a tremendous amount of complexity. And we're just going to spend a couple of minutes on it here today. So we, our soma, our bodies, all of our cells in our bodies, are diploid. We have two copies of each of our chromosomes in each of our somatic cells. We have one type of cell that is haploid, that is only one copy of each of our chromosomes in it. Those are our gametes, our sex cells. That's eggs if we're females, sperm if we're males. If we're plants, that's eggs if we're females, pollen if we're males. Otherwise, exactly the same.

- 5.51** These guys, the females are like that. The females are diploid. Their soma, their somatic cells, are diploid, and their gametes are haploid. But the males are haploid. All the males in the ants, bees, and wasps are entirely haploid. They don't have a diploid cell in their entire body. Now how does that work? That means that any gamete that they have, any sperm that they produce, is 100% related to them, but they can't make males. Males can't have sons because sons are also haploid. Sons are the parthenogenetic, the asexual product of queens, of females.
- 5.52** Where this gets even more complicated is that daughters, males can have daughters and females can have daughters, but the daughters, full siblings, are 50% related to their queen because they're diploid and their mothers are diploid, just like all of us are 50% related to our mothers and 50% related to our fathers. But because they got their entire genetic complement from their father, they got half their genetic complement from their father, but their father gave them his entire genetic complement. Daughters are not only 100% related to their dads, but they are, for reasons that I won't go into here, but you may be able to intuit if you think this way, 75% related to their full sister siblings. Whereas daughters are only 50% related to their moms.
- 5.53** That means they have, the full sisters have higher degrees of relatedness between each other than they do to either their mothers or any possible daughters they might have. So striking out on your own, if you are a female aunt bee or wasp, in order to make more daughters, isn't actually as good a genetic move than sticking with the hive and having your mom create many, many more sisters who are more related to you than your daughters would be. I'll just leave that there.
- 5.54** OK, segue to frogs. These are not poison frogs. These are *Physalaemus pustulosus*. Bret showed you a picture of a Trachops bat attacking these guys. The frog part of the story is also very interesting. This is actually on BCI, Kodak Pond on BCI, a male on the left. You can see his expanded throat sac calling with a female whom he is courting on the right. They are marvelous frogs, but they're not poison frogs, so we're going to skip forward and not talk about them today.
- 5.55** Poison frogs. They're poisonous. They're colorful, they're diurnal. They make terrific parents, all of which are rare in frogs. Most frogs aren't poisonous, most frogs aren't colorful, most frogs aren't diurnal. Most frogs are nocturnal, right? Most frogs spawn in these mass aggregations with external fertilization. They barely even touch their mate, and then they swim away or they walk away, they hop away, and they never even see their kids. That's kind of the opposite of making great parents. These frogs make great parents as well. We'll talk a little bit about what parenting they're doing, but what problems are they solving by doing these things? So I'm going to pose that now. We're going to come back to that, and I'll ask you once we learn a little bit more about what frogs are doing in general.
- 5.56** But I'm just starting off by saying all of these things. Most frogs aren't poisonous. Most frogs aren't colorful. Most frogs are awake and active at night, or at least in the

crepuscular hours, which is at dawn and dusk, as things are shifting from day to night or night to day. And most frogs put out massive numbers of gametes, and on average, over a lifetime, two of those will become successful frogs themselves, but they never know. They never meet them. They never know. So the questions will be, why are they toxic? Why are they colorful and active during the day? And why do they care so actively for their young? We'll come back to that.

- 5.57** First, what is a typical frog like? Again, yes, not poisonous, not colorful, not diurnal, don't take care of their kids. In the temperate zone, especially, as we are in, as most people who will be watching this are in, frogs tend to either be static during the winter or they even sometimes will bury themselves underground and freeze solid. I kid you not. You can dig them up and drop them and they'll shatter like glass, and then they'll emerge during the spring. That's not a typical thing for a frog to do, but being inactive during the winter, tucking yourself away, not doing much, is pretty typical. Emerging during the spring into these massive breeding assemblages called explosive breeding assemblages, because it can sound like that. If you've ever been near a pond where there's some spring peepers, for instance, or some other, usually temperate species of spring frog that has just come out for the season and is now thinking about sex for the first time in months and for the last, for only a couple of weeks, you know how loud that can be.
- 5.58** So those are explosive breeding assemblages. The males start yelling, the females start coming, and they all have basically a mass reproductive event for a few weeks, and it's over. And no one ever sees the kids, and very few of them survive, and that's what most frogs do. And they also, whatever body of water the eggs were laid in, that's where the tadpoles will develop. That's where the metamorphosis will happen. And then after metamorphosis, and we'll talk a bit more about metamorphosis later when we're talking about salamanders, after the metamorphosis happens, then the frogs are ready to come on to land.
- 5.59** But because they are amphibians, early tetrapods, but not being amnios yet, and they don't have that 4-part egg that allows them to be fully free of water, they will always have to come back to water in some form to reproduce. And also because one of the main respiratory organs of frogs, of all amphibians, is their skin. They also need to keep their skin wet in order to breathe. They have lungs. Almost all amphibians have lungs. A few salamanders have lost them. But they have lungs. They can breathe through their lungs, and they do. But their main respiratory organ is through their skin. So they're never completely free of water. They always have to be fairly wet. Even the toads, which are a type of frog, which have a lot of adaptations to deal with a dry environment, but they still need water more than they would if they developed this special egg as ambience.
- 5.60** OK, so this is the Amazon. I actually don't remember if this is the Rio Tiputini or Shiripuno, but this is a deluge in the Amazon, and frogs love this. But there's a lot of

ways that frogs have evolved to avoid needing to be in an actual pool or river of water for that very fragile stage when they are eggs and they can't do anything to defend themselves, except they kind of can—again, we'll come back to that later —and tadpoles, when they're also not very good at defending themselves. So some of the ways that frogs have dealt with the fact that, OK, we're dependent on water, but there's a lot of predators. There's a lot of those hostile forces, heterospecific hostile forces, in waterways because there's fish there and there's just all sorts of things that would eat delicious frog eggs and tadpoles that are undefended. Even if you're poisonous, you're undefended when you're eggs and tadpoles.

5.61 So here is a footprint on a trail in the Amazon. It has rained. The footprint has filled with water. A frog has come in and laid her eggs, and I don't know what species this is. I don't know if this was—this is probably not a frog with any pair bonding at all, but presumably, as generally happens, you don't have a female laying eggs and then hoping that some male will come by. There's been some attraction to the place. And again, there's external fertilization. These are fertilized eggs that are developing, and these eggs, all of them will die if it doesn't rain for the next week, if that footprint doesn't fill up again. They will just dry out and be gone. Or if someone steps in that again because it's a footprint on a trail. But it's amazing how many, even just footprints on trails, hold a tremendous amount of reproductive capacity in a place as diverse and productive as the Amazon.

5.62 So here's another way to be a frog. It doesn't look like it, though. This is the Amazon. This is that sea of green that I was talking about right at the top of the hour. That doesn't look like questions. That just looks like green. One thing you can see here is the cauliflory. Again, this is a word that Bret used yesterday. The cauliflory, which is fruits growing first flowers, which is why it's called cauliflory, and then fruits growing out of the same stem on actually the trunk of the tree as opposed to the terminal branches. But where's the frog in this scene? It can be very, very hard to find, presumably because frogs are pretty delicious, and they make great snacks. And again, we'll come back to that.

5.63 So here's another view, just a little bit ways down the trail. Same, this is Tiputini in the eastern Ecuador, in the western Amazon, by many measures, the most biodiverse place on Earth. Just extraordinary spot. Five species of wild cats, 10 species of monkeys, and I have no idea how many species of frogs, but it's a lot. There's a frog of some developmental stage in this picture, too. OK, look at the red flower in the lower left, and here's a closeup on that red flower. The frog isn't there, and in fact, it's not a frog. It's going to be a tadpole.

5.64 And check out the bromeliad in the lower part of that picture. Here we are in the middle of that bromeliad, and this is tough to see, but you could see the tail here of a tadpole. If you were looking closely at this picture without so much light, you can see the face of the tadpole as well. So how did a tadpole end up in there? That seems like

an unusual place for a tadpole to end up. And why aren't there a lot of them? That was an awful lot of eggs we saw in that footprint. Why aren't there a lot of tadpoles in this well? What actually is happening here? This is another way that frogs have evolved to avoid the problems of Darwin's hostile forces from heterospecifics, of being eaten by most likely fish.

5.65 OK, so here, from very close by, maybe the same species of tadpoles on this guy's back as what we just saw, but I don't know for sure—this is a dendrobatid. This is a dart-poison frog. And what they do broadly, there are lots of different variations on this theme, but what they do is they lay their eggs, they—the males call, and they attract females, and they have courtships, and the males have territories that they defend against other males, and the females range more widely. And females will come to calling males, will make a choice about whether or not he's the right one, and then they'll lay their eggs—again, this varies between species—but very often in a leaf litter, underneath the leaf litter. And the male will guard that against pathogens, against predators, until the eggs hatch out into being tadpoles.

5.66 But now you got tadpoles in the leaf litter. What are they going to do there? The leaf litter, it's a jungle, it's pretty wet. The dad actually sometimes will pee on the eggs to keep them wet. There are lots of ways to keep eggs wet in the leaf litter, but tadpoles generally need to swim. They need to move in order to get oxygenated enough to survive. So in some species, like this one, the dad backs up to the tadpoles, newly hatched. They wriggle onto his back, and then he carries them to a new home. That new home may be a bromeliad. Indeed, the new home may be multiple bromeliads. In many species, it's one bromeliad per tadpole.

5.67 This guy seems to have a couple of tadpoles that haven't made it already on his back. There are a couple that don't look to be alive anymore, but he's still got several that are alive. Who knows if any of them survive? But that's a lot of investment. After that happens, after the tadpoles are deposited into a well or wells, a bromeliad or a tree hole, like the buttress of a tree that is filled with water, then mom comes back to the picture, because so far we've heard from nothing about mom in terms of parental care. Mom comes back to the picture, and again, in many species, will feed her tadpoles unfertilized eggs. That's what the tadpoles eat until they metamorphose.

5.68 So you've got bi-parental care, you've got work between the mom and the dad on behalf of the young. Even though the mom and the dad don't care for the young at the same developmental period, they may never actually interact with one another again after that first reproductive event. But both of them are absolutely required in order to have successful frogs develop. So that's one way to not be a standard frog. And they're one type of the two types of poison frogs. So this is another Dendrobatid, another poison frog. That's a *Oophaga pumilio*, one of the orange morphs from the Bocas del Toro Archipelago in Panama.

- 5.69 And then we also have Mantella, which are entirely in Madagascar. This is actually an unnamed species. This is a picture that Bret took when we were in Madagascar the very first time. We thought, "This species doesn't even know. This is amazing." And it's true. It wasn't known at that point. There's one guy who had claimed the right to name it, but it hadn't been done. So this is just Mantella species. It doesn't have a specific epithet. And so we have the neotropical dart-poison frogs, about which I've already told you some. They very often have this very elaborate parental care, and they are avoiding, they're avoiding big pools or streams of water, which helps them avoid predators. And we have the Madagascan poison frogs.
- 5.70 So is that going to be a monophyletic group? The poison frogs of the New World and the poison frogs of Madagascar. Before I tell you what the evidence for and against is, I will say that in Bocas del Toro, in Panama, there's an island, which is where this one is from, Bastimentos, which is long and thin. And the first time that we were there would have been in 1998. And you could walk a cline, which is basically an ecological gradient, a cline of morphs of these guys. You could walk, and at one end, you'd see orange frogs with white toe tips, and then you see orange frogs with black spots, and then you see yellow frogs with black spots. And then you see yellow frogs with no spots. And there were also, interspersed in there, some blue frogs and some red frogs.
- 5.71 But there was a really nice gradient of these morphs that you could walk. And 10 years later, when I was back there the next time teaching a study-abroad program, it was all jumbled. The cline was gone. Most of the morphs were still there, but it was all jumbled, probably because these guys are so sexy that the pet trade loves them. And people were coming in and scooping them up, and then if they didn't need all of them, they would just drop them where they saw them. And so this is one of the problems with us knowing so much about these organisms that we can spread the news into the world that people say, "I want one of those." And if you're not careful about how you're collecting from the wild—it's an open question as to whether we should be collecting from the wild at all—but you can destroy what was a very interesting ecological experiment, a natural experiment.
- 5.72 OK, so we've got dendrobatids in the New World, including up through Nicaragua, I think, down to the Amazon, and across most of the eastern half and northern part of Madagascar, which is the wet half of Madagascar. Madagascar being the fourth largest island in the world, which is off the east coast of Africa. It's larger than California. It's gigantic, but it's also very, very poor. And so unless you've been there or you're a biologist, you have likely not heard much about the imports or exports or other activity around Madagascar. It happens to have a rate of endemism that is sky-high. Something more than 85% of all of its organisms are there and only there. So it's amazing if you're interested in seeing organisms that aren't anywhere else. All the lemurs of the world are in Madagascar. That's it.

- 5.73 OK, so characters that unite the poison frogs. Why might we think that the dendrobatids of the New World and the mantellians of Madagascar are sister to one another? Well, neither of them have teeth, which is actually rare in frogs. So this is again just like characters that you might use to build a hypothesis of relationship, an evolutionary tree—they don't have teeth. They have the identical skin toxins. They have these lipophilic alkaloids that is otherwise known only from one species of Brazilian toad and from the one species of poisonous bird, the pitohui, which is in New Guinea. So these are apparently hard to create, hard to derive.
- 5.74 It appears evidence, research that has been done since the lipophilic alkaloids were discovered, suggests that the diet of these frogs, which I think is the next thing here, they're myrmecophages, which means they eat ants and mites, but they almost entirely eat ants and mites. And there's something in the ants that these guys—this is again a Bastimentos morph of that same species of frog in Bocas del Toro, a New World dendrobatid—the ants that these guys eat and the ants that the Madagascar frogs eat are similar enough that they have the same building blocks with which to make exactly the same lipophilic alkaloids. So you take these guys into captivity, as many people have, and you start feeding them frog chow or whatever it is that you would feed them. You feed them crickets, actually. You buy them from the cricket farm. But you feed them crickets, which they'll eat, little tiny crickets. They don't lose their color. Their color's there at this point. It's not going to fade, but they do lose their toxicity.
- 5.75 So they're maintaining their toxicity, their poisonousness, with their diet, which they are building from the ants and mites that they eat when they're in the wild. So both, again, dendrobatids and mantellians are doing that. They both have aposematic coloration and diurnality. That is to say, again, they're poisonous and they are colorful. And that combination together, if predators can see you, is what's known as aposematic coloration. That's warning coloration. That is a sign to would-be predators, "You don't want to eat me. Yeah, I don't want you to eat me, but you don't want to eat me either. That's not going to work out well for you," and it doesn't.
- 5.76 So the diurnality, in answer to one of the earlier questions, is in lockstep, probably, almost certainly, with the coloration, because it doesn't do you any good to be brightly colored. Being brightly colored is expensive. It costs metabolic energy to make color and to maintain color. But if you're only awake at night, if you're nocturnal and brightly colored, what's the point? It doesn't make any sense. So if you're brightly colored, you're probably going to be diurnal. That's the conditions under which aposematic coloration works.
- 5.77 OK, now here's a surprising thing. Even their geography, remember New World tropics and Madagascar may suggest a shared history. How's that? It's weird. Here's a reconstruction of Gondwana from about 120 million years ago, which suggests the possible relationship, despite the fact that they have these disjunct distributions now. So the study of ancient landmasses, the animals and plants that lived on ancient

landmasses and how they moved over time, is the study of biogeography. When organisms move passively, with the landmasses they're on, this is vicariance biogeography. And so we have—it's a reconstruction. All of the reconstructions of Gondwana that are being done by the people who do this work are converging on this kind of answer. There are, of course, details that vary, but all the research is converging on this answer.

- 5.78** So Madagascar here, stuck between Africa, India, Antarctica, which was, of course, a very different place at that point, and South America here. There are Gondwanan distributions for other organisms. There are a number of organisms that are only in Gondwana, which is to say South America, Africa, Madagascar, India, Australia. Presumably, if we could figure it out, if we could drill down and get ice cores or something, we'd find fossils in Antarctica.
- 5.79** Even though in modern times, it seems like how could you possibly get there? Frogs don't cross oceans. They have wet skin, they breathe through their skin. You put them in saltwater, they don't do well. Modern frogs are not crossing oceans. But maybe these guys are relatives because their ancestors were in Gondwana. And even though there's no relatives now in Africa, maybe that distribution does suggest a relationship. Well, probably not, as it turns out. I've just given you a number of characters to suggest a possible relationship. Here's a number of reasons, or there are a number of other reasons which are too dry to go into.
- 5.80** But the osteological, the bone evidence, the soft tissue evidence, the muscles and tendons and cardiovascular and all of this, and the molecular evidence, mostly, clearly suggests that these two clades, the poison frogs in the New World and the poison frogs in Madagascar, are not each other's closest relatives. The rest of this doesn't matter, except to point out their most recent common ancestor was way back here, basically at the base of the tree of frogs, except for the *Ascaphus*, which is just a one little weird-tailed frog that persists in the Pacific Northwest to this day. But you have to go back a long time in history before you find a most recent common ancestor between these two guys.
- 5.81** So what we have is an amazing case of convergence. First, why are they toxic? Why are they colorful and active during the day? Why do they care so actively for their young? Maybe I'll just answer these questions. Frogs are delicious snacks, as I've said, unless you're toxic, which is a useful way to avoid predators. So be toxic. Once toxic, how do you warn them? Well, one way is to become colorful. And then once you're colorful, you need to be out in the day. So there we have our colorful and diurnal traits.
- 5.82** So toxicity almost certainly comes before—toxicity comes before colorfulness and diurnality. This is one of the things that is interesting to try to do in thinking about evolution. You've got a bunch of traits. Can you order those traits in a way that makes functional sense? And can you then test that ordering by, for instance, building a tree of a bunch of taxa or characters and seeing if you can figure out which came first?

Because just saying, "OK, they've got all these traits," isn't nearly as informative as saying, "OK, I think toxicity had to have come first."

- 5.83** Then the color and the diurnality may be kind of in lockstep. And then you have the problem, if you're well protected as an adult, your eggs aren't. There are no toxic frog eggs. They haven't figured out a way to do it, apparently. Or it's too dangerous for them. They toxify themselves. Regardless, it doesn't happen. And the tadpoles also aren't toxic yet. Basically, before metamorphosis, you don't have toxicity in frogs and toads. So they're still at risk for predation, but they're out there during the day, and they're being really non-cryptic. They're out there being bold in a way that frogs usually aren't bold, which means the predators of their kids could totally find them just by watching the adults. Like, "Where are you going? Let's follow you home. Let's eat your kids." Unlike most frogs, which make a living of being cryptic.
- 5.84** So maybe you move them from the ponds or streams where they're easy to find and become a little bit more cagey about where they are and make them be in small spaces and put them in phytotelmata, which, again, are these little pools of water found on plants. So bromeliads or bamboo wells, stems of plants that have fallen over and have left something that water can collect in, or the buttresses of trees. Those are all phytotelmata.
- 5.85** Now you've got new problems, though. How do you keep the babies watered? If you put your kid in a phytotelma, and it doesn't rain for a week and the water disappears, then your kid's going to die. So how do you keep them watered? How do you keep them fed? There's not a lot to eat in a little tiny pool. Whereas if you deposit your kids into a pool, there's probably a lot to eat. There's a lot to eat them, but there's also a lot to eat. So you solve one problem, just like with the ants. You solve one problem, you create new problems for yourself. And also, even though there are fewer predators in a bromeliad, for instance, than in a pool, there are going to be predators who find you, and you want to be able to protect yourself from them.
- 5.86** So I am now going to very quickly walk through the story of *Mantella laevis*, which is the frog that I worked on in northeastern Madagascar, and some of the things that I learned about what it's doing, because it turns out to be the poison frog in Madagascar that is the most highly convergent with the poison frogs of the New World. And it's remarkable the degree of convergence that has happened here.
- 5.87** So you have hostile forces that come in the abiotic form. These frogs, this is on a little tiny island, 510-hectare island, called Nosy Mangabe, off the east coast of Madagascar, in the Bay of Antongil. There is bamboo. There are bamboo stands along the coast of this island. And the bamboo is most of where the frogs are both being territorial, as it turns out, and also where they're making their homes for their babies. When bamboo falls over, between the nodes of the bamboo, if this bamboo broke there, you'd have several inches of water inside it, which is as soon as it falls over, it's dry. Then once it

rains, it's clean, pure water. No predators, no competitors, no food, but there's water. That is what these frogs turn out to be using.

- 5.88** There's a problem, though. What if it doesn't rain? You got a problem. What if the bamboo is exerting some chemistry into the water and turning the water highly acidic or basic? Well, I was shocked to find that actually in these cases, the frogs are actively choosing highly acidic wells, like with a pH of 4, which is crazy, actually. And they seem to be doing better in those wells than other things that are using bamboo wells can do. So whereas a more neutral pH around 7 might be easier to deal with for most organisms, these frogs have basically selected to be better at dealing with a highly acidic environment. So that's one way of avoiding their would-be predators and competitors.
- 5.89** There are a few, though. This is a boa. This is *Sanzinia madagascariensis*. Beautiful snake. Boas, incidentally, are one of these clades with a Gondwanan distribution. They're also in South America and Africa and Madagascar. I once saw a boa come out during a courtship and grab a female mantella in this boa's mouth and just hold it there for 20 minutes. All other activity in this bamboo stand—there was all this action. There was a courtship over here. There was a fight between these frogs over here. These frogs are incredibly dynamic, and they're tiny. These are small frogs. And as soon as this boa reached out and grabbed this frog in its mouth—I don't know the sex of the snake—everything stopped. The frogs kind of slunk away and didn't say anything. And the boa slowly chews on the frog for about 20 minutes, beginning to froth about the mouth, releases the frog.
- 5.90** The frog goes on to become a successful mother. I haven't yet told you about how these frogs are mothers, but the frog goes on to become a successful mother, and the boa, who'd been living under this bamboo stand for the entire time I'd been there, which at that point was a few months, left the next day. I never saw the boa again. So kind of a predator, but not really. So terrestrial predators on the adults don't work really well. Similarly, zonosaurus. These are lizards that are, again, endemic to Madagascar, as almost everything in Madagascar is—endemic again meaning just there and only there. And this is actually zonosaur sex. This is actually a picture of two lizards having sex, during which time they're not mostly focused on eating frogs.
- 5.91** But I did see, again once, a zonosaur take a frog, take a mantella, and run with it. My field assistant that year and I gave chase and tried to figure out what happened, and we don't know. So it's possible. We kept our eyes on that lizard and that frog for a good 45 minutes, running through the forest, but don't know in the end what happened. It's possible that zonosaurs have evolved a way to deal with the toxins, but there've been no other observations of that, so probably not. Probably that ended similarly.
- 5.92** But then you also have, in the wells themselves, I found these. This is again a lesson in how do you look? How do you learn to look at the scene that you've landed yourself in and try to figure out what it means? This is in the well. I started to see these my first

field season and went, "What is that? Is it a worm?" I have no idea. I had a little tiny field microscope with me. I looked at it, it didn't help. I don't know. I don't know aquatic invertebrates very well. So I took some home. I collected them and preserved them and took some home and showed them to some people who would know.

5.93 It turns out they're tipulids, they're crane fly larva. And crane fly larva are known in other groups, not in this group before, to preferentially eat the eggs of frogs. So these guys are living in the same little microhabitats that the frogs are preferentially depositing their eggs into, which is a real active force of predation. So you would expect the frogs to evolve a way to both detect the crane fly larva, the tipulids, and also to avoid them. And indeed, you do. So what I found was that wells with the tipulids in them, with the crane fly larva in them, were much less likely to have eggs in them. And if crane flies showed up after the eggs were already there, the the eggs disappeared. So the eggs got eaten whenever the crane fly showed up, almost always.

5.94 You also have hostile forces in the form of heterospecific competitors. If you can make sense of this photograph, it's gorgeous. This is one of Bret's photographs. This is the inside of a bamboo well. So this bamboo well is not that big, inside diameter. And this is the top of a frog. This is not a mantella. This is not a poison frog. This is a *Plethodontohyla notosticta*, is the scientific name. I don't think it even has a common name because there's not much that's been done in Madagascar. This is a dad. This is a male *Plethodontohyla*, inside a well that used to have mantella active in it. He's got one of his babies on his head. He's got another baby on the side of the well over there, and there're actually between 40 and 60 in this well with him.

5.95 So this is a big-bodied frog. This is a much larger frog than mantella, and they use exactly the same well as the mantella uses. And when they come in, a male and a female come in together, the female deposits eggs which get fertilized by the male, then the mom leaves and she never sees her kids again. It's the dad who does all the parental care. What he does is he hangs out while the eggs develop, and then as they hatch out into tadpoles, and as they metamorphose into little froglets, they're about ready to be done. He never leaves. He hangs out there the entire time and protects them by barking incredibly loudly if you look in a well with one of them, and it's shocking. And he doesn't eat the entire time. And once they're leaving, they're about ready to disperse at this point, then he goes. And he doesn't defend the well except when he's there with his brood.

5.96 Whereas mantella have these territories. This was solidly in the territory of one of my most successful male mantellas. And for the entire time, this *Plethodontohyla* dad was there, it's like, "Well, I'm out. There's no way I can use that well." Except that, yes, when this frog is already in a well, you don't get new courtships happening in that well. You don't get new mantella egg oviposition, mantella oviposition, into the well. So that's obviously a heterospecific competitor that's a hostile force. It reduces the ability of mantella, of the poison frogs, to do what they do, to reproduce.

- 5.97** But once you have one of these guys in a well, if they moved in and there were already mantella eggs in it, the mantella eggs have a better chance of surviving. They actually persist longer, once they have this apparently protective ability, this protective frog, giant frog that will bark at all the intruders, in the well with them. So is it a competitor? Yes. Is it helpful? Yes. Both of those things. It depends on when in the life cycle it shows up.
- 5.98** So this is a male. One of my field seasons, I marked them by tattooing them. This is D4 on the back. This is a male at the top of a well that he is defending actively. They sit at the top of their wells or in the areas around their wells, and they call. They call, and sometimes they get into incredible fights. This is actually male on male. You can't necessarily see it in this picture, but the frog on the bottom there is calling. He's got an extended vocal sac, and that's in part how you know for sure it's a male, because females don't call. Females can't call. They don't have distensible vocal sacs at all.
- 5.99** So you've got two males here, engaged in a fight that I don't remember this particular one, but might have lasted an hour and a half. They involve wrestling, and they'll go up against each other, belly to belly, and they'll knock each other off. They'll chase each other up the bamboo, knock each other off, climb on each other, ride each other around. At some point, through means that, even though I watched a lot of these frogs, almost 1000 hours of watching, doing focal watches on these frogs, I don't know exactly what decides, "You know what?" One of the frogs goes, "I'm out. I can't take it anymore. I'm done." Usually, if it's happening within the territory of someone, it's the frog whose territory it was. You have what is often called the bourgeois frog phenomenon. That thing that you already had, you're likely to keep.
- 5.100** So territory holders tend to keep their territories. Even if by the other metrics by which you might be a good acquirer of territory, you shouldn't be able to keep it anymore, if you already have it, you're more likely to keep it. And in part, that's because you know the landscape. You know where you can chase a competitor, a conspecific, maybe off a ledge and have him plummet. They don't die. These frogs are incredibly resilient. I mean, they do die, but they don't die when they plummet off of something in a fight like this. So it may be in part that the home court advantage is just that, that you know the landscape. But it's also true that, even if you are just at the edges of a territory, the guy who has already spent time defending that territory is much more likely to continue to have it. Similarities to humans, again, right?
- 5.101** OK, so here is actually a courtship. You could see here a little bit of the brightness that not only do they have these bright yellow backs and black venters, stomachs, they also have sometimes some bright blue on their stomachs, and their fingers and toes have these bright neon blue splotches, which are really quite beautiful in person. These courtships, like the fights, can go on for well over an hour. The male will call, the female will approach, and he does what I call chinning. He will rub his chin on top of her throat

while he emits a soft little courtship call. And at some point, if she says yes up to that point, he will lead her to a well.

- 5.102** And that's not enough, though. She has to check out the well. She has to thoroughly investigate the well before she decides yes or no. Sometimes if she rejects the well, he'll be like, "OK, but I got another one. Come with me over here. I got another one. This one might work." And sometimes that works. Sometimes she'll say, "OK, I'll check it out." And the next one that he takes her to is actually that she says yes to, and they actually have a successful courtship, and an egg is laid. But very often, if she rejects the first one, she's like, "No way, we're done."
- 5.103** If she accepts it, though, then she lays with him on her in amplexus, a single egg, a single egg in that well. So as the size of the clutch goes down, compared to those traditional, the common frogs that lay hundreds of thousands, or even tens of thousands, of eggs at a time and never see their kids, the fewer kids you have, the more likely you are to have parental care. Because you're not going to lay one egg and walk away and say, "Let's hope that works." You're going to end up investing in that kid.
- 5.104** And so what these guys do is, again, tattooed frogs. That's P6, a female. I don't remember if she has just oviposited an egg, and with K2, a male, following her—it's his well, it's his territory—or if she has rejected this well, and he's trying to convince her to go to a different one. But what happens then is that the male in whose territory that well is, defends it. Defends it against things like *Plethodontohyla*, against the big frogs, except you can't really do it against them. But he defends against everything else. There are other species of frogs that like to use these wells, and they're closer to the same size as mantella, and he can defend against them, and he can definitely defend against conspecifics, against other frogs of the same species.
- 5.105** Once those eggs, once that single egg, usually, very occasionally, too, once that single egg hatches and drops into the water below, because she lays it above the waterline—once it hatches and drops into the water below, mom starts coming back and lays unfertilized eggs for her tadpoles to eat. Highly convergent with what's going on with the dendrobatids in the New World, but clearly convergent. Not the same lineage. This is not homology. This is not similarity due to shared history. This is similarity due to solving a very similar problem with a very similar solution. And that, I think, is where we end. Actually, in this picture, you can see just at the base there, that's an egg that she has just laid for the tadpole that's in the well there, that you cannot see.
- 5.106** All right. Thank you, guys.

Lecture 6: Tactics and Trade-Offs

At a glance — Tactics and trade-offs in evolutionary strategy.

- 6.1 Albert Einstein once said that the trick in science is to make everything as simple as possible, but no simpler. Now, I realized yesterday, after I talked to you all about tent bats, that I had neglected to give you some of the nuance to the story, and that it's important to realize that, as clean as that story is, if you dig deeper, you find that there are all kinds of little gems hidden below the surface. So I thought I would start by just adding something to that story that was well worth knowing.
- 6.2 So when I started my work on tent bats, this paper was essentially the Bible. This was a review article by Tom Kunz and the other top people who had studied tent-making in bats, and it provided a review of everything that was known. It listed all the species that did it and all of the different evolutions. It listed all of the plants that they used. It listed every style of architecture that was known. And so it was a great place to go for information to figure out what I should be looking for when I got to the field myself.
- 6.3 Now, here is that central list in which the animals are listed, the plants they use are listed, and the architectures of tent that they have been observed to make are listed there. But when I started finding tents after initially not being able to find them at all, and then figuring out what they looked like and how to look for them myself, something started to bug me about what I was seeing. This animal is *Artibeus jamaicensis*, the Jamaican fruit bat. Never mind that its name is Jamaican. It's all over Central America. It's a very common animal.
- 6.4 And I did indeed find them in tents with some regularity, but it's not really like the other tent-makers. All of the other tent-makers exist only in tents. They don't roost anywhere else, and they're all small. That's not a small animal. That's a pretty big animal. In fact, it's big enough to hurt you if you don't handle it correctly. Here it is doing what it does best, eating figs. It will eat other fruits, but it does tend to specialize on figs. And here it is doing what I mentioned fruit bats do in the forest in order to limit their commuting costs. It's taking little bites of fig, and it's rolling the juice out of them, and then dropping the pellets on the forest floor. Now, of course, in the case of a fig, seeds are very tiny. It's bound to swallow some of them, almost no matter how careful it is. Here is an *Artibeus jamaicensis* mother flying with her baby. And then back to this list.
- 6.5 What started to bug me is that this animal, which didn't look like any other tent-maker, was too big, and it roosted in other places unlike other tent bats, every place that it was listed to use a plant and to build a tent of some particular architecture, some other bat was known to use that exact plant and that exact architecture. And I realized after a time that what I was looking at was an evolutionary answer to a different question. This

animal didn't have as big a problem with commuting costs as the true tent-makers because it's bigger. That means it flies more efficiently per unit of mass.

- 6.6 So making a tent probably just isn't worth it, but it does need to roost somewhere, and it does have the same problem, where fruit moves around the forest as this tree goes into fruit and that one goes out of fruit. And so it makes sense that it would opportunistically take advantage of tents made by other species, which is, in fact, a perfect match for the pattern that we see. So one of the lessons in biology is that you're dealing with the most complex thing there is. And that means that, to the extent that you've got a simple story that's mostly true, it may not be entirely true. And you always look for those added nuances to build the story, so you finally get an approximate picture of what's really going on.
- 6.7 Now, I will say that the diversity of plants used by these animals ends up being huge. I lived on Barro Colorado island for 18 months, and by the time I left, I had seen every understory leaf that was big enough to have a tent made in it turned into a tent, with two exceptions. And I will say, you have to have a standard. I saw a lot of things that could be tents, but I never classified anything as a tent until I had at least seen a bat roosting in it, or I had seen the same structure made twice in that species. Then I could classify it as a tent. Otherwise, I was never entirely sure.
- 6.8 This is one of the species that doesn't get made into a tent. This is a *Dieffenbachia*. The common name is dumb cane. It's named after the fact that it has a toxin on its surface that causes you to lose the ability to speak if you were to eat it. And then here's the other one, which doesn't get made into a tent for obvious reasons. This is *Monstera*, which is a *Philodendron*, and it has these holes in it that make it a very poor candidate for being a tent. Now, there is a part of me that wonders if it has those holes in it because otherwise it gets tents made out of it, right? That's a possibility, is that this is the answer to that question. How do you make a big leaf like that and not have all these little bats chewing it up all the time? But I don't know. It's just a hypothesis.
- 6.9 OK, and here is finally *Artibeus jamaicensis* doing the thing that no other tent bat does, which is roosting somewhere else. In this case, it's roosting in a hole up inside of a cave.
- 6.10 OK, onto a new organism. Now, that structure of this tree trunk here is the answer to an evolutionary question. Of course, it would have to be, right? But what is that question? All right. Well, in order to get there, let's talk a little bit about one of the central issues that all trees have to deal with in tropical forest, really in any forest, but especially in tropical forest. Now here I am standing in the middle of a stream, and it's pretty light. If I stepped into the forest, I would be in a very dark place. And this is not 100% obvious when you're there. And it's not 100% obvious because your eyes are much better than you think at amplifying light, which is why the light level out during the day tends to seem constant and then suddenly drops off very precipitously at the end of the day. What you've done is you've reached the edge of your ability to amplify, and then the amount of light collapses in real time.

- 6.11** In the forest floor, it feels like it's shady, but it doesn't feel dark. Now, when people take objective meters and they put them on the floor of a tropical forest, what they find is less than 1% of the incident light actually reaches the floor. The rest of it is either bounced off or absorbed and used by the plants above. And that creates a huge problem for canopy trees. Those trees start as seeds on the forest floor. And the first thing they do is they put up their cotyledons to start making a profit, using sunlight to turn carbon dioxide from the air into sugar and into cellulose.
- 6.12** We haven't mentioned cellulose much, but cellulose is the structural material of plants, and it's something that no animal can break down on its own, which is why you've seen the leafcutters partnering with fungi and why you see the hoatzin doing their foregut fermentation. These are tricks that animals use to get access. But the plants build that cellulose as part of their structural material. And actually, the sugar that they build and the cellulose that they build are really just mirror images each other—the one, readily digestible, the other one, completely indigestible by animals for some reason.
- 6.13** But in any case, on the forest floor, that seed germinates. It puts up its cotyledons, and the next thing that's going to happen to it, almost for sure, is it's going to die because it doesn't get enough light. That happens to almost every seed that germinates on the forest floor because there just isn't any light. What there are, are tiny little flecks, a little patch where the sun shines, but those don't stand still. If you put a camera out there, what happens is you watch those sun flecks sweep across the forest floor. Day after day, they sweep across the forest floor, and they may drag across that little seedling, and it may have a few minutes to make a profit, and then it's gone. And so that's not how trees get to the canopy.
- 6.14** The way trees in general get to the canopy is they get lucky, and some other tree reaches the end of its life and collapses. Now, when that happens, you get a light gap, and that light gap is hospitable to all kinds of things that otherwise can't survive in the forest. It's a very lively place. So then suddenly, at the point that there's a light gap, there's a race. All the little seedlings that landed in that spot are now in a race to take up that place in the canopy that's suddenly been opened up. So that's how all these trees have gotten where they're going.
- 6.15** They're reaching up towards that light. And you can get a sense from this picture that it's not a large fraction of the sky they're reaching for. Only when that area opens up in the forest do they get there. So that's kind of a bummer of a story if you're a canopy tree, right? That means you're just basically doomed at the point that you're a seed that germinates, unless you get very lucky and happen to end up in a light gap, which is the way it works. But there might be other ways.
- 6.16** So here are some plants that have dealt with this in ways that do not involve fighting for light gaps. This is an epiphytic orchid. This epiphytic orchid lives on the surface of another plant. That's what epiphyte means. It's an epiphyte. It lives on the branches and trunks of trees that have already gotten to the canopy, so it doesn't have to get

there. It has other problems. It now has to attract pollinators, and it has to get its pollen taken to the very same species of plant at some great distance. But the problem of how to get enough light is solved by this plant.

6.17 Now, I will point out trade-offs in everything. When you solve the problem of how to get to the canopy, you create a new problem. You've got plenty of light. It's always light up there, but you don't have enough water, which is counterintuitive because it's a rainforest. The water is always falling, right? But it's also very warm. What happens is the rain falls, it's wet up there for a little bit, and then it dries out quickly, and it can go dry for weeks sometimes. So these plants have to have a plan for water. And the orchid, you can see what its plan is. It has these very fleshy roots and leaves which store water. And it's a very hardy plant, and it does well in that environment. It's built for it.

6.18 Here's another plant that solves this problem very similarly. This is a bromeliad, which you'll remember from Heather's talk about poison-dart frogs. This bromeliad, unlike the one where we have the picture in the well on the ground, is way up in the canopy. This plant has solved the water problem a different way. It basically has a cup facing upwards, and it collects water when the water falls, which is why there are wells for those frogs to lay their tadpoles in. And some of those frogs have their tadpoles actually way up in the canopy. It's really quite a strategy.

6.19 And actually, whether or not you've ever seen one of these—I mean, you've surely seen them in flower shops and things—but whether you've ever seen one of these in the wild or not, you've definitely encountered that structure because this is the same family as pineapples. And this structure is the structure you see at the top of a pineapple, right? It's pretty clear. So anyway, the reason that it's structured that way is to capture water. And by capturing water, it basically is like an inverted plant. It gets its water from above, not from roots below.

6.20 Which brings us back to this structure. It's really one of my favorites. I love the answer to the question. And in fact, I can prove that it's a very good answer to the question of how you deal with the problem of light on the forest floor. Here we have it again. Now, I will point out, this plant is not that plant. This plant is not that plant. What we've got are strangler figs. Now, a strangler fig solves the problem of light on the forest floor by not starting there. What it does is, its seeds germinate in the canopy. That bat will have eaten a seed, or a bird will have eaten a seed. They'll crap it out while they're flying along. It will land somewhere where there's dirt accumulated in some other part of a tree, and it will germinate, and it will put up its cotyledons. And lo and behold, there's plenty of light, right?

6.21 Now, it does have a clock ticking on it, though. It's got a certain amount of water. If it's fallen in the right place, that spot in the tree may accumulate water and function like that bromeliad facing up. But eventually, if that's going to be a tree, it's going to have a water problem. It has to get to the ground. And so what they do is they encrust the

trunks of other trees, and they wind their way down. They are not sucking nutrients from that tree at all. All they're doing is encasing it. They're using it structurally. Instead of growing up towards the light, they're growing down towards the water. Now, when they hit the forest floor, they root. And eventually, the burden of that encrusted plant will kill it. And you'll end up with one just like this.

- 6.22** Here you're looking right up the trunk because the host tree here has died. It's gone. Like Heather said, rot takes place very quickly in these forests. So that plant is no longer in evidence. And what you have is this plant that has replaced it in the canopy, actually taken its very spot. That's an incredible thing for a plant to do. It's also a fantastic way to get your genes into the future. In fact, it's so fantastic that this behavior isn't limited to the neotropics. This behavior is found in the neotropics, it's found in Asia, it's found in Africa. It's a very good strategy that has spread very effectively.
- 6.23** Now, here, you're not looking at a New World version. This is an Old World version, and the Old World strangler figs have some tricks that the New World strangler fig don't. This one puts down extra trunks, right? It becomes like a grove. Now that has multiple effects. One, it makes a much bigger structure that holds that big place in the canopy and can use all of that space to make a profit. And with that profit, it can make seeds that it can use to try to disperse into places that don't yet have such a creature.
- 6.24** It also solves a structural problem because the outline of the tree here that is represented by the strangler is not nearly as strong as the tree it replaced. I mean, you can look at it. It is pretty strong given the amount of material, but there's so many holes. And so it's in danger of falling over, now that it's lost its host tree. But these Old World strangler figs have this extra trick. By building these extra trunks, they can actually support more weight, and they can become secure in the canopy.
- 6.25** The fact of dropping extra trunks is actually reminiscent of this plant, which doesn't do the extra trunk thing. It doesn't start at the top of the canopy. But these are aspens. And as you may know, aspens actually grow asexually underground. So that may be one organism you're looking at. It's many trees, but it's one organism, right? And then funny things happen, too, when these things intersect with people. Here you have of a temple in, I believe, Malaysia, and you have a strangler fig beginning to encrust it.
- 6.26** Now, here's the part I really don't expect about this story. So far, the story I've told you is surprising, but once you hear it, it all makes a good deal of sense. Here's the part that still throws me. There are trade-offs in everything. Strangler figs are masters of getting to the canopy by starting there, right? That's an incredible trick, and it comes with huge costs and burdens. But the same trees can do it the normal way, too. They can grow from a seed, and they can get lucky in a light gap, and they can reach up and grow, and they don't look like stranglers when they do that. The very same species. Somehow there is enough plasticity in this creature that it is capable of dealing with whichever circumstance it finds itself in, and it knows how to deploy its tools in the right way to deal with it.

- 6.27** OK. We're going to switch gears. We've talked a little bit about things like fig trees. Fig trees hire things like birds and bats in order to distribute their seeds. We've also talked a little bit about pollination, and we know that plants like the bee I showed you at first are hired by a flower using nectar and pollen in order to induce them to transport pollen to some other individual, so that they can mate across a distance that a plant can't cross. But that's not the only way to get the job done.
- 6.28** So this is an orchid in the genus *Ophrys*. And what they do is they produce a simulation of a female insect, in this case a bee. Not only do they produce a visual simulation of a bee, but they produce pheromones that are very similar to the ones that the female bee puts out in order to attract mates. And the fact is, these are insects. They are remarkable, but they're basically robots. And so those two signals are enough to induce a male to come to the flower and attempt to mate. And in attempting to mate, he picks up the pollen, often in the form of packets that actually are physically glued to the animal when he interfaces with it as if to mate. And then he flies off, and because it's not a vertebrate and they don't learn very well, he does it again with another flower somewhere else, and those pollen packets get deposited, and the orchid is fertilized.
- 6.29** And this has produced a huge number of morphs because, of course, there are many different kinds of bees that might do the pollinating. Now, I will tell you, Darwin was a very brilliant scientist. He actually had an example of this plant, a plant in this genus, local to him in Britain. He never figured it out. It was too much of a stretch somehow. I'm sure if he had lived longer, he would have. So basically, what you've got here is, instead of hiring a pollinator, it's hijacking a pollinator, which from its point of view is just as good. The point is, the plant doesn't really care about the insect's well-being. It cares about getting its pollen to where it needs to go. And the insect doesn't care about the plant's well-being. The insect is looking for food or, in this case, a mate. And so the fact that we see this shocking kind of indifference to a pollinator by a plant shouldn't shock us at all. We should expect it.
- 6.30** Now, I should also point out, though, orchids are a huge family of organisms, and I've just shown you a bunch of cheaters, right? But that's not characteristic of the family of orchids. Here, you have a very different orchid, doing what most plants that use animal pollinators do. It's paying a living wage. Here, it's hiring this hummingbird to pick up pollen and transport it to the next version of the plant. But you've got these two kinds of strategies. They both work, and you should expect the one that is more successful at bringing genes deeply into the future in the context of each of these organisms to be the strategy that you find, either hijacking a pollinator or paying them.
- 6.31** Hijacking is also not limited to orchids at all. Here, we've got a cadaver flower from Indonesia. This flower smells like rotting flesh, and its pollinator are flies. Those are flies looking to eat and perhaps lay their eggs so that their maggot offspring will have something to eat as they develop. Now, there's nothing for them there. It's a trick. It's just a smell. And these are little robots. They don't know any better. So they fly, land on

it. They look in vain for the cadaver. They can't find it. They fly off, and because they don't learn it, they go do it again. They find the next cadaver flower, and it gets pollinated. They hijack them.

- 6.32** Here's an entirely separate version of that exact same strategy—interestingly, from a similar forest, also Indonesia. So this flower smells like rotting meat. Looks like rotting meat, right? It's like a simulation of white fungus growing on a very old steak that's been sitting somewhere for a really long time. Smells atrocious. Nothing for the flies to eat, but they fly in, they go hunting around in between these structures. They come up in vain, and they fly off with the pollen and pollinate the next one.
- 6.33** It's not even just limited to plants. Here you have a stinkhorn, right? This is a stinkhorn, which is using this veil to distribute a scent into the air to attract flies. That's what this beautiful, very fragile veil is built for, is to distribute this scent. And here are the flies. They've come looking for something, and it's in vain. It's a simulation. Because these animals are orienting with an olfactory cue, they're being lured in to distribute the spores of this fungus.
- 6.34** OK, now we should talk a little bit. So far, what you should be getting the sense of is all of these creatures are trying get something done. Now, we know that ultimately it's the same thing. How do you get your genes deeply into the future, right? But they all have proximate problems that are very different, and they're trying to solve them—not only solve them, but as cheaply as possible. That's the key, right, because anything you can save while you're doing the job can be repurposed to something else. If you can not make a structure, then you have more material with which to make eggs. You have more energy with which to fend off pathogens. You have all of the things that you need in greater abundance. So being efficient is one of the key things that organisms are doing.
- 6.35** And one thing that you'll find, you can't see it, but if you study these creatures, what you'll find is that when you look at a big bush full of flowers and you think, "Oh, there's a lot there for the bees to eat," well, actually, the flowers differ in how much there is to eat, and different bushes put out different amounts. And the amount that they put out is actually part of a cryptic kind of competition, even between individuals of the same species. That is to say that those bushes which figure out how to pay the least and still get their pollen transported have extra resources to do other things with. So that's a competitive win.
- 6.36** And so there's a pressure to reduce the amount of reward. But then there's a point at which a plant is too stingy, and the animals don't come back. And in fact, one of the things that we know from bees is that they have a crude language for describing to their collaborators—as you learned from Heather, those are their sisters, three-quarter-related sisters—they have a mechanism for conveying where a good resource is. They can describe it exactly in space, how far, what angle, and in what direction. And they convey this to each other. And the point is, they are like a distributed sensory

apparatus that finds a reward worth it, and then comes back to the hive, and says, "Hey, you know where we can get paid? It's over there." So that's pretty cool.

- 6.37** But everywhere, you'll find this dynamic, where the plants are behaving. These are creatures that don't have any neurons. They're not thinking about how to do this, but they are behaving in such a way as to drive organisms to do their bidding. And sometimes they screw up, and they pay too little, and sometimes they pay too much, and then they suffer in competition with somebody who paid less. They also drive some really remarkable behaviors in some of the creatures that they hire.
- 6.38** So this, obviously, is a hummingbird. In this case, it's an Andean hummingbird. But what one finds, when you study hummingbirds, is that they do something called trap-lining. Now, the behavior called traplining is named that. It comes from the pattern that fur trappers used to have, where they would set out a bunch of traps, and they would circulate. They would do a loop. They would come back through their traps at the right periodicity to see whether they had captured anything. They didn't want to wait too long. They could lose what was in the trap to some a scavenger. Maybe it would escape, but they would come back in this line.
- 6.39** And if you monitor hummingbirds, what you find is that they do this, and that, in fact, if you stand at a particular, if you see a hummingbird and you see it come to a flower, stick around. Actually, look at your watch. You'll see it come back regularly. Now, it may not be the only one using that flower, which makes the game that much more complicated. But what's really happening, and this is absolutely measurable, is that this plant is putting out nectar at just the right rate. And this plant is in danger if it does too much, too fast, because if it pays the animal too much, the animal has no incentive to go anywhere else. The last thing this plant wants is to feed that animal enough that it doesn't have to go. It wants to give it enough that it's worth visiting and not too much that it sticks around.
- 6.40** So this drives the animal to do this trap-lining thing, and you've got a bunch of different plants on its circuit, and this plant doesn't know anything about the other plants on its circuit. So the best it can do is monitor how regularly the nectar is being taken out. It also doesn't know how many birds are visiting it. All of these dynamics play out in a very complex way that no one completely understands. Here, and I confess, I haven't 100% convinced myself one way or the other on this, I think this bird is cheating. I think this bird is taking nectar without going through the front door. And by not going through the front door, it doesn't get dusted with the pollen. So it gets the payment, and it shortchanges the employer, more or less.
- 6.41** It's possible that that's not taking place, and it's possible that the structure of this flower is built so that, by pointing its beak in there, it actually does get dusted with pollen. I don't know. It's one way or the other. But there are certainly plenty of creatures who take the nectar reward from a flower without doing the job. You'll often see them go in at the base of the flower. A flower will have a very long corolla tube.

That corolla tube is designed to exclude all of the animals that aren't good pollinators, right? It's designed to make it not worth it or impossible to get to the nectar. But of course, many animals have figured out that actually you can just chew through the side at the bottom, and you don't have to go down the long corolla tube to get the nectar. You can get it some other way. And so that dynamic unfolds.

- 6.42 Now, how much should the plant invest in making it thicker at the base, so an animal that tries to burrow in from the side either can't do it or it's not worth the effort? How do you discourage it? Can a flower discover that it's been penetrated from the side and stop producing nectar so that it's no longer worth it? And does that drive the animals that have been coming in through the side to go somewhere else?
- 6.43 OK, but here's the craziest part of all. You've heard Heather and me talk very fondly about convergent evolution. And the reason for that is because when evolution repeats itself, we have a good chance of figuring out what's going on. When you have one example of something, you don't always know. So for example, there are 11,000 or so species of birds in the world. They all have feathers, at least at some point in their life cycle. Does that tell us something about what feathers are for? No, because it all came from one ancestor that had feathers.
- 6.44 We can say how they function. We can say what they do in the present, but we can't say what drove them to be there because we have one and only one example. We have to work differently to figure out what it's about. But in the case where we have multiple examples of something, multiple clades that start building tents, multiple instances of farming, any of these things, we can say, "Well, what's similar about those things that caused the evolution of this answer? What is it that drove it?"
- 6.45 And here what we have—this is a picture Heather showed you a couple of days ago—this is a glossophagine bat. It's a nectar-feeder. And it is feeding on a vine called markea. Now markea, this particular vine, actually is very interesting. It ripens one flower per night, and the flower is open for a few hours. The bats visit, the flower dies, it drops off the next night. Another flower opens. The flower doesn't put out enough nectar to keep the bat here feeding all night on the one flower. The bat actually does exactly what the hummingbirds do.
- 6.46 In fact, I don't know why we don't call these hummingbats. We really should, to be consistent, because they are very much like hummingbirds. They hover, they do all the same behaviors. They trap-line. They really are like a nocturnal-mammal hummingbird. And the fact is they are in the same game theoretic dynamic with the plants that the hummingbirds are. Therefore, the same game emerges, right? You just don't see it because it's dark.
- 6.47 So this is an odd plant. This is *Angraecum sesquipedale*. And this plant, an instance of it—this is a plant that only grows in Madagascar, so it's an endemic, as Heather was telling you. Endemics only grow in the one place. This one only grows in Madagascar, a

place that Darwin never went. But there was an orchid grower, I believe his name is Bateman, who sent Darwin a specimen of this. He thought Darwin would be interested, given what Darwin was understood to think about. And Darwin was completely blown away by it. He was totally fascinated, especially this long structure here. That is a corolla tube.

- 6.48** Now, as I told you before, a corolla tube is a structure designed to exclude all of the pollinators the plant doesn't want having the reward. So this foot-long corolla tube is there to exclude a bunch of stuff. But it implies something that nobody knew about at the time. No one on Earth knew, or at least nobody connected to Western science. It's possible that people who had been living in Madagascar would have said, "Oh, yes, of course." But from Darwin's perspective, it was completely unknown, as for all of the other people that he was interacting with.
- 6.49** The fact is, that structure implies an animal. All right. This is *Xanthopan morgani praedicta*. It is named after the most famous evolutionary prediction ever. Darwin did not live to see his prediction tested, but he predicted that there would be a moth out there with a tongue long enough to get to the bottom of that corolla tube and that it would live in exactly the right place to pollinate that plant. Now, I will point out that Darwin is deploying the exact same logic that I formalized with that 4-part test for adaptation. That logic is robust. There's a reason that it works.
- 6.50** It probably, under ordinary circumstances, wouldn't even need to be said. Yes, it's obvious that the features of organisms are the result of adaptive evolution, once you know that adaptive evolution happens. But there was a long battle in the field of evolutionary biology in which people argued that drift could produce things that would fool us into thinking they were adaptations. And so a formalization like the one I came up with was necessary just simply to get past the question of, "Is that structure adaptive?", which it almost always is, to the interesting question, which is, "What does it do? How does it help that creature?"
- 6.51** So anyway, Darwin just simply reasoned, "There is no way in the world that there is going to be a plant evolved in competition with all of the other plants that's going to have a foot-long corolla tube that would be a complete waste of resources. Mind you, it would also prevent any pollinator from carrying the pollen of that plant to an appropriate mate. So it wouldn't make any sense to live on a planet that had the plant and didn't have the pollinator, unless the pollinator had recently gone extinct." So he predicted that it would exist. And then 21 years after his death, it was found. And there it is. And here you can see it actually engaged in pollination, as unlikely as that seems.
- 6.52** OK. You probably think you know what those are. You probably think those are coconuts. Those are a monstrosity. Those are not coconuts, even though in cartoons, the things they hang on the cartoon palm trees look just like this with the little dots on the end. This is a caricature of a coconut, even though what's on the inside is genuine coconut. That's a coconut. That's what a coconut looks like. A coconut is a large, woody

structure, much bigger than the thing you buy in the supermarket, just so much bigger. And it's incredibly hard to get into the damn thing. In fact, it's the kind of object that you need a machete to get in, but you don't want to hand a coconut and a machete to an amateur because they will hurt themselves.

- 6.53** It is not easy to get into that thing. It takes work. It can take, you know—somebody who's good at it could probably do it in 5 minutes. It would take me 20, right? It's a matter of getting that machete buried in the thing and using the leverage to pry that woody husk off. Which does raise kind of a question. Why the hell would that plant stick all that stuff on the outside of the coconut when the part that matters is there, right? That's the part that matters to us eating coconut, and that's the part that matters to the plant. That's the part that contains all of the food for the baby coconut palm, right? So that's the business part. Why do that? What the heck could that possibly be useful for? You want to say, say it loud.
- 6.54** To travel.
- 6.55** To travel. Yes. Stick a bunch of wood on the thing for traveling. He's right. But how does the wood help it travel?
- 6.56** Floats.
- 6.57** Floats. It floats. It's the most amazing dispersal mechanism. This thing doesn't need an animal to carry its fruit somewhere. For one thing, it lives at the beach. Those coconut palms are really good at the beach. They're not really very good at growing anywhere else without help. That means, when those woody objects drop off that palm, they are somewhat likely to find themselves washed into the sea, and they may well find themselves washed up on the shore of some other place. And maybe that's some other place doesn't have coconut palms. And maybe the thing will germinate, and it will put up a cotyledon, and it will grow into a beautiful palm tree.
- 6.58** This one, by the way, is too close to the water. It's not going to work, but good try, right? I mean, not bad. So that's coconuts, right? Coconuts, the woody part, is the answer to a question about how do you disperse if you don't have some large animal to transport a huge seed across vast expanses of saltwater, right? You just drop it into the saltwater and hope for the best, and it may float there, and it's fully capable of making a tree on the other end.
- 6.59** OK. What's going on here? You all know this thing. You all played with them as kids. You picked them up. Or if you didn't, I'm sorry for you, but you should have. You got it?
- 6.60** Somebody off right of the camera is making a wish.
- 6.61** Somebody? And did they get their wish? Can you tell?
- 6.62** I think so.

- 6.63** Yes, I think they did, too. No, there may be, in fact, there probably is somebody, a confederate of the photographer, is probably making a wish, maybe for a raise, I don't know what, but something. But what's going on from the plant's perspective? These are interesting structures, pretty improbable. It's not quite umbrella-like, but something with enough fine hairs and structure to be able to catch the wind? What is it?
- 6.64** Seeds.
- 6.65** Seeds. Good. So what does that imply has gone on here, from the point of view of the plant?
- 6.66** The wind is strong enough, it will fly.
- 6.67** Before that, I'm trying to lure you into a trap. If there are seeds here, then what must have just happened?
- 6.68** Pollinated.
- 6.69** Pollinated, and then what?
- 6.70** Germinated.
- 6.71** Germinated, right. Except that didn't happen. That's the incredible thing. It's exactly what you should think. In fact, those are the structures of seeds. But no sexual reproduction to place, in all probability, because that's a dandelion. And the thing about dandelions is that most of the time dandelions do not get fertilized. They have learned a trick, and their trick involves the production of seeds without sexual reproduction.
- 6.72** So all of the developmental stuff, the stuff that tells your mind you're looking at a plant distributing its seeds using the wind, your mind is working correctly. This plant has just found a strategy that's ingenious that doesn't require one step in that puzzle. And it's a little bit like I was telling you yesterday —there is a great advantage in being able to just simply clone yourself, which is not exactly what this plant does, but simply cloning yourself solves a lot of problems, right?
- 6.73** Especially if you're a plant and you can't march over somewhere and find a mate and convince them to reproduce with you, right? If you're stuck in one spot, the ability to reproduce yourself is awfully useful, and the offspring are highly related to you. So at least for a little while, it's a good way to get your genes somewhat into the future. But this plant prefers to do this without sexual reproduction. And there's probably a very good reason for that, too, because as I also mentioned to you yesterday, the thing about it is, even though any generation of females can get ahead, in fact by a factor of two, by cloning themselves rather than sexually reproducing, because the offspring that they produce, which will be equally expensive to raise, will then be twice as related to them as a sexually produced offspring, right? So that's a two-fold advantage.

- 6.74** So this plant somehow has a different calculus in mind. For some reason, in this case, the plant is producing seeds to distribute propagules, which contain only this individual's genes. Now, in all probability, the reason for that is that the way this organism spreads, it is much better not to require a mate. Imagine for a second the predicament of a dandelion that finds itself in a beautiful new habitat, perfect for raising baby dandelions, but there's no one to mate with. Is that not a tragedy? An animal that's beaten the dispersal game; it's found a beautiful new habitat; it could have spectacular evolutionary success and get way into the future, except there's no one to mate with, so it's going to die and no one will ever know, right? That's a tragedy, evolutionarily speaking.
- 6.75** Well, if you can figure out how to get past the problem of reduced genetic diversity, which then causes you potentially to be more vulnerable to pathogens or more vulnerable to environmental change—if you can somehow beat that problem, then the ability to reproduce without a mate is very advantageous, especially if you're going to try to get into new places, which these guys do, right? If you open up a piece of habitat and you strip it clean and you till it, you're very likely to have dandelions want to grow there. They're going to find their way in because these things are going to blow on the wind. And if they can get there and not have to find a mate, then they can get much farther than they otherwise would.
- 6.76** So it's forgoing reproduction, which now causes us a new problem. That's the new problem. Why are these plants producing those petals if they don't need to hire an insect to distribute their pollen because they're going to reproduce asexually? I don't know the answer to that question. I got a few starts on possible answers to that question. One of them is that they actually do a certain amount of sexual reproduction, that they need some sexual reproduction so that they can maintain enough genetic diversity that when they go asexual, it's not catastrophic, right? That's one thing.
- 6.77** And so the hypothesis then, which runs a little bit counter to that adaptive test I told you about, one possibility is that it can't afford to get rid of the petals. That if it got rid of the petals, then when it had to come back around to sexually reproducing, it would be missing one of the main ingredients. And so that then suggests, if that's true, that those petals are wasted effort that could be economized away, if only the plant could figure out how not to deploy the petals when it was going to be asexual or something along those lines. That's why I don't fully accept that story. But I will say there is a lot to be said for the fact that the creature does borrow some sexual reproduction in order to keep its asexual racket going. And it's not the only one that does that.
- 6.78** These are aphids. Now, aphids are famously asexual. These are plant parasites. They suck plant fluids out, and they reproduce asexually, and they can encrust your plants and do terrible damage to them. Now, the thing is, aphids reproduce sexually once a year. So they have a sexual reproductive phase, and then they go asexual.

- 6.79** And if you think about it, it's not that different from what you do, right? You will produce offspring sexually. And then after you've produced that zygote sexually, there will be a whole lot of asexual reproduction that will produce all of the 30 trillion cells that that creature will become when it reaches adulthood. So the idea of being asexual in a lot of your phases, and then sexual for one phase, is not so unthinkable. In fact, it's very standard. We just don't typically characterize it this way, and we don't typically see animals that look to us like individuals that were produced asexually. But there's really no reason that that's less valid.
- 6.80** Here we have another example. This is a whiptail lizard, and there are populations of whiptail lizards that are all female. They reproduce asexually. Now, it's interesting. Just like the petals on those dandelions, they do retain some weird sexual—it's not really structures, it's really a behavior. In fact, it is necessary for females to mount each other in order to stimulate the reproduction. These are lesbian lizards, and they engage in this pseudo-copulation, which triggers the production of asexual eggs, and it works. But again, every so often, especially at the edges of the range of the population that is asexual, there is sexual reproduction.
- 6.81** So this is something you should be thinking, which is, there's something very important about sexual reproduction. It is necessary. We know it's necessary for most organisms, and it is necessary for reasons that are a little hard to spell out, but that doesn't make them unimportant. However, there are lots of ways to circumvent it for a time or over some piece of territory, as long as there's some way to access the value of it somewhere else, right? So think of the population borrowing genetic diversity from the fringe populations that are not asexual, right? Or think of the aphids taking advantage of genetic diversity once in a season, rather than each time they produce new aphids.
- 6.82** There are lots of creatures we have now seen do a trick like this. I suspect it's a tiny fraction of the number that we will ultimately recognize can do the asexual reproduction trick. We've seen it in things like sharks. We've seen it in certain snakes. We've seen it in komodo dragons. It's never been seen in a mammal. It's been done in the lab in a mammal, but it's never been seen in a wild mammal or a captive mammal. But what I suspect is, if you think about what it takes for a creature to start a new population, there's a huge hurdle that can potentially be overcome if this one trick can be learned or can be picked up in some way that is equivalent to learning, some adaptive way.
- 6.83** The problem is, imagine, for example, a dragonfly that has been blown off-course in a storm and has, against all odds, found itself on an island where there are no dragonflies, but that has all of the things necessary for dragonflies to thrive. That individual is in a position to outdo what the average dragonfly does. Even a successful dragonfly will typically only produce two surviving dragonflies. How do I know it's two? Because it's sexually reproducing. And that means that, if the population was so small that there was lots of extra room in it, then the population would grow. Individuals

would produce more than two offspring until it reached what we call carrying capacity. That is, the limit of that environment to house that type of organism.

6.84 And at that point, then on average, an individual has to replace themselves with two others, each of which is half related to them. Two individuals produce two offspring. They have replaced themselves. The population isn't going up, and it's not going down. In the case that the population were to crash because, let's say, it had an inhospitable year, and then the following year was good, the population would grow, and individuals would produce somewhat more than two on average. And then at the point that things got bad again, if the population overshot the carrying capacity, then average individuals would produce something slightly less than two. But sexually reproducing organisms tend to produce two offspring.

6.85 A dragonfly that has blown onto an island that's hospitable to dragonflies in which there are no dragonfly competitors, well, who knows how many offspring might survive? But more importantly, think about the jackpot it has just hit from the point of view of getting its genes deeply lodged into the future. That dragonfly has everything it needs, except a mate. Again, it's a tragedy unless that dragonfly has a trick. And that trick is, if you really had to, could you reproduce asexually? Because if it can reproduce asexually, then it's got everything it needs. Then it needs to produce one generation of two sexes. There will be some reproduction. It'll be awkward because they're siblings, but they'll get through it. And the fact is, there might be a burgeoning population of dragonflies on that island that might allow that individual, if we looked several generations down the road, to have beaten the average expected reproduction of a dragonfly by a 1000 times, right? That's possible.

6.86 That's a huge evolution ordinary win, if only you have the trick. That's why I say, I expect that we will find that the harder we look, the more organisms we find that will have this trait, and it will be triggered by loneliness. That's the thing. And this is actually where we have typically seen this happen. Animals that have been brought into the pet trade or brought into zoos—we think of them as something that they're there for us to see them, but they are effectively isolated from all potential mates in many cases. And so whatever the equivalent of loneliness is in the species, it would trigger whatever it is that might allow an individual that found itself alone in paradise to produce more offspring so that it could create a new population.

6.87 So that's a prediction. We will see more of this in more taxa the harder we look, that it's actually not as uncommon as we think. It's uncommon because it's something that happens very rarely that produces populations from which many of the creatures we see are descendants. And you can see that something like that might have happened here.

6.88 This is a marine iguana in the Galápagos, and it's a very odd iguana. It actually looks a whole lot like its mainland counterparts, but it doesn't behave like its mainland counterparts. Its diet consists entirely of algae, which it gets by swimming to the shelf,

and it eats it off the rocks under there, and then swims back, and sits in the sun all day, and ferments. It's a very unusual animal. Now, it must have had ancestors that somehow got from the mainland, which has many iguanas on it, out to the Galápagos, which is a very rocky, inhospitable place in which there's not all that much to eat, but there is a lot of algae. I mean, the sea isn't isolated there.

- 6.89** And so not only did it somehow accomplish the trick of producing offspring that became a huge population of these animals—I mean, they're literally around the entire periphery of all of the islands in the Galápagos—not only did it do that, but it switched niches. It figured out how to make use of a resource that its ancestors didn't. Its ancestors were carnivores.
- 6.90** OK, I'm going to switch gears to some new organisms. These are kokanee salmon. Kokanee salmon are a special case of sockeye salmon. Sockeye salmon are anadromous, which you've heard means that they swim out to sea. The fry swim out to sea. They spend several years, like four years, out at sea, profiting from what's out there, fending off the hazards of the open ocean. And then they return to, in general, the streams where they were born.
- 6.91** Kokanee are a special case in which they never go out to sea. They are landlocked, and they live their whole life cycle in fresh water. And that's interesting. That probably evolved as a result of the fact that when we go out into the habitat where these Pacific salmon live, we see the opportunity to go to the sea and come back, and we think, "Wonder why the kokanee don't do that?" But the habitat is much more dynamic than we think, just not on a time scale we tend to think of.
- 6.92** During glaciation, many of the ways that salmon get out to sea and come back are blocked by ice. And so it makes sense, if you're going to try to get your genes deeply into the future, to have a plasticity that allows you to deal with the various different conditions. Can you complete your life cycle when the flow to the ocean is blocked? Can you complete your life cycle when it's open? If you can do both, then you are presumably at great advantage. So kokanee may be, in some sense, a hint at what happens during glaciation.
- 6.93** All right, here are your regular sockeye salmon. And here is a freshwater habitat in Idaho, where salmon breed. All right, there's an aggregation. Now, what I wanted to talk about was the phenomenon of strays. Now, strays are salmon that show up in the wrong place. Now, how does salmon find the right place—the right place being the stream or the watershed where they were born? They have a really cool trick. So they're born—let's say that we're talking about a population that's high up in the mountains—they're born up in the mountains. They spend a little bit of time there as tiny little fish, and then they move as fry down to the sea.
- 6.94** And in moving to the sea, they are passing—Heather was describing the sequence of rivers that brings you to the Amazon, tributary after tributary—well, each of those

tributaries in any watershed contributes a unique chemical complement. Every river has a slightly different composition of minerals and other things in it, based on what happens to be on its flanks, what washes into the stream during the rain. And so that signature can be read by the animals as they go out to sea. Every time they pass a creek or a river that contributes new water, some new component is added to this puzzle. At the bottom, where all of the waters have combined, you've got all of the rivers simultaneously giving their message.

- 6.95** But each of those things was added one by one. And so the animals, when they return four years later, they swim up and they can detect, "This is the mouth of the river where I was born, because it smells just like that place." And then they can swim up, and they can find the sequence of signatures that allows them to navigate back to the place where they were born, which is really cool. Now, the problem is, they don't always do it, and we don't have a very good understanding of the dynamics. But many ecologists have dismissed this as a kind of error. "Salmon are really good at finding the place that they were born, but they're not perfect."
- 6.96** Well, I don't think that that's the way it goes. What I suspect is that, if you look at all of the different watersheds that have salmon, there are vastly more watersheds than there are salmon species. That means that salmon got out of the places where they first evolved, and they took up residence in other places. Some animal had to do something that an ecologist might describe as a mistake, but it wasn't really a mistake because it resulted in a spectacular evolutionary profit. It resulted in an animal producing vastly more offspring than it would have, which, again, isn't really the point of evolution. But if they do it, if a particular salmon species finds its way into a new habitat that doesn't have any competitors, and it produces huge numbers of offspring, then it is that much more likely to get deeply into the future.
- 6.97** The bigger the sum total population of that species is, the more likely some member of that species will still be around 1000 years or a million years later. So there's a question about whether or not those stray salmon are errors, or are those animals gambling? My bet is that those animals are gambling, and that there are two ways to gamble. There's a principle—I don't know why it works, but in biology, if you find a pattern in space, the distribution of something in space, you can very often look for that pattern in time, and you will find it. Often, it will be named differently, but it will function similarly. Space and time are somehow alike in a way that is not very intuitive.
- 6.98** And in this case, strays can happen in space, as I've just described. They can also happen in time. You have animals who come back a year early, for example. Now, those animals are also potentially in a position to effectively colonize a new habitat because these rivers are dynamic. Things happen on them. You could have a mudslide that would take out an entire year's run of salmon. That would leave an empty year on the calendar. And if you came back at exactly the right number of years, the right

number, then you wouldn't find it, right, because you'd be coming back in the same year as all of your cohort.

- 6.99** But if you came back a year early and the watershed had been emptied, well, again, you would have the evolutionary equivalent of a bonanza, where you could produce huge numbers of offspring. You could make the population that much larger. Your particular lineage would be the dominant fact of that population, and therefore, your genes would be that much more likely to get deeply into the future.
- 6.100** And these kinds of explorations might occur in things that are very unlike salmon. Now, that is a penguin. That is a penguin in the Galápagos. This is an endemic penguin of the Galápagos, and the Galápagos is a very weird place for penguins. It's very warm, and penguins are famously not a warm-weather bird. And yet here it is. That's not a pool. That's just clear tropical waters. So the question is, did that animal's ancestor get washed off-course? Did it get very, very lost? Did it show up in the Galápagos, and it was all one big error? Quite possibly.
- 6.101** It's also possible that there's a certain amount of experimentation in nature, and that that experimentation ends up doing things like discovering archipelagos, like the Galápagos, which are so far away that you can't detect them from the mainland. So we have to ask ourselves the question. We don't know whether the error is the explanation for many of these creatures that have found these remote places, or if there is a strategy that involves finding remote places with enough regularity that that opportunity is discovered.
- 6.102** And the final thing I will say is that that question leads us directly to humans. And we will talk tomorrow about humans, but I want to put on the table the idea that what we call wanderlust is the desire to go to new unknown places, in spite of risks that may make it very hard to explain why that is a rational thing to do. Why does somebody get on an ocean-going vessel and cross an ocean when they don't know what's on the other side? It's not a smart move if survival is the thing you're trying to maximize. It's not a smart thing if finding a mate is the thing you're trying to maximize.
- 6.103** But it's a gamble. When those missions have been accomplished successfully, they more often result in a large bounty of some kind. So that's the question I will leave you with. Is it possible that human wanderlust is in fact an exploration strategy that is in some way in the background of all of the people of the Earth today? Did our ancestors explore in some way that made them more likely to find opportunities that allowed us to get here into the future? That's it.

Lecture 7: Adaptive Organisms

At a glance — Adaptive organisms of the Pacific Northwest.

- 7.1 Good morning, everyone. Today, we are going to start out in the Pacific Northwest. This is Mount Hood. It's taken from a plane, one of the many volcanoes that emerges from the spine of mountains that are the Cascades. Mostly, though, the organisms we're going to be talking about first in the Pacific Northwest are not up in the mountains. They are Pacific giant salamanders. *Dicamptodon* is the genus. *Dicamptodontidae* is the family. And we'll talk a little bit about what they are.
- 7.2 But the question we're going for with the Dicamps, the *Dicamptodons*, is some species always metamorphose. They're amphibians. Some of them metamorphose, as we expect amphibians to do. Some never metamorphose. They are obligate paedomorphs, and we'll talk a little bit about what that means. They don't metamorphose, but they become reproductively capable. And then some sometimes do, sometimes don't. Some individuals within a species will metamorphose, some individuals in the same species may not. Under what circumstances might you expect that to happen? So we'll come back to this question after we walk through a little bit what they are, what the dicamps are, and what metamorphosis and paedomorphosis and the conditions under which these things evolve happen.
- 7.3 This is an actual picture as opposed to an old scientific illustration of a *Dicamptodon*. There are only four species in this clade. It's a good clade. It's a monophyletic clade, sister to the *Ambystomas*, which include the axolotls, which are also obligate paedomorphs. They're called giant salamanders, but they're not really that big. They're about 30 centimeters, snout-vent length. So that's a third of a meter-ish. It's just bigger than many salamanders, but not terrifyingly big. There's nothing to worry about here. They actually only, like the frogs I was talking about yesterday, only lay a single egg at a time, which is rare in amphibians. Again, they have obligate maternal care in the form of just egg attendance. They don't feed them, they don't do anything else, but just hang out with the egg and make sure that it persists.
- 7.4 And they spend a lot of their lives at stream edges along fast-moving streams and in undergrowth and under logs. If you've spent any time in the Pacific Northwest, and if you've gone looking for amphibians, you know that flipping over logs is the way to find them. And it being a coldish, dampish place, you're very unlikely to end up flipping over a log and finding something that's going to hurt you. You're not going to find vipers under those logs. I don't recommend that practice in the jungles that we've been talking about, but in the Pacific Northwest, you can do that.
- 7.5 So they are nocturnal, as pretty much all salamanders are, and pretty cryptic despite being much larger than most salamanders. They metamorphose, some of them, all the time, and some of them don't ever, and some of them do sometimes. So the question

that raises is why metamorphose at all? What's metamorphosis for? Well, as we've talked about earlier in this course, amphibians are tetrapods that have emerged onto land, they're vertebrates that have emerged onto land, and have had to shift around a lot what they do in order to deal with, for instance, the different refractory index of air versus water in order to see. And they've had to change out their respiratory structures. They're amphibians, so they're breathing through their skin, but they also have lungs. Almost all salamanders have lungs. These guys do.

7.6 But they're also, having emerged onto land while still reliant on the water for their egg and larval stage, want to, in not trying to ascribe consciousness to evolution or forward-thinking to evolution, but the successful amphibians have been able to move successfully onto land as adults. So metamorphosis provides that capacity. So they can be aquatic as eggs, where they need to be aquatic in order to stay wet, and aquatic as larva, tadpoles if you're a frog, just a salamander larva if you're a salamander. And then the metamorphosis allows for a basically complete reorganization of your cells in order to become not a swimming organism that breathes with your gills, but a walking organism that breathes with your lungs and to some degree your skin.

7.7 And maybe you have the capacity to go back into the water, but you're not going to be as good in the water as you are on land if you fully metamorphose, because feet aren't as good as fins, for instance. And salamander larva never have fins, but they have webbing between their fingers and toes. They have this caudal fin along their back, which helps them move through the water as they undulate. So there's a lot of parts of their form, in addition to their physiology, like their circulatory system, that has to change as they move onto land. So the metamorphosis is effectively a way for them to live the amphibian lifestyle most completely. So they can be aquatic larva, and they can be terrestrial adults, and they can have both of those stages of their life successfully.

7.8 But sometimes metamorphosis isn't the best option. Sometimes pine martens show up. And if, as opposed to your previous environment or a previous moment in your same environment, if there weren't very many pine martens, and now there are pine martens, you might be really highly susceptible to predation on land, right? So maybe suddenly there are more predators on land or fewer predators in the water, or maybe there are now more competitors on land or fewer competitors in the water, or maybe there are fewer resources on land to eat or more resources in the water. So any of these 6 categories with regard to predators, competitors, resources, things that you might eat, might have changed in a more or less permanent way, such that what was good for you in terms of going onto land no longer is. And so under those circumstances, maybe you become a paedomorph.

7.9 A paedomorph is, for instance—here's your axolotl, *Ambystoma mexicanum*. It's a salamander that never develops adult characteristics in terms of lungs instead of gills. What you're seeing coming off his face are gills. Those are external gills. In larval *ambystoma*, you always have those. And in metamorphs, they lose them and develop

lungs instead. But in the paedomorphs, they keep the gills. They never pick up the lungs. They keep the webbing. These guys don't have that much, but they keep some of the webbing between their fingers and toes so that they can swim better, just like we were talking about with the gliders. Swimming and gliding are similar activities in regard to being able to move. And they keep the caudal fin on top of them so that they can swim more easily.

7.10 And so this is actually—I was in Mexico recently, and one of their currencies, one of their bills, has an axolotl on it. So this is just a picture of one of the bills. I don't remember what denomination it is, but this is some number of pesos in Mexico at the moment. So paedomorphosis—I've already effectively defined it, but it is the umbrella term for heterochrony—"hetero," changing, "chrony," time. So changing time of development, wherein—and you could be heterochronic in a paedomorphic way—wherein you retain larval characteristics, even though you're reproductively capable as an adult, or you could be heterochronic in the opposite way, and you could effectively get greater development. You could become sort of super-adult, hyper-adult.

7.11 And it's harder to track. It's harder to know exactly what that is because we don't know what more than adult would look like. There's no history of it, right? Whereas for juveniles, for childhood, there's a history of that, and you can say, "Oh, that organism doesn't look like it actually ever fully matured, even though its reproductive capacity did." So paedomorphosis is one of the names, is the umbrella term, for what people have probably heard of, if they've heard these organisms at all, which is neoteny. So neoteny is actually just a single example of—it's one of the mechanisms by which you could become paedomorphic.

7.12 So imagine if you've got a developmental time period and your reproductive parts are going as normal. There's nothing changing compared to what your closest relatives are doing. But the rest of your somatic body is developing less completely. Maybe you started developing later. That's post-displacement, I believe. Maybe you stopped developing early. That is hypomorphosis. I may have those flipped. And then neoteny is that the timing is the same, but the rate of change is slower. So these are just specific terms for mechanisms that we sometimes understand and often don't with regard to how is it that we ended up with a paedomorphic organism? Is it that they started developing late? Is it that they stopped developing early? Is it that they started and stopped developing at the same times as their closest relatives, but the rate of change is lower, and so you ended up with a less developed adult?

7.13 In any case, axolotls, obligate paedomorphs, they end up as in their larval form throughout their lives, even though their reproductive capacity continues on. We've talked about why you might end up being an obligate paedomorph, and why you might be an obligate metamorph, obligate metamorphs being the main thing that amphibians do. But why would you be a facultative paedomorph? Under what conditions might that be relevant? So maybe I won't ask you to try to answer that, but think on it. Under

what conditions would facultative —meaning it could go one way, it could go the other way—paedomorphosis be something that would be useful? Because the costs should be obvious. If you've got a developmental program that says, "OK, at this stage, you go metamorphosis," OK, that's pretty challenging as it is. But if there's a single developmental program, probably that can get locked in. It will usually work.

7.14 Similarly, if you've got a single developmental program of being an obligate paedomorph, it's going to get locked in, and it's probably going to work. Whereas if you keep your options open, there's a lot more that can go wrong. There's a lot more chance that it's just not going to work completely, because what you don't want to be is some intermediate form that's kind of good in the water, kind of good on land, not great in the water, not great on land. You're going to have a harder time getting away from the land and the water predators, and the land and water competitors, and probably have a harder time collecting resources as well, collecting food, because aquatic food that amphibians are eating and terrestrial food are very different, are going to require pretty different mechanisms of action to get them.

7.15 So being facultative is tough. It's a more challenging developmental process. So why do it? Well, when you need to keep your options open, or when it seems to be useful to keep your options open. This is a relatively new beaver pond. This is outside of Olympia, Washington. And the water level changes because it's not a totally static, it's not a completely established beaver pond, and so it's not a completely stable ecosystem. As a result, the stochasticity, the changing nature of this that is not predictable, means that under some circumstances, you might have conditions where it would make more sense to be on land as an adult, and under some circumstances, it will make more sense to be in water as an adult.

7.16 So when you have variable environments, you end up with just another term of art here, phenotypic plasticity. Phenotype is just the expression of what your genes, your genotype, told the phenotype to make, right? Told the body to make. And plasticity is the term that we use for the ability to be variable in response to things. So phenotypic plasticity, of which facultative paedomorphosis is an example, is a response to variable environments. There's a lot of ways to be variable. We're just talking about amphibians here today. But in any condition in which you might be exposed to lots of different things as an adult, that you're presumably totally fit for the current environment, parents might not have been exposed to, you might want to retain an ability to change over time. And when we're talking about humans later, you will see how humans do that, not so much through facultative paedomorphosis, but by other means.

7.17 So plasticity, again phenotypic plasticity, it's just a shorthand, plasticity is evidence of resistance to specialization. So we had a question yesterday about why would you want, why would you expect pollinators or plants to be so particular about the pollinators that they're after? In the case of pollinators, the answer that we generated was that a plant needs its pollinator to find another plant of exactly the same species in

the opposite sex. Depending on if the plant has two different bodies of sexes or has both sexes on one plant, it's monoecious or dioecious. But in general, specialization like that is a hazard. And so we have lots of examples of plasticity in the world. Here are some.

- 7.18** This is just a line drawing from 1909 by a guy named Woltereck, who formulated the idea of a reaction norm. A reaction norm is just the pattern of phenotypic expression of a single genotype across a range of environments. So these line drawings of *Daphnia*, of water fleas, show that some things are not variable. They always have one eye on each side. They always have two eyes. There's no phenotypic plasticity with regard to eye number, just like there isn't for humans. Occasionally, bad developmental things happen, but those are errors. Whereas the body shape of *Daphnia* is highly variable, even when you control for genes, even when you control for the genetic input, the way that the body looks in adulthood can vary widely. And that variation in expression, even when you hold genes constant, is called a reaction.
- 7.19** So Woltereck, over 100 years ago, was the first guy to describe this and start to think about, OK, what things aren't going to be plastic? In humans, bipedality, not plastic. We're bipedal. A very occasional developmental anomalies notwithstanding, we're bipedal. Whereas what is plastic about humans? Something to think about.
- 7.20** Tropisms. Tropisms just refer to what forces are plants growing towards or against. With regard to sunflowers, which is what this is a picture of, you have heliotropism, "helio" meaning sun. So sunflowers, over the course of a day, and you could find these time-lapse videos of sunflowers in a field basically moving their sun dials, effectively, with the sun, moving with the sun as the sun moves, as the Earth moves around them, but as the sun apparently moves in the sky around them. So tropisms is a way to retain plasticity in adult form, even in a plant which is supposedly sessile, which supposedly doesn't move at all.
- 7.21** Well, tropisms demonstrate that that's not purely true, as is the case for so much in biology. We can have—there's very little that's law-like in biology. Almost everything has exceptions. And then within humans, variation in aorta. Aorta anatomy is extraordinary. So the aorta is the major artery going off the heart that goes into the systemic system. So we have a four-chambered heart, and what that means is that as blood comes into the heart from the body, it's deoxygenated. It goes into the heart, comes out pressurized, but still low oxygen, goes into the lungs, gets oxygenated, comes back into the heart, now high oxygen, low pressure, comes back into the heart to get pressurized, now comes out high pressure, high oxygen, goes into the body and perfuses oxygen into the body.
- 7.22** So the aorta, which is the main artery off the heart that sends oxygenated high blood, high pressure blood into the body, would seem to be important. It would seem to be really, really important. If it fails, you're dead. And yet we have incredible plasticity in terms of our anatomy of our aortas. Why is that? Why would selection have let that be

something that's super-variable? Well, the answer in part is going to be that it's remarkable how much variability there is actually in circulatory and also neural anatomy that, to some degree, with regard to both blood and nervous signals, it needs to get everywhere. And so being the most efficient at getting there is not necessarily important because there will be multiple ways to get everywhere.

7.23 Whereas with muscles, you have highly conserved anatomy within species. Yes, some of us have larger muscles or smaller muscles, but every one of us has the same muscles that are attached to the same origin insertion points. Whereas with regard to circulatory anatomy, even with regard to one of the most major vessels in our body, we have incredible diversity.

7.24 So plasticity across all of these things, and also *Spea*, which are spadefoot toads, develop—*Spea* tadpoles develop into cannibal morphs when in the presence of conspecifics. So if they're growing up in a situation where everyone around them is another *Spea* tadpole, they develop into a morphology—the tadpoles develop into a morphology that has bigger teeth, bigger heads, and they are then more able to eat their own kind. And if they are developing in the absence of conspecific tadpoles, in the absence of other *Spea* tadpoles, they develop into kinder, gentler tadpoles. They don't have big teeth, they don't have big heads, they tend to munch on plants.

7.25 So you've got this plasticity that is responding to what is in the environment. How are they telling? There's some chemical senses by which they can tell what else is in the environment. It's obviously not conscious. These are amphibians. They're not making choices for themselves. None of this plasticity is conscious. We have no—no individuals have control over it at this level. But there are many examples, like the *Spea* example, where tadpoles, when exposed to different things, will turn into either bigger, fiercer, more carnivorous morphs, or slightly smaller-headed, kinder, more vegetarian morphs.

7.26 Here we have another one. Early life stress in children is correlated with reduced size of brain centers used in socioemotional processing. So the things that children are exposed to can affect their brain development. At one level, that's completely obvious, and at another level, it would seem to be something that a lot of people who would say, "You know what? Parenting doesn't matter. It's all about genetics. You do the best you can, and it's fine," would argue, "Well, no, that's not going to happen. What possible effects could early learning have on the children's brains?" Well, the answer is a lot. And of course, of course it is. So hopefully, given what we've been talking about through this course, that will be obvious, but it would come as a surprise to some.

7.27 OK, a few more examples of plasticity. You have sex for life with temperature-dependent sex determination. So we, mammals, have genetic sex determination. Almost all of us—we'll be talking about this a little bit later. And birds also have genetic sex determination, albeit an opposite system where the females are heterogametic and the males are homogametic. And there's a few other cases of genetic sex in vertebrates as well, a few other evolutions of genetic sex determination, but a whole lot of

vertebrates have environmental sex determination, and usually it's by temperature. So if you are an egg layer, if you're viviparous, if you carry your baby on board, it's tougher to have temperature sex determination. Because at the very least, you're probably going to have all one clutch, all one kid, all one group of kids at a time, that's either male or female, because your body is just one temperature at a time.

7.28 But some crocodilians develop into females at low temperatures and males at high temperatures, and some lizards and snakes as well. Some do it the other way around, and some develop into—try to remember, it's males at both extremes and females at the intermediate temperatures, or it may be the opposite way around. But there's at least three different ways to do temperature sex determination in various reptile species. And are those contiguous with one another? It's not clear, but they will respond to changing environmental conditions, which also means, of course, that if the environment around them changes, they may be in trouble because suddenly, if it's a lot colder than it used to be, suddenly they may be having all males or all females, depending on what the species is and what the particular changes are happening in the environment.

7.29 There is sex for now, sequential hermaphroditism. This is an anemone fish, also known as a clownfish. This particular species has what's called protandrous hermaphroditism. "Pro," first, "and," male. So they are first males. They are born as males. All individuals of the species are born as males. And then some of them, under some circumstances, change into females. There are a few fish, there are a few reef fish, that can actually change back. Mostly with sequential hermaphrodites, they are only able to do it once, and there are very particular social and behavioral constraints that predict when they will change. And once the anatomical and physiological changes happen, so too does the behavior change. So that's rare in vertebrates, completely unknown in birds and mammals, but somewhat common in reef fishes, which, of course, raises the question, "What's going on with reefs?" Why is it important in reefs, as opposed to other places? I'll just leave that as an open question for now.

7.30 You have hatch or be eaten. I've shown you this frog before. This is the red-eyed tree frog in Central America. In some amazing research in the 1990s, a woman named Karen Warkentin discovered that what was already known about these frogs is that they lay clutches of eggs. They're more typical frogs. They don't have a lot of parental care. They don't need parental care. The males and females meet once for basically combining their gametes, and then the male leaves and the female leaves, and they never meet their kids. But they lay their eggs on leaves over hanging ponds. And sometimes those ponds are seasonal, sometimes they're permanent, but they lay them on leaves over hanging ponds, the advantage of which seems to be that eggs are particularly susceptible to predation because they can't defend themselves at all. Tadpoles aren't great at defending themselves, but they can at least make an attempt at swimming off.

- 7.31 So if you lay your eggs on a leaf over a pond, and it's the rainforest, and they only spend 5 to 7, at the most 10, days in egg form, you're probably not going to dry out, and you're safe from aquatic predators. And then as you hatch, you basically hatch and drop into the pond below. So you end up with leaf-born, somewhat terrestrial eggs, aquatic tadpoles, and then they metamorphose, and they become arboreal frogs. OK, so that's what was already known about these guys. What Warkentin discovered is that when the cat-eyed snake, *Leptodeira*, when the cat-eyed snake comes and tries to eat the eggs, because frog eggs are delicious, as we've talked about. They are undefended by any poisons, and these frogs aren't poisonous anyway. When the cat-eyed snake comes and discovers lunch for himself—if these eggs, which normally wait to 7, 8, 10, 9 days to hatch out, if the snake arrives at 4, 5, 6 days, the eggs will preferentially hatch early.
- 7.32 Now, they hatch early and they escape from the predation by the snake because the snake is not aquatic. The snake does not go into the water below and find the tadpoles. But well, if they could do that, why don't they just do that anyway? Why isn't that happening all the time? And it's not happening all the time because there's a benefit in staying in the egg form for longer. Even when they take 10 days to hatch out, they've still got a yolk sac attached to them. They've still got a lot of development to do. They don't eat for a few days. They don't have all of the aquatic ability yet. And so the younger you hatch, the less capable you are, and the more likely you are to get eaten in the water. But having this phenotypic plasticity in terms of when being attacked by something that will eat you for sure, hatch out early and drop into the water below as a tadpool, those frogs that do that and become tadpools earlier, do better. They have a higher chance of survival. They have a higher rate of survival than those who didn't.
- 7.33 OK, and finally, when growing up is a choice. We're back to our *Dicamps*, our *Dicamptodon*, which this is actually—I couldn't find a picture of the right species, but as I said, there's one species in *Dicamptodon* that is a facultative metamorph, that lives in, apparently, a stochastic-enough environment where sometimes metamorphosing is the right move and sometimes not metamorphosing is the right move, and they're keeping their options open at the species level. Under what conditions does facultative paedomorphosis make evolutionary sense? We've kind of arrived there, right? When environmental conditions are sufficiently stochastic, that some adults are more likely to succeed on land and some adults are more likely to succeed in water.
- 7.34 OK, so the next thing we're going to talk about before we begin talking about humans is mixed species aggregations, sometimes called mixed foraging flocks, mixed species flocks, mixed species troops, depending on the species you're talking about. Everyone will recognize this as a scene from the African savannah. Many ungulates of lots of different species hanging out together. What problems are they solving by doing so? Well, the most common answers, especially for something like this, is safety in numbers. It's just more eyes, more eyes to see the threat when it comes, and also,

statistically, you're less likely to be the one who gets picked off if there's more of you. If you're hanging out in a group of 3, and a predator comes, and he's really hungry and really intent, you have a higher chance of being the one eaten than if you're in a group of 30 or 300.

- 7.35** And then also, they're concentrating at a limiting resource. So for these guys, for giraffes and gazelles and wildebeests and such, there may not be any communication necessary. There may not be any willingness necessary. There may not be any decision at all to hang out together. They may just end up in the same place, and then they get this extra benefit of their safety in numbers. "We're here, we're drinking the water," which is highly patchily distributed, especially during some times of year when the water sources are drying up and becoming fewer and far between. And then they get the benefit of being, of having the safety in numbers. And this requires no inherent buy-in from any of the members.
- 7.36** Primates also often hang out in mixed foraging troops, mixed species aggregations. And the kinds of explanations, in addition to safety in numbers, that will be attributed to primates in general, is sharing or parasitism of knowledge about local food sources. So OK, if you're two species of frugivores, and fruit is, as we learned with the tent-making fruit bats, both highly concentrated in time. And yes, once you know where a fruit tree is, that fruit tree isn't going to move, but it's only going to be fruiting for a short period of time each year. So you want to know where all the fruit trees are, and it may be by hanging out in a mixed species group that sometimes this troop, this species, knows more about which species is now fruiting, and sometimes it's this one. So there's a possibility for sharing or for parasitism. Maybe it's always one species that actually knows where the fruit is, and the other species is just hanging on and not actually doing any good for the others.
- 7.37** And then there's maybe—I'm not as sure about this one, but this is one of the things that is proposed in the primatological literature—increased insect capture rate by flushing prey. So if you've got a lot of monkeys in a tree, as opposed to half or less than that, depending on what the sizes of the groups of species are, got a lot of monkeys in a tree, you may get greater flushing of the insects because frugivores are basically never entirely frugivores. They always eat insects as well, and they will eat them opportunistically, and you may get greater flushing of insects if you've got more individuals around. So that's just again going to be an advantage of greater numbers. The first one here is actually the social knowledge of the monkeys may be something that is beneficial, and sometimes it's going to be about more monkeys means more flushing of insects means potentially more access to the insects that are being flushed.
- 7.38** OK, but these guys, capuchins, in the genus *Cebus*, and squirrel monkeys, as you've already heard about, have a particular relationship. And there are a couple of capuchin species. I'm just going to be talking about them as if they're all one species here for today. But there's some terrific work done in Manú National Park in Peru, looking

specifically at the relationships, the mixed foraging aggregations, the mixed species aggregations, of these guys. And what John Terborgh and his colleagues found, that Cebus, he already do this, they're larger. They live in smaller groups. So squirrel monkeys are little gorgeous monkeys that occasionally leap on bat tents and fling them to the ground, fling the bats to the ground. And they live in groups of 30 or 40. Whereas the capuchins are larger monkeys, and they live in much smaller groups of 8 or 10 or 12. That's variable, of course, but that's roughly average.

7.39 So the Cebus monkeys, the capuchins, are larger. They live in the smaller groups, and they're the leaders. When these two species are together, they're always the leaders. Very occasionally, Saimiri, the squirrel monkeys, will go off leading, but then they always check back. They always check back. And if the capuchins aren't following, they go back. Whereas the capuchins, when they lead, they don't care. They don't care if the squirrel monkeys are following them or not. They just proceed off, and the squirrel monkeys either follow or they don't.

7.40 And the Cebus monkeys, the capuchins, also emit alarm calls that the squirrel monkeys respond to. And there's a lot of variation in the alarm calls that monkeys will give. And so the alarm calls are not just like, "Oh, my God, I'm panicking. There's something bad here." It will tend to have some information in it about the nature of the threat. Is it coming from the air? Is it coming from land? Is it coming up the tree? And capuchins emit these. Squirrel monkeys don't tend to, and furthermore, squirrel monkeys attend to, attend to the alarm vocalizations of capuchins. So squirrel monkeys are getting led to fruit trees by capuchins, and they're also listening to the alarm calls of the capuchins, and presumably being better able to avoid predation as a result of being around these larger-bodied monkeys who are more likely to make alarm calls.

7.41 And so Terborgh and his team conclude that, on balance, the capuchins are just mildly parasitized actually by the squirrel monkeys. It's not actually this "everyone's happy and doing wonderfully here." The capuchins don't really try to kick the squirrel monkeys out of their vicinity, but they do have dominance over them. So if a squirrel monkey is sitting somewhere eating something that the capuchin wants, they can displace them. So they've got the displacement ability that is one of the common marks of dominance. And the capuchins mostly seem to put up with the squirrel monkeys, but they don't actually seem to be benefiting particularly from them.

7.42 OK, one more example of a mixed species aggregation. This is not a mixed species aggregation. This is one species. This is, in fact, just one individual. This is a chickadee, a black-capped chickadee. Again, they're all over North America, but we're going to be talking about a particular study done in the Pacific Northwest, again in Washington State. Why are chickadees and these other birds they hang out with together? Well, again, it seems to be, at least what's in the literature before the study I'm going to talk about, is it's about interspecific eavesdropping. Interspecific eavesdropping is usually the explanation because chickadees give these, again, very precise alarm calls that

allow other birds that are hanging out with them warning about potential predators of the area.

- 7.43** That's all well and good, but are chickadees like the capuchins in the last example? Are chickadees hanging out with a bunch of other birds and not getting anything from it? Well, that's what a couple of my students actually asked. So a number of years ago, Adam Hagler and Drew Schneider did a study to figure out if, when chickadees are in mixed foraging association with these beauties, golden-crowned kinglets, and a few other species as well, but golden-crowned kinglets, is there an advantage that the chickadees get? Or is the advantage only being accrued by the kinglets? What these guys do, what the kinglets do, and what all the other species that are hanging out with the chickadees do, including a couple of other species of chickadees, or at least one other species of chickadee, the chestnut-backed, is they listen to those highly specific alarm calls of the chickadees, and they respond.
- 7.44** So again, are they parasitizing the chickadees, or are they actually gaining from them? Well, this is another golden-crowned kinglet, and he seems to be yelling about something. What is he talking about? I've said they don't really give alarm calls. Why they don't give alarm calls is interesting. Maybe because they're always able to find chickadees to hang out with, but that seems like a risk. So it's not an alarm call that he's talking about. What else might birds have to talk about? Any thoughts? Yeah?
- 7.45** What about mating call?
- 7.46** Mating call. In this case, it's not, but yes, that's one of the main things that birds have to talk about. Absolutely. This does not turn out to be—and the description, the distinction that ornithologists tend to use is song versus call. And so they're both vocalizations. When it's about mating and when it's a songbird of Passerine and has much more harmony and range in it, it's a song. And when it's simpler and less harmonious, it's a call. These are calls, so I'll tell you that, which I didn't tell you before. Anything else? Yeah?
- 7.47** Defending their territory.
- 7.48** Defending their territory. Great possibility. No, not in this case. So they are not at this point, at the point that they are in these overwintering, mixed foraging aggregations, territorial. They are moving around in these big groups with chickadees, with kinglets, with other kinglets, other chickadees, and occasionally a few where there are very little Passerines, little songbirds, that are traveling together, and they're mostly just trying to make a living. They're trying to survive the hostile force that is winter.
- 7.49** Signaling the presence of chickadees or other resources?
- 7.50** Oh, I love this. So signaling the presence of chickadees. No, they're not. But why wouldn't they? If you had some kinglets who are so clearly benefiting from being around chickadees, and you've got kinglets without chickadees, someone finds a

chickadee, might say, "Hey, I've got him. I've got a chickadee." We don't have any evidence that it's happening, but great possibility. Anything else? Yeah?

- 7.51** OK. Yeah. Maybe it's the reverse where they're imitating a chickadee call in terms of trying to have a less competitive environment in terms of food resources.
- 7.52** So maybe they are trying to mimic the chickadee. Also a fascinating possibility. We don't see any evidence that they're doing that, but very interesting. There are, of course, bird mimics in song, things like mockingbirds, most famously. But no, we don't see any evidence that they are mimicking chickadees in this case. The thing that they are doing together is foraging. They are eating food, and what the kinglets do is have foraging calls. The chickadees don't talk when they find food, they just eat. The chickadees talk when they see a problem, and then they have an alarm vocalization.
- 7.53** The kinglets talk about the fact that they've found food. And so if the kinglets attend to the alarm vocalizations of the chickadees, well, the chickadees attend to the foraging vocalizations of the kinglets. So in this case, it's not clear what the value of each of these is, but both of the bird species in this particular work are getting a value out of being in each other's company. And what my students specifically did was they played kinglet vocalizations, foraging vocalizations, and also song sparrow foraging vocalizations, which are sympatric, meaning they live in the same area, but they never hang out in the same groups.
- 7.54** And they looked to see to which vocalizations the chickadees, if any, did the chickadees care at all? Did they not respond at all? Did they respond to the song sparrows and the kinglets both, but they didn't discriminate between them? Or did they discriminate between them? And if so, which one did they listen to more? Would have been totally shocking if they listened to the song sparrow more, although maybe it would have been like, "Hey, what's that? We never heard that before." But what the chickadees clearly did was attend more to and orient towards the foraging vocalizations of the kinglets. So in this case, unlike with the capuchins and the squirrel monkeys, there does seem to be value obtained by both of the species. All right, so mixed species flocks have distinct advantages for the different players when there are advantages to both of the species.
- 7.55** OK, for the rest of my time, I'm going to give you a brief introduction to mammals, primates, humans, and set you up to talk more about humans in the second half of the day. So this is the segue slide, but this is also a Yumbo trail in the Andes of Ecuador. The Yumbo were a people who pre-existed the Inca in Ecuador, and they're walking these trails that—they were walking these trails for so long and so persistently that moderns can now walk them. And so it's truly remarkable to literally walk in the footsteps of people who are no longer on the land, but who are walking these trails for hundreds of years.
- 7.56** We've already done some of this, where we're going to quickly go through what are some of the diagnostic, what are the characteristics, what are some of the

synapomorphies, shared derived characteristics, of mammals, and then of primates, and then of humans? OK, so mammals, of course, have mammary glands. That's our eponymous trait. That's what we are named for. And not all mammals actually have nipples, but mammary glands. So the basal mammals, the echidnas and the duck-billed platypus, actually don't. And also the monotremes, the echidnas and the duck-billed platypus, at the most basal extent, mammals, also lay eggs. So whereas all the rest of the mammals, the placentals like us, and the marsupials like kangaroos and wombats, have gestation and live birth.

- 7.57** But a very few mammals at the base of the tree are different, and they're different in lots of ways. They also have not quite a million, but close to a million sets of sex chromosomes. They're really bizarre, like 8, 9, 10 sets of sex chromosomes to our one set. Mammals have obligate maternal care because once you have a mammary gland, that's your food for your baby, and so you've got obligate maternal care. This is a sloth baby and mom from BCI, from Barro Colorado Island in Panama. And once you have gestation and lactation, as the vast majority of mammals do, you have the precursor—you have it set up so that you are likely to get even more maternal care. Once you've got some, you're likely to have more.
- 7.58** And so you get extended developmental periods, and you get the teaching of cultural traditions. Once you have maternal care at all, you get more, and the maternal care ends up being about more than just nourishment and about safety, about keeping the kid close. But now it's also going to become about childhood is longer. "Let's spend more time learning how to be and teaching you how to be, letting you learn from mom and dad and from peers and from the environment, and let and let us teach you and let you learn from all of these things." So that happens in mammals and also separately in birds to a remarkable degree.
- 7.59** OK, a few more mammalian synapomorphies. This is a—I did not take this picture. This is a porcupine mother and juvenile. We've already talked about a few of these. Hair, right? The diaphragm, the muscularized transverse septum, that allows us to breathe more. I think I don't have on this list the non-nucleated red blood cells. We have genetic sex determination, a separate evolution from that in birds. And again, aside from those wacky mammals at the base of the mammal tree, the echidnas and the duck-billed platypus, we have a single evolution of genetic sex determination, which never varies, which is males are heterogametic, that is to say, XY, and females are homogametic, XX.
- 7.60** That is what determines the sex, but that is not what makes males and females the sex they are. That's about the gametes themselves. Individuals who produce or could produce or will produce eggs are females. Individuals who do or will or would or could, but for some developmental anomaly produce sperm, are males. And the genetic sex determination, the sex chromosomes that we have, are what determine what sex you will be. And in mammals, because it's males who are heterogametic, it's the sperm that

determines if the zygote into fetus, into baby, into human being, is going to be a male or female.

- 7.61** We are heterodont, meaning "hetero," again different, "dont," teeth. We have lots of different kinds of teeth. We have incisors, canines, and premolars and molars, and that's new with mammals. And you've got a lot of different kinds of tooth formation, tooth arrangements, in mammals. But the having different teeth for different jobs is a mammalian innovation. We have a single lower jawbone, this dentary, which makes us, allows us to have actually a stronger bite force than we would otherwise. We know we do not have a lower jawbone made of multiple bones that are fused together, unlike our most recent common ancestors with reptiles. And some of the bones that had been the lower jawbone of reptiles have moved up into our middle ear and formed our middle ear, which allows us to hear as well as we do.
- 7.62** We also have a four-chambered heart, again separately evolved, independently evolved, with that of birds, that of actually Archosaurs. Archosaurs seem to be the group that have the four-chambered heart, which suggests that —well, birds and dinosaurs, Crocodylians, have like a 3.5-chambered heart, which is strange. And they're probably on their way to getting a four-chambered heart, but they're just not there yet.
- 7.63** We have endothermy and homeothermy, the distinction being endothermy, we generate our temperature inside. We regulate our temperature inside, rather than needing external environment to collect our heat for us. Again, separately, independently evolved, convergently evolved with birds. And we have homeothermy, meaning, we keep our temperature the same, "homeo," at all times. And we can use fever, behaviorally or physiologically, to deal with pathogens, for instance. But fever is a response to something. It's not, we don't tend to fluctuate our temperature, by and large. And all the mammal species are going to have a somewhat different set point. But humans, by and large, have a set point, and that's 98.6 Fahrenheit, variable depending on the individual, but that's being homeothermic.
- 7.64** And we have a large brain, especially the forebrain, especially what's called the telencephalon—the front of the primitively 3-part or 5-part, depending on how you divide it, brain, which becomes enlarged in mammals and was historically associated with smell. So the biggest part of our brain was historically about smell, olfaction. And as such, you find that with most mammals, they do smell exceedingly well. Mammals are mostly using smell to navigate their world. Birds, in contrast, separate evolution, independent evolution, convergent evolution of so many of these traits, do some things very, very differently, obviously. They fly, for instance. But they also are much more sighted. They are much more visually focused, and they have a very poor sense of smell.
- 7.65** OK, so that's mammals. There's lots more, but those are mammals synapomorphies. Now let's go into the smaller nested group within mammals. So all primates are mammals. All mammals are fish, and all fish are vertebrates. We could do these nested

sets and go closer and closer. Primates include several species we've already talked about. In the monkeys, we have New World monkeys here, which includes squirrel monkeys and capuchins. And we have the apes here, which include the gibbons and the siamang and all the great apes, which includes us. And we've mentioned lemurs some. So this is just excluding a lot what the primate tree looks like, as far as we can tell.

7.66 And some of the characters that we use, some of the morphological characters—so if I give you a list of the molecular characters that we use to build this tree, it's not interesting, right? It's just a bunch of Ts, As, Gs, and Cs. But major innovations of primates at the morphological level, things that we can actually see in the space that we live in, at the scale that we live in, we've got opposable thumbs and big toes. Humans have lost the opposable big toes, but that's a primate synapomorphy. That's an innovation of primates. And we have nails as opposed to claws. And along with the nails come pads, the fingerprints. In some cases, in the group of neotropical, of New World monkeys that have prehensile tails, as Bret talked about, they actually have what's effectively a tailprint, a fingerprint on the back of the tail as well. But we have, all primates have finger and toe pads. And although a few of the more basal primates have things that are a little bit more like claws, most of us have nails rather than claws.

7.67 We have unfused long bones, which means, if you do this, well, you can do that. You can rotate your thumb around the axis of your forearm, and you can also do it at your leg. Whereas if you're a canid, you can't. If you're a dog, you are more stable in a long distance run, but you can't climb very well because climbing requires this, right? So we've got these unfused long bones —primates do—which allows us to be better climbers. And that probably points to us basically having adaptive radiation in the trees, early primates, because we got really good at climbing at the point that we both got finger and toe pads and those unfused long bones that allow us to both turn our arms and our legs to grip a tree that we might be on and also to grip it with our pads.

7.68 We're more visual, less olfactory. We've got a movement away, we've got the shrinking noses, movement of processing, of memory processing, of scenario building, of all sorts of other kinds of neural processing moving into that primitively olfactory part of the brain, and a reduction in the part of the brain that is actually being used for smell processing, and an increase in the part of the brain that's being used for visual processing.

7.69 We are brainier in general, just as with mammals, the brains are getting bigger relative to body size. Primates have, compared to the rest of mammals, larger brain-to-body size ratios. And of course, humans have even larger brain-to-body size ratios. We have, with primates again, longer pregnancies and smaller litter sizes. So we're cooking the kids for longer. They're spending more time in development, and we're having fewer at a time, both of which points to, again, having more parental care, having more to learn, having more to teach, having basically a more complex social environment and world in

general. And that lengthening childhoods, which is part and parcel of the overall, which follows from the smaller overall litter size. OK, that's primates.

7.70 And then monkeys, again, is a nested set within primates. It excludes the lemurs and lorises and galagos and tarsiers. So monkeys do all of that, but even more so—everything I just described. They don't unfuse their long bones more. But everything on this last slide that could be a continuum, longer pregnancies, smaller litter size—the monkeys continue those trends. And almost all monkeys, many of the non-monkey primates, but almost all monkeys, with the exception of, I think, just one species, are diurnal. We're basically moving into the light, and this is also part and parcel of becoming more visually oriented, as opposed to smell oriented. If you're going to be visual, you're going to want light to see.

7.71 We tend to have monkeys, all monkeys, tend to have singletons or twins, rather than litters. So again, a decrease in litter size. This is actually remarkably important. We tend to have individual cycles rather than breeding seasons. So for non-monkey primates and for many mammals, there is a time of year when you're going to breed, and there's a time of year when everyone is having kids, and everyone is in estrus, and then having kids. And that means that everyone is basically the same age. Everyone was born around the same time of year, and everyone's at the same level of susceptibility at the same time. And it allows for—that, having breeding seasons—allows for a concentration of—it allows for males to basically monopolize female reproductive success more easily than once you have breeding, once you have individual cycles, and females are effectively in estrus multiple times throughout the year. You have a much more individual approach to reproduction, and also the chance for a much, even greater than in other species, a much greater social interaction between males and females of breeding age.

7.72 All monkeys have a social group. There are very occasional non-monkey primates that seem to be solitary, although I even doubt that. But the primatologist claimed that a few of them are solitary. They'll meet up to mate and then maybe hang out together for a little bit, but then split up. I'm not even sure that's true, but definitely all monkeys have a social group. Apes have, of course, a lot more synapomorphies, a lot more shared derived characteristics that define them. But one of the ones that's most interesting, I think, is that we have more facial muscles than our most recent common ancestors. And what those more facial muscles allow us to do is have more facial expressions. We can communicate more emotion across distance without making noise than our closest relatives can.

7.73 One of the other things that we can do, which I'm seeing in the audience right now, is also we have whites of our eyes, and our closest relatives don't. If you think about, if you've never given the thought to the whites of your eyes, which most people haven't, think about what it allows you to do. You can all see where I'm looking even if I don't move my head. That means that you can communicate across distance without making

a sound, which is going to be very useful in circumstances like, you're hunting, for instance. So these eye whites of humans—and this is a human characteristic, rather than an ape one—is remarkable and is new to us.

- 7.74** All right. So what do humans have? We've got an even more expanding frontal lobe, the front part of our telencephalon that is involved primarily in memory and in looking forward, in thinking through what it meant about what you did and what you might be doing in the future to grow, to build on what you've done in the past. So our frontal lobe continues to expand. We're highly physically altricial. Altricial is the opposite of precocial. Precocial is the same root as precocious, but it doesn't mean exactly the same thing. A precocial bird, or a zebra is precocial. It's born and it can run around within a half an hour, an hour. An altricial animal, like humans, can't even roll over for 3 months. Really completely helpless. Sometimes I misspeak and say hopeless, and they're not hopeless, but they seem that way occasionally. They can't do anything, babies, newborns. We have extreme physical altriciality.
- 7.75** In a part, that's because our bipedal gait has made us somewhat unstable. Basically, the width of women's hips is already as wide as it can go and remain stable, standing upright. And so a bunch of the development that has to happen has to happen outside the body. You can't cook the babies any longer inside the body and have women continue to be able to walk around and not fall over.
- 7.76** So we have obsessive curiosity in childhood. As anyone who's been around kids or has kids knows, or has been a kid, everyone's been a kid, you remember being curious, and you hopefully don't remember too many times when you were told, "Don't ask that, don't do that, don't explore that," because that's what we're supposed to be doing, is asking, doing, exploring. We have generational overlap, as do some other species. The list of organisms that are highly socially complex tends to be ones that are long-lived, have generational overlap, have long childhoods. And there are things like elephants and toothed whales, which includes dolphins and orcas, wolves, and parrots, and corvids, which is like crows and jays. So all of those organisms have some of, well, certainly this trait, generational overlap.
- 7.77** And we live in extremely large groups. Another trend across primates is, the more derived you are, the more unlike your basal state you are as a monkey, the bigger the groups you live in are, to some degree. It's not a perfect, it's not a linear change. But baboons, for instance, live in extremely large groups. They're not apes, but they're extraordinarily smart Old World monkeys. And we, of course, live in groups that can be in the tens of millions. And there's a few other things, of course, that distinguish us, things like language and menopause and just a few other traits.
- 7.78** What do we have, but we're not unique about? Because we do like to talk about all the things that we do that are so amazing to us. And some of the ways that we've been taken down a peg in the 20th century, and to some degree, the 21st century, but a few of the places that we were taken down a peg with better behavioral ecology of other

organisms was, oops, there's tool use in other organisms, and a lot of it. Oops, there's war. Not that we wanted that to be shared by other organisms, but Jane Goodall famously discovered what really can only be described as war between neighboring troops of chimps. So there's a lot of stuff we do that is not unique to us, that seemed for a while to be unique to us. And even language, you can argue that some other organisms have not just vocalizations that are fairly precise, but they actually use syntax.

7.79 Here are some of the things we've talked about in this course, though. We've got plasticity, but so do some other things, like *Dicamptodon*. We've got agriculture, but oops, so do ants. Not a lot of them, but enough to make a difference. And they've been doing it for 50 million years. We've only been at it for 10 to 12,000. We've got elaborate parental care, but so do poison frogs and a lot of other species as well. We've got a tendency to explore, but so do salmon and a whole lot of other organisms as well. We've got specialists who tend to hang out with other specialists and thus make ourselves stronger because we bring our sets of traits, our sets of skills, together and become stronger by being together. But then it seems like the chickadees and the kinglets are doing that, too, and some other species as well.

7.80 So there's an awful lot that humans do that in aggregate makes us, obviously, the most successful species on the planet. But so much of what we do, if you take it one by one, character by character, isn't actually distinct to us. So complex sociality across taxa, again, the long-lived generational overlap organisms that I've talked about, display traits that, again, we will think are going to be distinct to humans, but not necessarily. We've got an elephant's emotion, clear displays of emotion, including grief. And we have grief in other organisms as well, clear displays of grief in, for instance, domestic dogs. Of course, domestic dogs is effectively a coevolved species with us. We and they have coevolved together.

7.81 We have the rapid communication of spatial information in echolocating dolphins. We don't echolocate, but we can obviously rapidly communicate spatial information, in part by looking at each other with the whites of our eyes. We have cultural transfer of social information, but this is not a staged photo. This is Corvids of some sort. I don't remember which. This is some crow, in which one bird is telling the others how it's going to be. This happens in crows, especially. They do a lot of cultural exchange of information. Obviously, we do it more, but they do, too.

7.82 And we have theory of mind. Theory of mind refers to the ability to infer what someone else is seeing or thinking or understanding about the world, even if you are inferring or seeing something else. So there are these famous psychological studies with babies. At what point, if you put a piece of food down in front of a baby or any object down, and then you obscure it for them, but the adult who can also see it can't see that you've obscured it, at what point do they understand that what the adult can see and what the baby is seeing is different? It takes a while. At the point that they

can't roll over, they also don't have theory of mind. They're really, really new at that point, and they can't do most of the human stuff.

7.83 But at some point, pretty early on, we all develop theory of mind and our ability to say, "OK, I know that I'm having this experience, and you're having that experience. And even though we might share watching a third experience, your and my perspectives are going to have been different." And it's not just about we're standing in different places, but it's standing literally in different places and metaphorically being in different places because of our development up until now and what it is that we bring to the world with our biases and such. So baboons also have extraordinary theory of mind. And basically, so do all of the primates in which it's been looked at. But baboons are the ones in which it's been looked at most.

7.84 We're successful. We thrive in almost every terrestrial habitat on Earth. We tell stories to each other, fact and fiction. Sometimes we don't distinguish between those two too well, but we should be trying. We domesticate other organisms for food. Ants do that, too, some ants, but we also domesticate other animals, other organisms, for textiles, for labor, for transport, for protection, and for friendship. That's unusual. We harness the productivity of the land, sea, and sky for our own ends, and we have unlocked many of the universe's secrets. We're remarkable, right?

7.85 We're also compromised, though. Our gait restricts movement and provides back pain. We're good at it. We're not that good at it because it's really hard to be good at being bipedal. Birds do it better. Our linguistic diversity severely limits to whom we can talk with, to whom we can talk. Our children are dependent for a very long time. I'm not saying much more about any of these because, for each of these, we've already talked a little bit about what the flip side is. So, the final point here is going to be that all of these things are a trade-off. We are gossipy, sentimental, and superstitious. For each of these, think about—I'm framing this on the negative, with a negative valence. We're gossipy, sentimental, superstitious. What's the positive valence on that? What's the positive on being bipedal, on having linguistic diversity, on having really long childhoods, on building extravagant monuments to fictional gods? Why do we do that? What is that and what's the good of it? In all of these things, there are trade-offs.

7.86 This is a view. This is a picture that Bret took when we were at Carnaval in Cuenca, in Ecuador, in 2016. There are as many trade-offs in that picture as I can imagine. I'm not going to enumerate all of them. But the point is, in all of our traits, there is a trade-off whether or not you can see it yet. And so even the parts of us that seem unlikely or not good, there's probably a flip side. All right, I'll stop there. We'll take 15 minutes and come back.

Lecture 8: Omega Principle

At a glance — The Omega Principle, the final module.

- 8.1** All right, folks, it's time for the last module. I don't know how long this is going to take. I'm hoping that we have a bunch of time at the end for you to ask Heather and me questions about whatever has come up for you in any of these modules. The last module is going to focus on people, and it's a little hard to know how to convey this. There's a ton that can be said, but there's also a few core things that will really change the way you understand yourself, if you can grasp them. And so I'm going to try to get them across in a way that they will take hold so you can play with these concepts and continue to discover where they lead once this course is over.
- 8.2** So this is really about what we are as human beings and how we got to the future, which is where we now live. Now, in order to get there, I wanted to take us backward a little bit, and I wanted to tell you a few things that you should now be able to see about this hummingbird that will help you see where you are when we get to that point. This hummingbird, you learned from Heather, who introduced you to phylogenetic thinking, is a dinosaur. It's not descended from dinosaurs. It is a dinosaur. And the reason that it is a dinosaur is because, if you clip the tree of life so that all the dinosaurs fall off on one branch, you get every single bird that has ever lived. They are dinosaurs.
- 8.3** So most of the things that we think about when we say the word dinosaur went extinct at the K-T boundary. But from the point of view of getting into the future, that large range of different morphological experiments had one branch that made it. That is a dinosaur. That's how dinosaurs got into the future. In all of those experiments, it was the birds that did it. Now, that, too, is a temporary form. That's the form that birds take at the moment. Now, in the unlikely event that birds get lucky again and that, 65 million years from now, their descendants are still on the Earth, how radically different could they look from that animal?
- 8.4** And the fact is, it's all valid ways of getting to the future. The form that the animal takes is simply a means to an end, which is also true of you. What you are, all of the fantastic things that are so incredibly special about humans—and I do not mean for one minute to underrate the specialness of human beings—but all of that special stuff is means to an end, and it's the same end that all organisms are built for.
- 8.5** So Heather led you through some of the phylogenetic steps that bring us towards being human. I want to lead you through some of the ways in which creatures deal with dynamic circumstances, so that you can see how special we really are. And I want to take you back to a couple of things that we've already looked at. Try to ignore the very charismatic sea lion over there and look at the kelp.
- 8.6** Now, you know that the kelp has a problem with respect to light, which is that it lives in a tidal environment where the water is rising and falling regularly and that the water

breaks down the sunlight. So if the thing had a trunk like a tree, that trunk would either be sticking out of the water and vulnerable, or it would be submerged and lose a lot of potential light that it could photosynthesize with. Instead, it's got floats in it. It sits there at the surface. That also protects it from the trunk breaking in the water. It's a very good solution to the problem of a dynamic water level and dynamic water flow.

8.7 The strangler fig has a different problem, which is that the light that reaches the forest floor is scarce, and it needs to get to the canopy. And so it has developed a mechanism for moving down from above. It's a very good solution to that. It does build a trunk, but it starts at the top and goes down, and it has plasticity. It can also start from the ground and go up, in the rare case that it ends up in a situation where that's possible. So those are two non-animals that are nonetheless capable of dealing with a stochastic environment, an unpredictable environment, with a clever strategy.

8.8 Animals have a much greater range of mechanisms for dealing with stochasticity. Here we have an arthropod. This is a spider. And that spider is capable of deploying strategies much more rapidly than either the kelp or the strangler fig or almost any plant. There is one or maybe a few plants that are capable of making rapid motion, like a Venus flytrap or the so-called sensitive plant that can close its leaves when herbivores touch it, but the spider can reach out and grab prey that comes to this rock. That said, it is deploying almost entirely a built-in program. It's simply deploying it automatically. And that built-in program leverages the sensory apparatus, and it leverages the muscles, and allows it to actually successfully capture prey.

8.9 It may also have done a certain amount of choosing. It may have chosen that rock based on the fact that that rock had characteristics that made whatever program it was using think it was a good bet, or it may simply be the result of selection. That rock may be a good bet, and there may have been spiders on a lot of rocks that weren't as good a bet. And I saw this spider because this spider happened to survive long enough to be seen. So we don't really know. But the point is, it has a high capacity to deal with a dynamic environment. It can actually grab prey in real time, but it's limited in terms of how flexible that program is.

8.10 It's not quite fair to say, though, that arthropods cannot learn. That's sort of shorthand that we use, but we know it's not true, because here we have a bee, and we know that bees can learn. Well, how do we know? We know because a bee, when it returns to the hive and it tells its sisters that there's a particularly worthy patch of food, and it describes the direction of the patch of food, the height of the patch of food, and the distance of the patch of food, it has to have recorded those parameters. It couldn't convey them if it hadn't remembered them in some sense.

8.11 On the other hand, it's three parameters. It's not a highly complex description of the world. It's just enough information to convey to some other creature how to get to a particular location in space. It learns, but not much. In general, arthropods don't learn very well. And the reason that this bee orchid is capable of getting itself pollinated, or

getting another individual pollinated with its pollen, is that they don't learn. If, after the bee had attempted to mate with this plant and had discovered that it was not, in fact, an animal, it had decided not to do that again, then there would be no pollination. It is only because the animal fails to learn the lesson that this pollination strategy works. So it is a rule of thumb, but the rule is arthropods don't learn, at least very well.

8.12 Now, we can't say that all invertebrates are this way. And I felt compelled to put this slide in here, even though it's going to very much complicate the story I'm trying to tell you. Because this cephalopod mollusk—this is an octopus—is spectacular at things that you wouldn't think an invertebrate could do. They appear to be highly intelligent. In fact, they must be highly intelligent to solve some of the problems that we've seen them solve. They seem to have a kind of consciousness, though it's impossible to prove that it is there. But nonetheless, they are the exception that proves the rule about invertebrates and learning and intelligence and consciousness.

8.13 They have many of these characteristics, even though their life history does not suggest that they should. They're not social. They don't meet their parents. The mother will actually die in the place that she has laid her eggs, and she will be gone, and they will consume her at the point that they hatch. So the things that we tend to associate with high levels of intelligence do not accompany this life history strategy, and yet the individuals have characteristics we don't expect. Hopefully, we will find out the answer to why it is the exception to those rules. But nonetheless, at the moment, let's just say it is for some reason.

8.14 So arthropods, not great at learning. Here we have an animal that can learn. This is a vertebrate. In this case, this is a gecko. This one happens to be from Ecuador. And this is an animal that can figure out idiosyncratic features of its habitat and can exploit them. It can have a favorite place to hide. It can have places that are very productive for hunting. It can do all of these things. It can recognize that a certain kind of animal is a threat to it and deploy a different strategy than it might to some other animals. So this is a fairly intelligent creature that has a high capacity for a dynamic response to its environment.

8.15 But it does have a problem. There's a limit to what it can do with that intelligence because it doesn't meet its offspring. This animal will end up laying eggs somewhere, somewhere carefully chosen because it's hospitable to those eggs surviving. And if the eggs survive, they'll hatch and they won't meet the adult. Now, that does tell us something pretty amazing. And this is also true of the arthropods. The program necessary in order to be a successful gecko, it's in the egg somehow. Somehow in the genes that are transmitted in that egg, everything it needs to know is there.

8.16 Now, what it needs to know might amount to, "here are some things you should try out and then remember what happens and refine your own program." But nonetheless, this creature self-deploys its own dynamic behavioral response without the need to meet its parent. You could take it to some place where this species doesn't exist at all,

and it would still figure out what to do if the right conditions were present. So it's all there in the egg somehow, and that is a really remarkable statement. But nonetheless, the limit comes from the fact that it doesn't meet its offspring.

8.17 Same here. These are creatures that are good at learning. We know that because of the story I told you about how it is that they make it back, in most cases, to their natal birthing ground. They do it by remembering a pattern of basically the chemical signature that grows as the animal swims downstream, and then years later it reverses the pattern, and it finds just the right set of tributaries to bring it back to that place it was born. That's obviously learning. It's pretty impressive. But again, that animal will be dead before its offspring hatch. They don't meet, and so it can't convey anything other than through the genes.

8.18 All right. Now, I wanted to talk about a couple of caveats that have to be made. We're going to discuss human beings and the very special fact of the relationship between different generations. The fact that we meet our offspring has huge implications, but it's not sufficient in itself. And I wanted to make that very clear. In both cases here, we have a caiman on the left, and we have leafcutter ants on the right. The caiman carries its offspring to the water. The leafcutter ants brood their offspring in a nest. They meet, but that doesn't mean that they convey anything important about how to be such an organism. They don't behave as if that's the case, right? Especially in the case of the leafcutter ant. Mostly what they do is deploy an automatic program that makes them successful at doing what leafcutter ants do.

8.19 But then we get to mammals like these tent bats. Here again, we have a mother and her offspring. Here, we have a mother, a putative father, and another offspring. And this kind of intensive relationship opens the door to a very different kind of adaptation, a cumulative kind of adaptation. And that's really the first thing you have to understand about what's special about humans. And again, as Heather said, most of the things that are special about humans aren't completely without precedent, but they are so extreme in humans that they really are as if new.

8.20 Here we have a case of a primate. This happens to be a squirrel monkey. You see a mother and her offspring. And in this case, you've got the clear opportunity for that adult to convey things in many ways to this offspring. The offspring, just simply by observing its parent, can figure out what to be worried about and what not to worry about. It can figure out how to forage effectively. It can learn what kinds of branches can support its weight, what kinds of gaps can be leaped. All of these things can be conveyed just simply by sticking together as they go about doing the business of being a squirrel monkey. And that conveys a tremendous amount of information, not just about how to be a squirrel monkey, but how to be a squirrel monkey right here.

8.21 And if there's one thing we know, it's that that baby is going to end up being an adult somewhere near where that mother taught all those lessons. Those lessons are likely to be highly relevant, and they may be very attuned to the particular patch of habitat and

its idiosyncrasies—the presence of certain animals there, the absence of other animals, the presence of certain places that one can cross between an unproductive patch to another one in relative safety. All of those things can be conveyed.

8.22 I could say the same thing about these sea otters. In the sea otters' case, you've got this coastline that has all of these nuances, little coves, outcroppings of rock, all sorts of things for which there might be very specific lessons that the adult picks up, either from trial and error or from their parent, and then passes on to the offspring, that the offspring will be more successful for having. Here we have another example. Eagles. They have these territories that have certain opportunities in them, and they learn to exploit those opportunities very particularly. They're very strategic creatures. They hunt a wide variety of things that require different strategies to be successful, and they convey these things to their offspring. And that is the nature of a highly intelligent vertebrate. But it's not yet where humans are.

8.23 The thing that makes humans so different is language. And, yes, there are antecedents in other creatures, no doubt about it, even up to syntax. However, the thing that we do with language is absolutely extreme and extraordinary, and as much as I think that it is almost impossible to overrate its importance in the human story, even I, I'm sure, am underrating it. It's just that significant. So what language does is it allows us to convey information with great precision, but not just information. The bee conveys information with precision. It's important information. "Where's the food?" But it can't convey abstractions, especially novel abstractions.

8.24 If I come up with some thought that no one has ever thought before, there's a pretty good chance that I could vibrate the air molecules between us in just such a way that you would end up with a facsimile of that thought in your mind. That's a miracle. Every time it happens, you do it every day. You don't think about it. But this is a miraculous capacity, right? If I say, "A ping-pong ball sitting on the tip of a Sharpie marker," right, and then I were to ask you to draw that, you'd all draw something in the neighborhood. We could collect the drawings, and we could look at them, and we could be pretty darn certain that I had successfully conveyed this concept, even if no one's ever said those words before or thought that concept, right? Conveying that over open air and abstraction, and then being able to be certain that it has been conveyed, is a remarkable capacity.

8.25 Now, in this picture, we have !Kung Bushman. We have three generations of them. And this begins to hint at the thing that really makes human special, the ability to transmit abstract ideas between minds, not just by showing, by telling. That then sets the stage for cumulative culture. You all sitting here today are not much modified from your hunter-gatherer ancestors. But in terms of how you understand the world and how you make your way in the world, you're unrecognizable. They literally would not understand what they were looking at if you teleported them into present and let them observe you. Where's all the food production? How are you folks feeding yourselves?

You don't seem to be all that concerned about finding tubers or ungulates or anything of the kind. You're very different creatures.

- 8.26** And yet you're almost unmodified in terms of your physique. That's a stunning fact. And it is owing to you having a collection of mechanisms that allow you to change evolutionarily, adaptively, over time, without your genes being terribly, deeply involved. It's an amazing statement, right? You evolve, and not only a little, but radically different. You become that way with changes in things that aren't genes. And this allows a lot of things to happen.
- 8.27** Human beings are effectively a generalist robot. Think of your body as a robot. Now, I don't mean a metal robot. But I mean, you are an elegant machine, and you are not built for specific tasks. Right? Humans aren't best at almost anything physical—maybe long distance running being one of the few things we can point to. Manipulating objects we've become pretty good at. The point is, you are a generalist robot. You can do a lot of things, and if you can't do them with what you've got, you might be able to make a tool that allows you to do them. Right? That's amazing.
- 8.28** But that generalist robot has something that makes it into a superorganism. The thing that it has is a very powerful computer. That computer is generated by a genetic program, but that computer is not overly programmed from the factory. When you're born, as Heather points out, you can't even roll over. You've got a lot of room to learn how to be a human, and that's not one answer, right? If you were to take a modern person and teleport them into the past, into the Pleistocene, and you were to have them raised by hunter-gatherers, they'd pick up the hunter-gatherer program. Of course they would, right? If you were to transport a baby who was born in Manhattan and you were to take them to Beijing, they would speak Chinese, just like a baby that was born in the hospital there.
- 8.29** These programs get loaded on after we are born. They're not entirely loaded on by other individuals. There's a lot of trial and error involved, a lot of exploration. But nonetheless, cumulative culture, right—the ability to pass the program that you have to your offspring, and though it's a little painful to say this, their ability to ignore the part of your program that's no longer useful—all of that creates an organism that moves through time and is capable of being up to date without the genes having to do the catching up, right? Freeing us from the genes was a bit of genetic genius, as it were.
- 8.30** And it creates the opportunity for a massive division of labor, because as much as human beings are a species that is very good at dealing with the idiosyncrasies of the particular habitat it lands in, that's not where it stops, right? You can have different people specialize on tiny little pieces of what it takes to live as a human being. And by collaborating and by exchanging the products of what they produce, the point is everybody gets what they need to get into the future, even though most people are doing something narrower than would be necessary on its own. Now, again, we have to

go with the caveats. Biology is so complex, as Heather said, it's very rare to find a true rule.

8.31 Here, we have another species with a clear division of labor. In fact, it's a morphological division of labor, right? Ants have different castes, and the castes have different shapes, depending upon what role they play. In this case, what you have is a leafcutter ant carrying that leaf out of the canopy on the way towards the fungus farm underground. And there's another ant riding on the leaf. Is that a freeloader? Nope. What that is, is defense for the leaf against syrphid flies. Syrphid flies, which try to lay eggs on the leaf to get into the colony to parasitize the product of the agriculture. This guy is going to defend that leaf from the syrphid flies. That's a division of labor.

8.32 But the point is, that division of labor didn't arise because somebody said, "Hey, you know what would be a good idea? I'm sick of these syrphid flies. What we're going to do is we're going to have some of you ride on the leaves and fend them off." That's obviously not what happened. There's a genetic program, and that genetic program takes genetically similar ants and causes them to end up with a different morph, right? Just like the tadpoles that end up with a different morphology if they end up in a puddle with something meaty to eat, rather than just a bunch of muck, right? So it's a kind of plasticity that results in different cast of ants doing different roles in the colony.

8.33 Ok. It is time that we talk about omega. Omega is a principle. Those of you who have read our book have run into omega before. Omega is important. We chose omega, a Greek letter, in order to evoke pi. The reason that we wanted to evoke pi is that pi specifies an unalterable relationship between the diameter and the circumference of a circle. And the whole point is unalterable. Show me a circle —pi. It's there. Omega is the unalterable relationship between our genes and the epigenetic phenomena that ride with them. Epigenetic means above the genome.

8.34 And it is important to say that much of what we have talked about does involve epigenetics of various kinds, but that word has gone through a transition. It used to simply mean above the genome. And then we discovered all of these incredibly powerful molecular mechanisms that regulate gene expression—the methylation, the histone wrapping, that adjust which genes are turned on where. And those things are incredibly important. In fact, you couldn't be an organism if they didn't do what they do, right? Because the fact is, all the cells in your body have the same genes, right, except your red blood cells, which have lost all their genes, but never mind. All the other cells have the same genome, but they look different and they work differently, and they agree to work differently.

8.35 There's no disagreement in the body about whether or not your liver cells would prefer to be eye cells. Your liver cells, once they're in the liver, they're perfectly happy doing liver stuff. All of that depends on the genes that are not relevant in the particular tissue in question being turned off, and they willingly turn themselves off using these epigenetic mechanisms. So the ability to be something other than a blob of cells is all

contingent on those molecular mechanisms. Again, it would be impossible to overrate their importance.

- 8.36** But the problem is that the word epigenetic got radically narrowed to just those mechanisms at the point we figured out what they were. And that means that what we used to mean by that term, which would certainly have included culture, which also regulates the expression of the genes that you bring to the table, those things are now not thought of. When people say epigenetic, we instantly think of those molecular mechanisms. That's a mistake. Omega is the relationship between the epigenetic mechanisms, including culture and the underlying genome. And, strangely, the relationship is the same, whether we're talking about the molecular mechanisms or we are talking about the cultural ones.
- 8.37** So with that in mind, let's talk about what this unalterable principle might be, and at the very end, we'll talk about whether or not we can alter it. The omega principle has two parts. I don't know if that's hard for you to read, but it says, "Epigenetic phenomena, including culture, are more rapidly adapting and therefore superior to genetic phenomena." Does that mean they're a competitor to genetic phenomena? Nope. The genes have offloaded the work to the cultural layer because the cultural layer is faster to adapt. It's more dynamic. So part one, epigenetic phenomena, including culture, are more rapidly adapting and therefore superior to genetic phenomena.
- 8.38** Part two. "Epigenetic phenomena are subordinate to genetic phenomena which retain control over the objective." Now, we've been talking about this since the beginning. The genes are in charge. The whole point of the exercise, all of it, all that dynamism, all those epigenetic regulators, all of their refining you into a marvelously capable machine, all of that stuff is for one mind-numbing purpose, and that purpose is to get your genes as far into the future as you can. If you can manage it, you want to be like that hummingbird. You don't want to go extinct with the other dinosaurs. You want to escape through the bottleneck. Whichever form you have to take is the one you should take.
- 8.39** So that's it. The genes have created a layer that is more dynamic than they are, and it has offloaded as much work as it can to that layer, but it has retained control over the objective. You are the most marvelous thing that has ever occurred in the known universe, and I'm not exaggerating. But your purpose sucks. Your purpose is simply to get your genes into the future. And what's more, from your genes' perspective, it matters. The difference between you and the person you're sitting next to, the fact that your genes are spelled differently, even if their genes are spelled better than yours, even if they have an enzyme that does something more efficiently than your version of that enzyme, your enzyme wants to get into the future, and it wants to exclude that better version of the enzyme. That's how petty the genes can be.

- 8.40 So in any case, I won't defend them, but we need to recognize that this is what happened. We are in a cosmic spelling bee. It is a terrible game that we have been put to, but it has created the most marvelous machinery that you can imagine, including all of our best characteristics, our ability to collaborate, our ability to be compassionate. All of these things, our ability to produce beauty and insight, all of these things are a means to that stupid end. So anyway, the real question at the end of this is going to be, can we take control away from the genes, and can we repurpose ourselves, and can we give us an objective that's actually worthy of the marvelous machinery that selection has built for us? That's it. That's the deepest philosophical question I know.
- 8.41 OK, the upshot of the omega principle. "Culture is the primary mode by which we adapt." Why? Because it's so great. You can actually have something desirable. An adaptation can arise, and it can spread horizontally across a population, right? The Boxing Day tsunami that Heather mentioned, that tsunami was a terrible tragedy, but it caused the world to suddenly become aware of what a tsunami was. That knowledge spread across the Earth in a matter of 24 hours. Everybody understood that there was this hazard. They understood how they could recognize it. That's an amazing thing. It's an insight spreading literally, physically, across the surface of the Earth. And a powerful one, a life-saving one. So culture is the primary mode by which we adapt. But propelling our genes into the future is still the objective of the exercise. Everything that is special about humans is a means to that same end, and we share it with all other creatures.
- 8.42 OK, it's time for us to talk about campfire. Campfire is a metaphor that Heather and I use, but it's almost not a metaphor. We're alluding to something that was very real and very physical and happened millions of times. So the argument here is that human beings actually have a brand new niche, brand new with our species. And it involves culture, which we've talked a little bit about, and consciousness, which we've barely mentioned. So the argument is that humans expand through space and across time by deliberately exploring new niches. This is what we've been referring to as an explorer mode. And I'm going to try to explain in relatively simple terms how this explorer mode works so that you can see how it is manifest through history, and maybe even more importantly, prehistory of humans.
- 8.43 What has happened is that our niche has become niche switching. That's a funny thing to say. What is a niche? A niche is an opportunity that can be exploited productively by biological organisms. A niche is something that you're good at dealing with enough that you can make a profit in it so that you can produce offspring, so that you can get at least one generation into the future. That's a niche.
- 8.44 And niches fill up. If you have a niche, if you happen into some valley and you have the skills to deal with that valley, well, how many people can that valley hold? Your population is going to rise to that level. That's what we call carrying capacity. And then it's going to stabilize, right. That's what's going to happen. So that's a niche. But our

niche, I am arguing, has turned into something else. Our niche has become escaping niches, escaping niches into new niches, some niches that have never been inhabited before. So that is the ultimate explorer mode, and it involves this campfire metaphor, culture and consciousness.

- 8.45** So let's start with what consciousness is. People will tell you it's impossible to define. And the problem is they're trying to define something too big. They're trying to find a definition that suffices to deal with all of the various uses that people put that term to. And I'm going to argue there's a better way to do this. The better way to do this is to listen to what people say when they talk about consciousness, and then figure out what the epicenter is, right? We can infer what we must mean by that term by figuring out where the center of gravity is when people tend to talk about it. That means you throw out the outliers, the people who believe that the particles might be conscious, right? We don't need to deal with that. That's a fringe idea.
- 8.46** What we need to do is get at the thing that we all mean when we say consciousness, right? And what Heather and I argue is that consciousness is that fraction of cognition that is packaged for exchange. It is that fraction of what you think that if somebody said, "Hey, what are you thinking?" , you could tell them, right? And it's not the part that is, you know, controlling your rate of breathing, right? You couldn't report how that thing works to save your life, right? If you picked up an object and you threw it so it would hit the wall in the back of the room, and I said, "Well, how did you contract the muscles to get that to happen?", you couldn't tell me. It's not conscious. It's neurological, but it's not conscious.
- 8.47** The conscious part is things that you can pass on to somebody else. That almost always gets passed on between members of the species. And I say almost always, and of course, into my mind leaps my dog, because I can convey a certain amount of stuff to my dog, and my dog can convey a certain amount of stuff to me. But that is, of course, the result of a special case, a case of domestication that goes back three times as far as the next nearest competitor, as far as we know. Dogs have been human partners for 30,000 years. That's 30,000 years of getting used to how the other thinks.
- 8.48** And so we can do a certain amount of cultural exchange there that would have to be rated as conscious, but it's an exception to the rule. So everybody have it? Consciousness is that fraction of cognition that is packaged for exchange. And we've already talked about what culture is. Culture is information that is transmitted between individuals.
- 8.49** So the argument about campfire is that the apparatus that we have that allows us to go do ecological things during the day, to figure out how to make a profit from the environment that we are in, is very, very powerful. And nighttime isn't a good moment for it, right? As Heather said, trade-offs in all things. We happen to be very diurnal. If we tried to do the stuff that we do during the day at night, we'd be very vulnerable to predators, we'd be very vulnerable to injury, and we wouldn't be very productive at it.

You can't see well enough to do all that human stuff during the night. So you've got this amazingly powerful computer riding around on your shoulders, and it's got nothing to do.

8.50 So what do we do? Well, we gather. And the campfire—it's too simple to say that it is the place of consciousness because it's also the place of culture. What we do at the campfire is we exchange relevant information. Now, it may be that the relevant information is just passing on what we've learned about our niche, or it may be that what goes on around the campfire is about talking about the fact that our niche isn't as good as it was before, that there might be some reason to change some facet of it up. And the argument that we make is that consciousness is about exchange because its fundamental nature is not individual.

8.51 We all tend to think that consciousness is an individual phenomenon, that it is my subjective experience. And consciousness certainly involves subjective experience. If you want to convey to somebody what it is you're thinking, the first thing you have to do is figure out what it is, which involves feeling it, right? It involves looking at your own thoughts before you can transmit them. So there's no question, you have this subjective experience. But the consciousness is a mystery. What is the point of thinking all of this stuff in any sort of abstract way inside of your mind? How does having a subjective experience of a thought make it any better?

8.52 Well, it's not obvious that it does. But what does make it better is when you pool it with the thoughts of other people. And remember, human beings have this incredible capacity for a division of labor, which means that when you gather around the campfire, you're not gathering with dozens of people who think like you do. You're gathering with dozens of people who think somewhat like you do and somewhat differently than you do because they've had a different experience. They may have a different role in the society, and so they have different strengths and different weaknesses and different blind spots.

8.53 And so if you're puzzling over something, the best thing you can do is hand it off to a bunch of other people who don't have your blind spots. You hand it off to a bunch of people, and then what they can do is they can parallel process those thoughts. And the parallel processing of thoughts—shared consciousness—is what creates the emergence that allows human beings to be unlike any other species, right? That's because the sum of those thoughts, asymmetrically parallel process by different people with different strengths, is greater than the sum of all of the individuals thinking about it in isolation. Not only do we have cumulative culture, but we have cumulative computing power, and then some. It's emergent. Our thoughts are better when we talk about stuff, which is why we are so obsessed with doing it.

8.54 Now, consciousness is not the tool that you should use for things that are relatively straightforward, right? And this is a little confusing to us because, living as moderns, we have to point our consciousness at a tremendous fraction of what we encounter. That's

not a normal state of affairs. That's actually a pathological state of affairs. The normal state of affairs would have your conscious mind reserved, riding more or less as a spectator until it encountered something that was new, something that your unconscious mind, which is actually much better at doing things you're good at, couldn't handle, right?

8.55 Now, by analogy, you should think about an elite athlete, right? Let's say a tennis player. That tennis player is not thinking their way through the tennis game. In fact, when they start thinking, they choke, right? That tennis player is trying to get in the zone and stay in the zone. And what being in the zone means is that their conscious mind is not in control. The part of them that really knows how to play tennis well is in control. But imagine that tennis player is on the court and their opponent, who they've never played before, keeps putting the ball somewhere that they can't get to it. What is that person doing? What are they seeing about my behavior that is conveying to them where they should put the ball that I won't be able to get to it? That's a new problem. That's novel. And the fact is, your unconscious mind isn't so good at those problems.

8.56 Your unconscious mind needs your conscious mind to weigh in and say, "Well, what is it that they are noticing? What happened each of the times they put the ball where I couldn't get to it?" And maybe you figure out an answer, and then maybe you test it. Your conscious mind says, "Well, if that was true, then I'm going to do the thing that causes them to put the ball over there, and before they deliver it there, I'm going to start going." And you see if that works. And then if it does, you drive it into the unconscious layer, and your unconscious mind learns to play that new trick, right? That's the way it's supposed to work.

8.57 So the punchline is, the purpose of consciousness is to deal with novel circumstances. The rest of the time, your conscious mind should stay out of it. It doesn't mean you should go to sleep. It should watch passively, and it should wait for things that are in its area of specialty. For us, it's all reversed because you're encountering new things every day. Your conscious mind is puzzling over the most mundane stuff, right? When you think about a hunter-gatherer wondering whether or not they should eat something, the answer is, if that something is in the hunter-gatherer's habitat, they know whether they can eat it. But you're there in the supermarket looking at the label on a package, and you don't know what half the words even mean. Should you put it in your mouth? Who knows, right? Your conscious mind is bewildered by the choice and bewildered by the fact that you even have to make a choice, and that there are 20 different versions of the item with slightly different ingredient lists. That is a waste of consciousness.

8.58 OK, so the result of our ability to parallel process information and to have cumulative culture is that somehow it allows us to behave like many species at once. Think about this. If I say to you, "What does a leaf-tailed gecko do?" Well, if you know what a leaf-tailed gecko does, you'll tell me that it hides by flattening some appendages around its surface against a tree so you can't see it, and it waits for prey to wander up the tree,

and then it grabs it. You'll tell me that. So, all right, that's true. That sounds like a species has a niche.

8.59 But what's our niche? Aren't there people who hunt and gather? Aren't there people who farm? There are people who gather birds' nests from high inside caves to make bird nest soup. The fact is, human beings, even before you get to modernity, have done hundreds or maybe thousands of distinct things simultaneously. It wasn't like those were a sequence of things that humanity passed through. It's doing all these things at once. That is, again, the result of the fact that you have a computer riding around on your shoulders that can absorb many different programs. You can absorb a program that is relevant to your habitat.

8.60 And what program are you going to get? Well, under any normal circumstances, you get the program of your elders, your ancestors. They knew what to do in your habitat, and they pass it on to you when you're young. And that all works great. Except for a couple of things. The fact is, you have dynamism over both space and time. People can exploit a niche, and that niche can cease to function. And then what do you do? Even when your niche is functioning, all of these niches have a spatial dimension, and there are people at the edge of that niche. Once the carrying capacity has filled up that niche, it literally has people on its edge who aren't doing so great, because the stuff that works in the middle of the habitat doesn't work so well at the edge. Right?

8.61 So the claim is that when people find that, for whatever reason, either because the ancestral wisdom has gone out of date or because it is relevant to somewhere that isn't quite right here, there is a pressure. There is a pressure to discover what new thing we might do that we could substitute for the wisdom of the ancestors that isn't working for us. Now, in order to understand what that looks like, we're going to use the example of the Beringians, which is really the greatest story never told. Now, I don't want to pretend that we understand it better than we do. What we know is that there was a population of something like a couple thousand people living on a continent that is now submerged, called Beringia, in the Bering Strait. It's now submerged because sea level has risen. This was an Ice Age continent, and although you might think it would be covered in ice, it wasn't. It was a cold grassland.

8.62 And so the idea is, as water that falls on the land remains frozen, sea level drops, and there's lots of stuff around the world that emerges from the sea because it's very shallow. Beringia is such a place, and the Beringians lived there for quite some time, probably a couple thousand years. At some point—and there is a lot of contention over when that point was and exactly what happened; we really, truly don't know—but at some point, the descendants of these Asians who had been living in Beringia found their way under the ice sheets that were covering what is now Canada and in North America. And the amazing thing is that they became the ancestors of all of the New World native populations. All of them, right? This tiny group of people that made it under that ice sheet are the ancestors of all of the populations that were in the New

World, with the exception of the Inuit, at the point that Europeans rediscovered America.

8.63 All of those populations had a bottleneck right here. And then they founded all of these ways of being. And all of these ways of being were terrifically diverse. There were something like 50 million people in the Americas at the point that Columbus rediscovered this side of the world. And they were doing amazing stuff. Some of them were still hunting and gathering the way their Beringian ancestors had been. Others had created complex societies, like the Maya. And the Maya were truly sophisticated.

8.64 The Maya, not only were they able to do things like predict the location in the sky of Venus, a truly difficult astronomical problem, they also had higher math, including a concept of zero, which even the Greeks didn't have. And they had intensive agriculture in which they were modifying fragile soils chemically in order to make them more productive. There was a lot going on in the New World, including a written language. That is also something that the Mayans had. So all of that is the result of this population of people who walked into the Americas. Yes, they had the ability to control fire. That is certain. They had language, but not written. They had no agriculture. It was invented in the Americas, separately from the examples of it being invented in the Old World.

8.65 How could that all have happened? How could simple hunter-gatherers, in a matter of 15 to 20,000 years, produce all of that complexity? Well, the answer is, they had a tool for bootstrapping their way into new niches. And what they did was they filled up niches. And as they filled them up at their edges, people must have figured out how to deal with the next niche. Now, Heather and I are both very fond of the idea that they probably came in along the coast because a lot of the things that would have applied in the coast of Beringia probably also applied in the northern parts of North America. But the coast is a line. It's got a lot of edge.

8.66 There's a lot of profit that can be made just a few miles from the coast if you could figure out how to go from either gathering shellfish or hunting fish or hunting mammals in the sea to hunting mammals on the land, figuring out what plants might be productive. That's all just waiting to be discovered. And the moment you do discover it, if you can figure out how to do it well, then the amount of reproduction, the growth in the population, and most importantly, the chances of your descendants getting into the future goes spectacularly up.

8.67 So what we are suggesting is that people at the edges of all of the niches that would have been found first here would have innovated the next niche over, and they would have quickly spread across the Americas and discovered all of these different ways of making a profit. OK, I hope everybody has the model. There'll be a chance for you to ask questions shortly. I said that the conscious mind in the modern world is occupied with mundane things that really shouldn't concern it, and that this has caused us individually to have a very confused view of what it is to be a human being. I want to

point out a couple of things about what it would be like if you lived in an environment that made sense.

- 8.68** These are stairs. You walk down them all the time. You have no idea how complicated the program is that allows you as a biped to walk down those stairs without falling and breaking your neck. It's really hard. Ask people who have programmed bipedal robots to do this, and they'll tell you how hard a program that is. But you do it without a thought. I have walked down the stairs in my house, putting my shirt on, right? Pulling it over my head, blind. I can walk the stairs. I know where the stairs are. It's not my conscious mind that's walking those stairs, right? It's very familiar, right?
- 8.69** Now, you can mess somebody up. If you build stairs and you put one a couple inches high, wow, people will trip on that thing because their automatic stair program doesn't work in that case. But the basic point is, you are very good at doing highly complex stuff without your conscious mind getting involved, right? Walking down the stairs is one of a dozen, 100 activities we could name in which your unconscious mind is perfectly capable of doing it, even though it's really tough. So there's that.
- 8.70** And then there's the question of emotion. And I was spurred to put this slide here by the question about whether or not it takes courage for these flying squirrels to leap out of the tree. I thought that was a very interesting question, and I puzzled on it for a while. And what I came up with was, well, there are a couple of things here. When we say that these guys glide in order to avoid predators, mostly the predators they're avoiding are on the ground. So the point is, you have to think, yes, there is something called courage, but it might even be that this is the coward's way out. They're leaping from tree to tree, so they don't have to go to that frightening place called the ground where they could get nabbed. So something along those lines.
- 8.71** But in any case, the underlying point is this animal has emotions. We can be relatively certain of that because of the group that it belongs to. And those emotions likely dictate when it feels ready to jump and how it navigates the world. And the argument I want to make is that when you live in an environment where you are well adapted, and therefore your unconscious mind is in control of a lot of things, what that actually feels like is walking around in the zone, the same zone that tennis player would be in. So you would walk around and you would just feel your way. You would intuit your way through things. Something would seem too dangerous to be worth your while. Something else would seem appealing to you. You wouldn't have to think so carefully about whether or not there was something that you were missing.
- 8.72** That's presumably how an animal like this flying squirrel navigates its world. It feels it's time to jump. It's time not to jump. It's time to hide. In fact, I will tell you, these guys are not so rare. They're hard to see, though, because they're so quiet unless they're fighting. We have twice seen them when they were bickering with each other, which called our attention to the fact that they were flying squirrels gliding over. And both times, the same thing happened, which is an owl heard them bickering and flew in, and

suddenly the animals became invisible. They went quiet, and they didn't move, and they didn't glide. And anyway, that tells you something. That was probably not a matter of them thinking, "Oh, geez, owls, time to hide." It was a matter of them having that program on board, and the emotion of terror causing them to do exactly the right thing because they were built for it, because they're in exactly the environment for which they were programmed.

- 8.73** So our problem is that we are an environment for which we were not programmed. Well, why didn't our parents program us for it? Because it didn't exist. They don't know anything about it. And this is the issue of hyper-novelty. I do not know what this young woman is doing. She is riding a one-wheel personal electric vehicle and juggling some balls that are glowing. Is that adaptive? I mean, it mostly passes the adaptive test, right? It's complex, it's expensive. She seems to be doing it a lot more than some other people. The one thing it doesn't do is pass the test of time. We don't know if that's adaptive. If 500 years from now, everybody's riding one of those around and juggling those things, we'll know. But we don't know.
- 8.74** And this is true for virtually everything in modernity, right? Virtually all of the stuff, even the stuff that you think is probably all right, you don't know. We will know at the point that we've had the huge experiment of people being exposed to it and some people not doing so well after they did. And we keep injuring ourselves because we don't understand this lesson. We will know if it's good for us at the point it's passed the test of time. Until then it's a lot of speculation. What we are living in is what Heather and I call the hyper-novel world. The hyper-novel world is a world that is changing so rapidly that the world you were born into doesn't exist when you become an adult. And what that means is that you're dealing with everything consciously because you've got no other choice.
- 8.75** You didn't build a program for this world because this world didn't exist. Literally nobody knew it was going to happen. So that is not a healthy situation. It's making us sick in every conceivable way, psychologically, physically, socially. We are not doing well because the rate of change is above the rate that we are capable of dealing with evolutionarily, even though we are the fastest evolving creatures that have ever existed, the fastest adapting because so much of what we do has been offloaded to this cultural layer which can adapt on a dime. This is still too fast for us, and we are going to have to rein that in if humanity is going to make it another 1000 years.
- 8.76** OK, so the final question I want to put on the table. I've been very clear about the fact that, as far as we can tell, all creatures have the same purpose. And even though we have been endowed with this amazing capacity to adapt by offloading all of what is ordinarily done by genes in most creatures to a cultural layer that is capable of changing dramatically and rapidly, even though that is the case, we are still set about the same mind-numbing purpose that every single one of our ancestors has been on about for 3.5 billion years. Now, we are amazing creatures. We have all of these

marvelous capacities, and many of them are truly beautiful and laudable. That's for real, even though they were built for this purpose.

8.77 The question is, is there any conceivable mechanism for somehow getting out from under omega so that we can take the amazing thing that we are, and we can use it for something worthwhile, that we could make a better, fairer, more sustainable, more decent world, that we could preserve the opportunity that exists here on Earth for as many generations as possible? Because at this rate, what we're going to do is we're going to compete ourselves into destroying the planet we live on, and we're going to deny lots of human beings endowed with those same marvelous characteristics—we're going to deny them the ability to live on this marvelous planet, and that's not right. So the question is, can we escape omega, or is it game over because omega is inviolable?

8.78 And the last thought I will offer you is that there is enough flexibility inside omega in order to do this. And it's not even a technical violation, because if you think about that hummingbird, right, that hummingbird was the right plan to get the clade Dinosauria into this future. It's a very strange plan, but it was the right one. Now, my claim is, if we don't wise up about the problem of omega and the problem of hyper-novelty, we'll destroy ourselves. That means our genes won't get any farther into the future. So there is a way in which if you really understand the lesson of these things, what it says is that the problem needs to be abstracted so that we can understand just how dangerous it is, and we can take proper action to deal with it. And if we take proper action, then we are, in fact, living by omega, because that is the only way to get our genes deeply into the future.